Volume-independent local estimates in the full SU(2) Wilson vacuum

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1. Original Hamiltonian, physical state, and geometric input

Fix an integer $L \geq 2$. The vertices are $\mathsf{V}_L = \{-L, \dots, L\}^3$. A physical link $e = (n, i)$ is oriented from n to $n + \mathbf{e}_i$ whenever both endpoints are vertices. Its independent variable is $U_e \in SU(2)$; an inverse traversal contributes U_e^{-1}. Every contained elementary face $p = (n; i, j)$, $i < j$, is included, with

$$W_p = \operatorname{tr}\left[U_i(n) U_j(n + \mathbf{e}_i) U_i(n + \mathbf{e}_j)^{-1} U_j(n)^{-1}\right]. \tag{1.1}$$

Write $\mathsf{E}_L, \mathsf{P}_L$ for these complete sets, and

$$N = 3(2L)(2L+1)^2, \qquad M = 3(2L)^2(2L+1).$$

The Hilbert space is $\mathcal{H}_L = L^2(SU(2)^{\mathsf{E}_L}, dU)$, with product Haar probability measure and inner product conjugate-linear in its first entry. The differential operators and all physical constants are

$$T_a = -i\sigma_a/2, \quad X_{e,a} f = \frac{d}{dt} f(\dots, e^{tT_a} U_e, \dots)\Big|_{t=0}, \quad E_e = -\sum_{a=1}^{3} X_{e,a}^2,$$

$$H = \kappa \sum_{e \in \mathsf{E}_L} E_e + b \sum_{p \in \mathsf{P}_L} (2 - W_p), \qquad \kappa = \frac{2g_{\mathrm{YM}}^2}{a}, \quad b = \frac{1}{2g_{\mathrm{YM}}^2 a}, \quad \xi = \frac{b}{\kappa} = \frac{1}{4g_{\mathrm{YM}}^4}, \quad a, g_{\mathrm{YM}} > 0. \tag{1.2}$$

In the original metric $c(X, Y) = -\operatorname{tr}(XY)/2$, the orthonormal frame is $2T_a$, so $\Delta_e^c = 4\Delta_e^T$. Thus the electric term in (1.2) also equals $-g_{\mathrm{YM}}^2 \sum_e \Delta_e^c/(2a)$. No coefficient or energy origin is changed. In particular the scalar $2bM$ remains in H.

The original geometric condition remains

$$\operatorname{Im}\tau \left(\operatorname{Im}\beta - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau}\right) \neq 0. \tag{1.3}$$

The input magnetic patch uses $L_T = \operatorname{Im}\tau_T$, $q_T = -\operatorname{Im}\beta_T, m_T = \operatorname{Im}\mu_T$, $D = L_T q_T + 6m_T^2 > 0$ and $\theta = 2\pi a^2/D$. Its exact projected small-angle state, proved in the parent source, is

$$\chi_\theta = -\frac{2}{3}\theta^2 v_{w^{\mathrm{nat}}} + O_{H^1}(\theta^4), \qquad w^{\mathrm{nat}}_{(n,1)} = n_2^2, \quad w^{\mathrm{nat}}_{(n,2)} = w^{\mathrm{nat}}_{(n,3)} = 0. \tag{1.4}$$

The angle remainder in (1.4) is a fixed-regulator input, not a bound claimed uniform here. The result below concerns its actual electric state with the exact original weights. Sections20–21 additionally retain the entire finite-angle product. Section21 proves its negative-sector weight and vacuum-coefficient bounds at every positive coupling. Sections22–23 control the full product norm and energy remainders on the prescribed cusp at fixed $0 < \xi \leq 10^{-16}$, with its denominator, spectral comparison and fixed-coupling energy escape.

The product Laplacian on this compact connected manifold has domain H^2, form domain H^1, and compact resolvent. Its smooth bounded potential preserves these domains. A lowest form minimizer can be chosen nonnegative by the gradient inequality for the modulus, obtained at its zeros by approximation with $\sqrt{|f|^2 + \epsilon^2}$.

Elliptic regularity and the strong maximum principle make it smooth and strictly positive. Write it as $\psi > 0$, with $\int \psi^2 dU = 1$, and write its physical energy as $\mathcal{E}$. Integration by parts using $H\psi = \mathcal{E}\psi$ gives

$$\mathfrak{q}_{H-\mathcal{E}}[\psi F] = \kappa \sum_{e,a} \int \psi^2 |X_{e,a} F|^2 dU. \tag{1.5}$$

A second ground eigenfunction divided by ψ would have all derivatives zero in (1.5), hence be constant. This proves simplicity. The gauge action $U_e \mapsto g_{s(e)} U_e g_{t(e)}^{-1}$ preserves the operator, form, and positive unit vector; consequently ψ is gauge invariant. All states below are formed from this same vacuum.

For each link set

$$r_e = \#\{p : e \in \partial p\}, \quad W_e^\star = \sum_{p \ni e} W_p, \quad t_{ef} = \#\{p : e, f \in \partial p\}, \quad \gamma_e = \langle \psi, E_e \psi \rangle, \quad v_e = (E_e - \gamma_e)\psi.$$

Here $2 \le r_e \le 4$, including boundary links. Define the actual matrices

$$C_{ef} = \langle v_e, v_f \rangle, \qquad \mathcal{N}_{ef} = \langle v_e, (H - \mathcal{E}) v_f \rangle. \tag{1.6}$$

They are real symmetric positive-semidefinite matrices. Smoothness of ψ supplies every operator and form domain used in (1.6).

2. A local projection and an exact comparison operator

Let $S \subset \mathsf{E}_L$ be a nonempty set of links. It need not be connected. Define

$$\mathsf{J}(S) = \{p : \partial p \cap S \neq \varnothing\}, \quad m_S = |\mathsf{J}(S)|, \quad W_S = \sum_{p \in \mathsf{J}(S)} W_p, \quad H_S = \sum_{e \in S} E_e,$$

$$H_{\mathrm{ext,S}} = \kappa \sum_{e \notin S} E_e + b \sum_{p \notin \mathsf{J}(S)} (2 - W_p). \tag{2.1}$$

The last operator acts on the exterior tensor factor only. Its bottom energy is $\mathcal{E}_{\mathrm{ext,S}}$; if that factor is empty it is the zero operator on $\mathbb{C}$. The exact decomposition is

$$H = \kappa H_S + H_{\mathrm{ext,S}} + 2b m_S - b W_S. \tag{2.2}$$

Every face meeting S, including faces crossing its boundary, is present in W_S. Every other face is present in $H_{\mathrm{ext,S}}$. The constant terms total $2bM$ in both (1.2) and (2.2).

The map

$$(P_S f)(U_{S^c}) = \int f(U_S, U_{S^c})\, dU_S, \qquad Q_S = I - P_S \tag{2.3}$$

is understood with its output lifted as a function constant on S. Fubini proves $P_S = P_S^* = P_S^2$. Its range is $\mathbb{C}1_S \otimes L^2(SU(2)^{S^c})$, and its kernel consists of functions of zero conditional Haar integral on S. It commutes with all link Casimirs and with exterior operators, and preserves smooth functions and Sobolev domains. Each plaquette in W_S contains a link of S exactly once in the fundamental or inverse fundamental representation. Integration of that link is zero: the substitution $U \mapsto -U$ changes its matrix entries' signs. Therefore

$$P_S W_S P_S = 0. \tag{2.4}$$

This is an operator identity, not a statement that the interacting vacuum has a Haar marginal. Haar invariance under endpoint left/right multiplication also proves that P_S, Q_S intertwine the gauge action.

Take as a trial vector the constant on S tensored with an exterior ground vector. Equation (2.4) makes its W_S expectation zero. The full variational principle gives

$$\mathcal{E} \le \mathcal{E}_{\mathrm{ext,S}} + 2b m_S.$$

Consequently the exterior self-adjoint operator

$$K_S = H_{\mathrm{ext,S}} + 2bm_S - \mathcal{E} \quad \text{satisfies } K_S \geq 0. \tag{2.5}$$

No gap of K_S is assumed or needed. Put $B_S = \kappa H_S + K_S$. It is a sum of nonnegative operators on different tensor factors. Their spectral resolutions strongly commute; the operator domain of their sum is their domain intersection. The full smooth vacuum belongs to that domain, and its exact equation is

$$B_S\psi = bW_S\psi. \tag{2.6}$$

This identity retains the original interaction and exterior vacuum dependence. It is not a free-vacuum eigen-equation.

The single-link eigenvalues of E_e are $j(j+1)$, $j = 0, \frac{1}{2}, 1, \ldots$, with the constant as its unique zero vector. Indeed the Pauli matrices give $-\sum_a T_a^2 = 3I/4$ on the fundamental representation, and the spin raising/lowering matrices give $j(j+1)$ on each higher representation. Completeness of their matrix coefficients under product Haar measure gives the joint spectral decomposition. Thus $H_S \geq (3/4)Q_S$. On $Q_S\mathcal{H}_L$, $B_S \geq 3\kappa/4$. Define

$$R_S = Q_S(B_S|_{Q_S\mathcal{H}_L})^{-1}Q_S.$$

Joint spectral calculus gives, without a Neumann expansion,

$$\|R_S\| \leq \frac{4}{3\kappa}, \qquad \|E_eR_S\| \leq \frac{1}{\kappa} \quad (e \in S). \tag{2.7}$$

For the latter assertion the multiplier is $\lambda_e/(\kappa\sum_{f\in S}\lambda_f + k) \leq 1/\kappa$ for $\lambda_f, k \geq 0$. This also proves that the displayed composition has a bounded extension on the full Hilbert space. Applying Q_S to (2.6) now gives the exact local-resolvent equation

$$Q_S\psi = bR_SW_S\psi. \tag{2.8}$$

3. Local electric moments at every positive coupling

Theorem 3.1. For every box $L \geq 2$, every $a, g_{\mathrm{YM}} > 0$, every link e, and every nonempty finite link set S,

$$\|E_e\psi\| \leq 2r_e\xi, \qquad 0 \leq \gamma_e \leq \frac{16}{3}r_e^2\xi^2, \tag{3.1}$$

$$\|Q_S\psi\| \leq q_S\xi, \qquad q_S = \frac{8}{3}\left(\sum_{e\in S} r_e^2\right)^{1/2} \leq \frac{32}{3}\sqrt{|S|}. \tag{3.2}$$

The constants have no dependence on the number of exterior links or faces.

Proof. Use $S = \{e\}$ in (2.6). Its local interaction satisfies $\|W_S\| \leq 2r_e$. For the joint nonnegative spectral variables of κE_e and K_S, their sum is at least the first variable. Squaring this scalar inequality and integrating against the spectral measure of ψ proves

$$\kappa\|E_e\psi\| \leq \|B_S\psi\| = b\|W_e^\star\psi\| \leq 2br_e.$$

Since each nonzero eigenvalue of E_e is at least $3/4$, the spectral inequality $E_e \leq (4/3)E_e^2$ gives the second bound in (3.1). The commuting projections onto nonconstant individual link factors obey $Q_S \leq \sum_{e\in S} Q_{\{e\}}$ in their joint zero/one eigenvalues. Also $Q_{\{e\}} \leq (16/9)E_e^2$. Taking expectations and using (3.1) gives

$$\|Q_S\psi\|^2 \leq \frac{16}{9}\sum_{e\in S}\|E_e\psi\|^2 \leq \frac{64}{9}\xi^2\sum_{e\in S} r_e^2.$$

This proves (3.2). All estimates used the actual normalized vacuum; no estimate on its full-box distance from the constant was used. □

For later use, let F be any bounded multiplication operator depending only on S, with full Haar mean $\overline{F}$. With $\eta = P_S\psi,\ u = Q_S\psi,\ q = \|u\|$, Fubini gives $\langle\eta, F\eta\rangle = \overline{F}\|\eta\|^2$. Expanding all four terms of $\langle\eta + u, F(\eta + u)\rangle$ and using $\|\eta\|^2 = 1 - q^2$ therefore gives

$$\left|\langle\psi, F\psi\rangle - \overline{F}\right| \leq 2\|F\|q + 2\|F\|q^2. \tag{3.3}$$

This is a proved bound on a marginal expectation. It does not replace the marginal by Haar measure.

4. An edge-star resolvent with an explicit state-vector remainder

Take

$$S_e = \bigcup_{p \ni e} \partial p, \qquad m_e^\star = m_{S_e}.$$

Each face at e adds at most three other links, so $|S_e| \le 1 + 3r_e \le 13$. Each link meets at most four faces, hence $m_e^\star \le 4|S_e| \le 52$. Equation (3.2) gives $q_{S_e} \le 32\sqrt{13}/3 < 39$. These counts hold also at every boundary of the open box.

Theorem 4.1. Simultaneously for all links and all boxes, on the explicit interval $0 < \xi \le 1/64$,

$$\left\| E_e\psi - \frac{\xi}{4} W_e^\star \psi \right\| \le 4200\xi^2. \tag{4.1}$$

This compares two vectors in the same full interacting Hilbert space.

Proof. In Section 2 take $S = S_e$ and abbreviate its operators as P, Q, K, B, R. Write $\eta = P\psi,\ u = Q\psi$, with $\|u\| \le q_S\xi$. Applying P to (2.6) and using (2.4) gives

$$K\eta = bPW_S u, \qquad \|K\eta\| \le 2bm_S q_S \xi. \tag{4.2}$$

Since $E_e P = 0$, equation (2.8) gives

$$E_e\psi = bE_e R W_S \eta + bE_e R W_S u. \tag{4.3}$$

Its second term has norm at most $2m_S q_S \xi^2$, by (2.7). For a face not containing e, $E_e W_p \eta = 0$. For a face containing e, all four boundary links lie in S, and

$$E_e W_p \eta = \frac{3}{4} W_p \eta, \qquad H_S W_p \eta = 3 W_p \eta.$$

Here the four fundamental Casimirs contribute separately, with exterior variables untouched. Multiplication by such W_p commutes with K. The bounded map $\zeta \mapsto W_p\zeta$ from the exterior space to the local four-fundamental sector intertwines $3\kappa + K$ with B. Its image is in $Q\mathcal{H}_L$. Resolvent intertwining and (4.3) thus give the exact identity

$$bE_e R W_S \eta = \frac{3b}{4} W_e^\star (3\kappa + K)^{-1} \eta.$$

The exterior inverse exists because $K \ge 0$. Subtracting $(\xi/4)W_e^\star\eta$, while retaining K, gives

$$-\frac{\xi}{4} W_e^\star (3\kappa + K)^{-1} K\eta,$$

whose norm by (4.2) is at most $r_e m_S q_S \xi^3/3$. Replacing η by ψ in the comparison term has the additional norm cost $r_e q_S \xi^2/2$. We have proved the more detailed bound

$$\left\| E_e\psi - \frac{\xi}{4} W_e^\star \psi \right\| \le (2m_S + r_e/2) q_S \xi^2 + \frac{r_e m_S q_S}{3} \xi^3 \le 4134\xi^2 + 2704\xi^3. \tag{4.4}$$

For $\xi \le 1/64$ the last coefficient is $4134 + 2704/64 = 16705/4 < 4200$. This proves (4.1). Equations (4.2)–(4.4) exhibit every discarded-in-an-estimate term as an exact retained term with a bound; none is set to zero. □

5. Uniform local Wilson and connected electric covariances

For elementary faces p, q, Haar integration gives $\int W_p W_q dU = \delta_{pq}$. For unequal faces a link occurs in only one support; its central sign change annihilates the integral. For equal faces Haar invariance reduces the integral to $\int_{SU(2)} (\operatorname{tr} U)^2 dU = 1$. In quaternion coordinates the latter is $4\int u_0^2 = 1$, since the four coordinate second moments on the unit three-sphere equal $1/4$.

The function W_pW_q has norm at most 4 and uses at most eight links. Equations (3.2)–(3.3), with $q_S \leq 32\sqrt{8}/3 < 31$, prove, for $0 < \xi \leq 1/64$,

$$|\langle W_pW_q\rangle_\psi - \delta_{pq}| \leq 248\xi + 7688\xi^2 \leq 370\xi. \tag{5.1}$$

This remains valid for faces at any separation; no finite-range assertion about their true correlations is imposed.

For an individual face choose any one boundary link e and decompose $\psi = P_{\{e\}}\psi + Q_{\{e\}}\psi$. The first-first matrix element of W_p vanishes by integration over e. Equation (3.2) gives $q \leq 32\xi/3$, and the other three terms give

$$|\langle W_p\rangle_\psi| \leq 4q + 2q^2 \leq \left(\frac{128}{3} + \frac{32}{9}\right)\xi < 47\xi.$$

Consequently the connected Wilson moment obeys

$$|\langle W_pW_q\rangle_\psi - \langle W_p\rangle_\psi\langle W_q\rangle_\psi - \delta_{pq}| \leq 410\xi. \tag{5.2}$$

Indeed (5.1) and $47^2\xi^2 \leq (2209/64)\xi$ give a smaller constant than 410.

Theorem 5.1. On the same interval, simultaneously for all boxes and links,

$$\left|C_{ef} - \frac{\xi^2}{16}t_{ef}\right| \leq 42500\xi^3. \tag{5.3}$$

Proof. Set $u_e = E_e\psi$ and $a_e = (\xi/4)W_e^\star\psi$. The preceding bounds give

$$\|u_e\| \leq 8\xi, \quad \|a_e\| \leq 2\xi, \quad \|u_e - a_e\| \leq 4200\xi^2.$$

Using the exact difference $\langle u_e - a_e, u_f\rangle + \langle a_e, u_f - a_f\rangle$, its absolute value is at most $42000\xi^3$. Also

$$\langle a_e, a_f\rangle = \frac{\xi^2}{16}\sum_{p\ni e, q\ni f}\langle W_pW_q\rangle_\psi.$$

There are at most sixteen ordered pairs in this sum. Its difference from $\xi^2 t_{ef}/16$ is at most $370\xi^3$, by (5.1). Finally (3.1) bounds the exact centering term by

$$|\gamma_e\gamma_f| \leq \frac{65536}{9}\xi^4 \leq \frac{1024}{9}\xi^3 < 114\xi^3.$$

The sum $42000 + 370 + 114 = 42484 < 42500$ proves (5.3). No odd Taylor coefficient has been asserted nonzero: this is an absolute real-coupling remainder bound. The fixed-box evenness theorem is compatible with (5.3); without an additional uniform analytic argument it does not improve this bound to order ξ^4. □

6. Energy entries and a volume-independent weighted numerator estimate

Put $A = H - \mathcal{E}$. The product rule, with the original sign of E_f, gives on the smooth vacuum

$$AE_f\psi = [H, E_f]\psi = b\left(\frac{3}{4}W_f^\star\psi - 2D_f\psi\right), \qquad D_f\psi = \sum_{p\ni f,a}(X_{f,a}W_p)X_{f,a}\psi. \tag{6.1}$$

The electric sum commutes with E_f. The potential contribution is $b[E_f, W_f^\star]$, retaining its first-derivative term in full.

For any one occurrence of a link in a simple Wilson loop, cyclically basing the holonomy at that occurrence writes its derivative as $\mathrm{tr}(T_aU)$, or its negative, with a possible adjoint rotation of the index a. Quaternion coordinates then give

$$\sum_a |X_{f,a}W_p|^2 = 1 - W_p^2/4 \leq 1. \tag{6.2}$$

Pointwise Cauchy–Schwarz, followed by integration, bounds each plaquette contraction in $D_f\psi$ by $(\sum_a \|X_{f,a}\psi\|^2)^{1/2} = \sqrt{\gamma_f}$. Thus (3.1) gives

$$\|D_f\psi\| \le r_f\sqrt{\gamma_f} \le \frac{4}{\sqrt{3}} r_f^2 \xi \le \frac{64}{\sqrt{3}}\xi < 37\xi. \tag{6.3}$$

Theorem 6.1. For $0 < \xi \le 1/64$, all boxes, and all links,

$$\left| \mathcal{N}_{ef} - \frac{3\kappa\xi^2}{16} t_{ef} \right| \le 27000\kappa\xi^3. \tag{6.4}$$

For every real edge weight w, including weights depending on L and ξ,

$$\left| w^T \mathcal{N} w - \frac{3\kappa\xi^2}{16} \|\mathsf{I}w\|_2^2 \right| \le 351000\kappa\xi^3 \|w\|_2^2, \qquad (\mathsf{I}w)_p = \sum_{e\in\partial p} w_e. \tag{6.5}$$

Proof. Since $A\psi = 0$, centering does not change $\mathcal{N}_{ef} = \langle E_e\psi, AE_f\psi\rangle$. Insert (6.1). Replacing $E_e\psi$ by a_e in the first term costs at most $6 \cdot 4200\kappa\xi^3 = 25200\kappa\xi^3$, because $\|W_f^\star\psi\| \le 8$. The comparison of its remaining Wilson products to Haar costs at most $1110\kappa\xi^3$ by (5.1). The full derivative term is at most $2\kappa\xi(8\xi)(37\xi) = 592\kappa\xi^3$. Their sum is $26902\kappa\xi^3 < 27000\kappa\xi^3$, proving (6.4).

To prove the weighted conclusion without extending the entry bound to an unjustified operator bound, expand the double commutator:

$$\mathcal{N}_{ef} = \tfrac{1}{2}\langle\psi, [E_e, [H, E_f]]\psi\rangle = \frac{b}{2} \sum_{p: e,f\in\partial p} \langle\psi, [E_e, [2 - W_p, E_f]]\psi\rangle. \tag{6.6}$$

The first identity follows from commutation of E_e, E_f, the vacuum equation at both endpoints, and reality of the two equal cross terms. In the second, a face absent from f has zero inner commutator. A face containing f but absent from e has coefficients and derivatives commuting with E_e. This proves exact support on shared faces. A link has at most $1 + 3r_e \le 13$ partners, counting itself. The remainder in (6.4) has the same support. For its real symmetric matrix R,

$$|w^T R w| \le \sum_{e,f} |R_{ef}| \frac{w_e^2 + w_f^2}{2} \le 13 \cdot 27000\kappa\xi^3 \sum_e w_e^2.$$

Finally $t = \mathsf{I}^T\mathsf{I}$ by its definition. This proves (6.5), with no factor of M, N, or L in its constant. □

7. Actual states, arbitrary weights, and the entire leading kernel

For real w write $\Gamma_w = \sum_e w_e E_e$, $v_w = (\Gamma_w - \langle\Gamma_w\rangle_\psi)\psi$. These operators and states preserve the gauge constraint; all products are taken on smooth ψ. Equations (5.3) and (6.5) give simultaneously

$$\left| \|v_w\|^2 - \frac{\xi^2}{16} \|\mathsf{I}w\|_2^2 \right| \le 42500\xi^3 \|w\|_1^2, \tag{7.1}$$

$$\left| \mathfrak{q}_A[v_w] - \frac{3\kappa\xi^2}{16} \|\mathsf{I}w\|_2^2 \right| \le 351000\kappa\xi^3 \|w\|_2^2. \tag{7.2}$$

The ℓ^1 norm in (7.1) is not replaced by an ℓ^2 norm. The long-distance covariance entries are retained. Equations (7.1)–(7.2) are genuine full-vacuum inequalities at fixed nonzero coupling, not coefficients interpreted outside a finite-box perturbation disk. Section 12 proves an additional ℓ^2 upper bound by a convergent full-vacuum cluster construction. Sections 13–14 subsequently prove absolute distance decay and the stronger two-sided ℓ^2 remainder (14.5). Equation (7.1) remains valid on its original larger interval.

On the entire real kernel $\mathsf{I}w = 0$, a sharper order statement follows from the vector calculation. Summing (4.1) gives exactly $\Gamma_w\psi = \sum_e w_e(E_e\psi - a_e)$, since $\sum_e w_e W_e^\star = \sum_p (\mathsf{I}w)_p W_p = 0$. Projection perpendicular to ψ is a contraction, so

$$\|v_w\|^2 \le 17640000\xi^4\|w\|_1^2. \tag{7.3}$$

In the sum of (6.1), the same cancellation gives $A\Gamma_w\psi = -2b\sum_e w_e D_e\psi$. Using (6.3) and (4.1), without dividing by a possibly small state norm, proves

$$0 \le \mathfrak{q}_A[v_w] \le 310800\kappa\xi^4\|w\|_1^2. \tag{7.4}$$

The constants are $4200^2 = 17640000$ and $2\cdot 4200\cdot 37 = 310800$. These estimates preserve the already calculated nonzero fourth-order kernel forms. They do not assert a new lower bound on those forms. The uniform fourth-order identification and a positive lower bound on the entire incidence kernel are proved below in (16.21)–(16.24), using the retained coefficients and a smaller explicit coupling interval.

There is also a uniform, quantitative local Rayleigh result. For any single link e and

$$0 < \xi \le \frac{1}{680000}, \tag{7.5}$$

(5.3), $r_e \ge 2$, and $42500/680000 = 1/16$ give

$$C_{ee} \ge \frac{r_e}{32}\xi^2 > 0.$$

Equations (5.3) and (6.4) therefore imply

$$\left|\frac{\mathfrak{q}_A[v_e]}{\|v_e\|^2} - 3\kappa\right| \le \frac{4944000}{r_e}\kappa\xi \le 2472000\kappa\xi. \tag{7.6}$$

The numerator in this calculation satisfies $|\mathcal{N}_{ee} - 3\kappa C_{ee}| \le 154500\kappa\xi^3$, which supplies the constant in (7.6). This is the energy of the actual smooth nonzero gauge-invariant local trial vector. By the spectral variational principle its quotient is an upper bound on the physical finite-box excitation gap, not a lower bound on that gap. The explicit range (7.5) is equivalently $g_{\mathrm{YM}}^4 \ge 170000$, retaining $\xi = 1/(4g_{\mathrm{YM}}^4)$. Neither a nor L appears in this range. Section 15 proves the matching volume-uniform lower threshold for the physical operator and every nonzero signed electric state; together they bound the attained minimum over the entire actual electric-state family.

8. Propagation to the original native magnetic state

For the unchanged n_2^2 weights in (1.4), set

$$\mathcal{A}_L = \frac{L^2(2L+1)}{15}(24L^4 + 24L^3 + 8L^2 - 4L + 3),$$

$$S_{1,L} = \frac{2L^2(L+1)(2L+1)^2}{3}, \qquad S_{2,L} = \frac{2L^2(L+1)(2L+1)^2(3L^2+3L-1)}{15}.$$

The exact identities are $\|\mathsf{I}w^{\mathrm{nat}}\|_2^2 = 4\mathcal{A}_L$, $\|w^{\mathrm{nat}}\|_1 = S_{1,L}$, and $\|w^{\mathrm{nat}}\|_2^2 = S_{2,L}$. To check them with all boundary terms, direction-1 links have $2L$ first coordinates and $2L+1$ third coordinates, so the latter two sums are $2L(2L+1)\sum_{t=-L}^{L} t^2$ and $2L(2L+1)\sum_{t=-L}^{L} t^4$. The exact sums are $L(L+1)(2L+1)/3$ and $L(L+1)(2L+1)(3L^2+3L-1)/15$, respectively; their formulas follow by taking the difference under $L \mapsto L+1$ and checking $L = 0$. For face sums the 12 profile is $t^2 + (t+1)^2$, the 13 profile is $2t^2$, and the 23 profile is zero. Counting their allowed lower corners gives

$$\mathcal{A}_L = 2L(2L+1)\sum_{t=-L}^{L-1}(t^2 + t + 1/2)^2 + 4L^2\sum_{t=-L}^{L} t^4.$$

The first sum is $L(4L^4+1)/10$, checked by the two added endpoint terms and its value at $L=1$; substitution gives the displayed polynomial.

Equations (7.1)–(7.2) now yield the explicit new remainders

$$\left| \|v_{w^{\mathrm{nat}}}\|^2 - \frac{\xi^2}{4}\mathcal{A}_L \right| \le 42500\xi^3 S_{1,L}^2, \tag{8.1}$$

$$\left| \mathfrak{q}_A[v_{w^{\mathrm{nat}}}] - \frac{3\kappa\xi^2}{4}\mathcal{A}_L \right| \le 351000\kappa\xi^3 S_{2,L}. \tag{8.2}$$

All native weights are nonnegative, so on each face $(\sum w_e)^2 \ge \sum w_e^2$; summing faces gives $4\mathcal{A}_L \ge \sum_e r_e (w_e^{\mathrm{nat}})^2 \ge 2S_{2,L}$. Consequently the relative error in the native energy numerator is at most 936000ξ, independently of L. In particular on $0 < \xi \le 1/1872000$ it is at most one half of that numerator's positive leading term. This is a proved fixed-coupling extensive energy estimate, not a volume-shrinking perturbative statement.

The upper error-to-leading-term ratio furnished by (8.1) alone is $170000\xi S_{1,L}^2/\mathcal{A}_L$. Its L^3 growth is visible in the exact polynomials. This bound does not give a uniform relative error for the native denominator. No step through Section 8 proves summability of the separated electric correlations, and the fourth-order leading kernel is not discarded. The proved local covariance and weighted numerator estimates therefore do not yet establish the fixed-coupling minimum over every extensive weight or its continuum limit. This statement records the scope of the inequalities actually proved; it is not an assertion that the remaining correlations have no relation to the native state. Sections 11–12 below prove a uniform full spectral lower bound and an extensive covariance upper bound on an explicit smaller nonempty interval. Sections 13–14 then prove a uniform two-sided remainder by controlling the separated correlations. In particular (14.8) replaces the L^3 relative loss by the explicitly vanishing $8\epsilon_\xi$, and (14.7) applies to the native electric-state quotient. These later estimates do not change the original coefficients in (8.1)–(8.2). The gauge-equivariant calculation in Section 15 additionally proves the uniform signed-state minimum (15.21) and improves the native extensive upper bound to (15.17); the stronger lower threshold is not obtained by discarding a small native or signed covariance denominator.

The full physical coefficient conversions are

$$\xi^2 = \frac{1}{16 g_{\mathrm{YM}}^8}, \quad \xi^3 = \frac{1}{64 g_{\mathrm{YM}}^{12}}, \quad \kappa\xi^2 = \frac{1}{8 g_{\mathrm{YM}}^6 a}, \quad \kappa\xi^3 = \frac{1}{32 g_{\mathrm{YM}}^{10} a}.$$

Thus, for example, the right side of (8.2) is exactly $351000 S_{2,L}/(32 g_{\mathrm{YM}}^{10} a)$. The actual vacuum energy includes $2bM = M/(g_{\mathrm{YM}}^2 a)$ throughout. The proof does not exchange any spatial, coupling, cusp, or continuum limits.

9. Primary literature, exact dictionaries, and provenance

The local projection method is related to the Feshbach–Schur reconstruction map, not claimed as a new general projection principle. Dusson–Sigal–Stamm, *The Feshbach–Schur map and perturbation theory*, arXiv:2105.02058v1, Theorem 1.2, equation (1.11), and Lemma 2.1 give the complement inverse and eigenvector reconstruction mechanism (primary source). Their finite-rank perturbation theorem is not used as a volume-uniform hypothesis. Here $P = P_S$ has infinite rank when the exterior is nonempty. For an exact operator dictionary, set

$$A_S^Q = Q_S(H - \mathcal{E})Q_S|_{Q_S\mathcal{H}_L} = B_S|_{Q_S\mathcal{H}_L} - bQ_S W_S Q_S.$$

On its domain, when $2m_S\xi < 3/4$, the bound $A_S^Q \ge \kappa(3/4 - 2m_S\xi) > 0$ proves its bounded inverse. The Q_S block of the full eigen-equation gives exactly

$$Q_S\psi = b(A_S^Q)^{-1} Q_S W_S P_S \psi. \tag{9.1}$$

The retained block is then

$$\left[K_S - b^2 P_S W_S Q_S (A_S^Q)^{-1} Q_S W_S P_S\right] P_S\psi = 0. \tag{9.2}$$

Equations (9.1)–(9.2) follow directly by substitution, including the minus sign and complete exterior operator. Their domains are the exterior domain of K_S and its image under the bounded reconstruction correction. The off-diagonal factors are bounded, so the effective correction in (9.2) is bounded on the exterior Hilbert space. Conversely a vector in that domain solving (9.2), inserted into (9.1), lies in the full domain and solves the original zero-eigenvalue equation. Applying P_S to this reconstruction recovers the original exterior vector; its kernel is therefore zero. This proves the exact bijection of the two kernels. The estimates in Sections 3–8 instead use (2.8), whose inverse needs no smallness condition on b; (9.1) explains the precise relation between the two complement-resolvent presentations.

The canonical corpus also routes Henheik–Teufel–Wessel, *Local stability of ground states in locally gapped and weakly interacting quantum spin systems*, arXiv:2106.13780v3, Section 2 and Theorem 7 (primary source). It permits infinite-dimensional single-site spaces and unbounded on-site operators. Its locality constants are existential in the statements read; none is assigned a numerical value in this note. The exact model dictionary can be given without changing physical coordinates. Associate the edge (n, i) to the indexing site $x(e) = 2n + \mathbf{e}_i \in \mathbb{Z}^3$, retaining this map's inverse on its image: the unique odd coordinate identifies i, and then $n = (x - \mathbf{e}_i)/2$. Attach $L^2(SU(2))$ and on-site $h_{x(e)} = \kappa E_e$ to that site. The on-site gap is $3\kappa/4$. Assign a face $p = (n; i, j)$, $i < j$, to its edge (n, i). Its other three indexing sites have ℓ^1 distance 2 from $x(n, i)$. At most two faces are assigned to one edge, so $\Phi_{x(e)} = b \sum_{p \text{ assigned to } e} (2 - W_p)$ has norm at most $8b$ and range 2. The tensor-permutation unitary defined on elementary tensor functions by this bijection intertwines the complete Hamiltonian with $\sum_x h_x + \sum_x \Phi_x$, including all scalar terms. The indexing map is not a physical rescaling: the original midpoint is exactly $a\, x(e)/2$. The source's small-interaction theorem is not invoked to supply the explicit constants (3.1)–(8.2); those were proved above for this precise model.

The local source identities and conventions were read in full in:

- `magnetic_translation_true_vacuum.md`, Sections 1–9, especially (1.1), (1.4), (5.3), and (8.1)–(8.5);
- `local_energy_current_true_vacuum.md`, Sections 1–6;
- `wilson_variation/ELECTRIC_COVARIANCE_UNSIGNED_INCIDENCE_20260908.md`, Sections 1–8, especially (2.8)–(2.9), (5.7), (6.5), and (7.13);
- `wilson_variation/BULK_PLAQUETTE_EXTENSION.md`, Sections 1–10, especially the explicit box-dependent disk and conditional-form maps.

Their common root is `../ym_gap_primary_20260908`. The complete finite incidence-kernel and second-order coefficient calculations are preserved there. This note does not repeat those calculations or promote their O_L constants to uniform ones. Its new input to that chain is the independently proved quantitative local-resolvent control and the resulting estimates above. The machine-readable lane route records the canonical query results, primary TeX locators, exact reading coverage, and remaining source reads.

10. A uniform score bound and coercivity of the actual conditional operator

The full vacuum equation also controls the conditional density without a small-coupling expansion. This gives a second volume-independent estimate useful in the exact elimination construction. It does not presuppose a spectral gap for the full interacting system.

Use the product Riemannian metric for which the three T_a fields at each link are orthonormal; write ∇_e, Δ_e for its link gradient and Laplacian, and $\Delta = \sum_e \Delta_e = -\sum_e E_e$. This is precisely the metric already used in (1.2), not a new physical metric. In quaternion coordinates $U = u_0 I + i \sum_a u_a \sigma_a$, its metric is four times the metric of the unit three-sphere: the tangent vector T_a at the identity has quaternion Euclidean length $1/2$, while its specified Riemannian length is 1. Left multiplication is orthogonal on quaternions, so this identity holds at every point. Consequently a link factor is isometric to the round sphere of radius 2. Its sectional curvature is $1/4$, Ricci tensor is $g/2$, and diameter is 2π. The curvature coefficient also follows from the second fundamental form of the radius-2 sphere: its normal derivative has magnitude $1/2$, the Gauss equation gives sectional curvature $1/4$, and summing the two transverse sectional curvatures gives $1/2$. These computations retain the factor four relative to c.

Put $u = \log\psi$, a globally smooth real function because $\psi > 0$. Dividing the eigen-equation by the same positive ψ, with its original unit norm left unchanged, yields the exact nonlinear equation

$$\Delta u + |\nabla u|^2 = \frac{V - \mathcal{E}}{\kappa}, \qquad V = b \sum_{p \in \mathsf{P}_L} (2 - W_p). \tag{10.1}$$

For each fixed link e define $w_e = |\nabla_e u|^2$, and let $\nabla_f \nabla_e u$ denote the covariant derivative, in factor f, of the factor-e gradient. On the product manifold the partial Bochner identity is

$$\Delta w_e = 2 \sum_f |\nabla_f \nabla_e u|^2_{\mathrm{HS}} + 2\langle \nabla_e u, \nabla_e \Delta u \rangle + w_e. \tag{10.2}$$

Here is its coordinate derivation. Take orthonormal geodesic frames in every factor at the point under consideration. Applying each second derivative in Δ_f to the sum of squares of the three components of $\nabla_e u$ gives twice their squared first derivatives plus twice their contraction with the second derivative of the vector. For $f \neq e$, the product connection has zero mixed curvature, so that second derivative commutes with ∇_e. For $f = e$, commuting the covariant derivatives of the differential du gives

$$\Delta_e^{\mathrm{rough}} \nabla_e u = \nabla_e \Delta_e u + \mathrm{Ric}_e(\nabla_e u).$$

This sign can equally be read from the positive sectional-curvature Gauss formula above. The Ricci contraction in the differentiated norm is $2\,\mathrm{Ric}_e(\nabla_e u, \nabla_e u) = w_e$. Adding the factor equations proves (10.2). The scalar smooth Bochner identity, with the same sign of Δ, is given in C. Villani, *Optimal Transport: Old and New*, Chapter 14, equation (14.28), printed p.388 (PDF p.394 in the indexed local edition). The partial-factor and vacuum substitutions used here are calculated explicitly, rather than inferred from a dimension-independent constant not stated there.

Let $\mathcal{D}_u = \Delta + 2\langle \nabla u, \nabla \cdot \rangle$. Differentiating (10.1) and using the symmetry of the full Hessian gives

$$\mathcal{D}_u w_e = 2 \sum_f |\nabla_f \nabla_e u|^2_{\mathrm{HS}} + w_e + \frac{2}{\kappa} \langle \nabla_e u, \nabla_e V \rangle. \tag{10.3}$$

In detail, the derivative of $|\nabla u|^2$ in factor e is $2(\nabla^2 u\, \nabla u)_e$. Its contraction with $2\nabla_e u$ in (10.2) is minus $4\nabla^2 u(\nabla u, \nabla_e u)$. The drift applied to w_e is plus that same quantity. This proves the cancellation with every mixed factor f included.

Theorem 10.1. For all $L \geq 2$, $a, g_{\mathrm{YM}} > 0$, and each link,

$$\|\nabla_e \log\psi\|_\infty \leq 2 r_e \xi, \qquad \|\nabla_e \log\rho\|_\infty \leq 4 r_e \xi, \qquad \rho = \psi^2. \tag{10.4}$$

In particular

$$0 \leq \gamma_e \leq 4 r_e^2 \xi^2. \tag{10.5}$$

This improves the constant in the earlier valid bound (3.1). All estimates (4.1)–(8.2) remain valid with their displayed constants.

Proof. Equation (6.2) gives $|\nabla_e W_p| \leq 1$ for every face at e, and all other face derivatives vanish. Thus $|\nabla_e V| \leq b r_e$, without any factor M. The smooth nonnegative function w_e attains its maximum on the compact product. At that point its full gradient vanishes and $\Delta w_e \leq 0$. Equation (10.3) therefore gives

$$0 \geq w_e - 2 r_e \xi \sqrt{w_e}.$$

If the maximum is zero the conclusion is immediate. Otherwise dividing by its positive square root gives $\sqrt{w_e} \leq 2 r_e \xi$. This proves the first bound everywhere. Since $\log\rho = 2u$, the second follows with its exact factor two. Finally integration by parts gives $\gamma_e = \int |\nabla_e \psi|^2 dU = \int \rho |\nabla_e u|^2 dU$; its probability integral and (10.4) prove (10.5). □

To apply this to the actual conditional operator, fix a nonempty set S of eliminated links and write $r = U_{S^c}$, $s = U_S$. Define

$$\rho_{S^c}(r) = \int \rho(s, r)\, ds, \qquad p_S(s \mid r) = \frac{\rho(s, r)}{\rho_{S^c}(r)}, \qquad \omega_S = 8\pi\xi \sum_{e \in S} r_e. \tag{10.6}$$

These are the marginal and conditional density of the same vacuum, not a postulated Wilson Gibbs density. Holding r fixed and joining two S-configurations one link at a time by minimizing geodesics, (10.4) and the diameter 2π give

$$\sup_s \log p_S(s \mid r) - \inf_s \log p_S(s \mid r) \le \omega_S. \tag{10.7}$$

The denominator in (10.6) cancels from this oscillation. No differentiability or uniform limit of the exterior marginal is assumed.

The product Haar Poincare inequality on S, with its exact single-link spectral coefficient, is

$$\inf_{c\in\mathbb{C}} \int |F-c|^2 ds \le \frac{4}{3} \sum_{e\in S} \int |\nabla_e F|^2 ds. \tag{10.8}$$

To prove it, expand F in the complete product matrix-coefficient basis used in Section 2. Removing the constant leaves at least one nonzero spin, so its total electric eigenvalue is at least $3/4$. Parseval gives (10.8) first for finite sums and then by form closure for every H^1 function. For each exterior r, let $p_{\max}, p_{\min}$ be the maximum and minimum of its smooth positive conditional density on S. Minimize over c, first bound the density above, apply (10.8), and then bound it below in the energy integral. This gives the full comparison calculation

$$\begin{aligned} \mathrm{Var}_{p_S(\cdot|r)}(F) &\le p_{\max} \inf_c \int |F-c|^2 ds \\ &\le \frac{4p_{\max}}{3} \sum_{e\in S} \int |\nabla_e F|^2 ds \\ &\le \frac{4}{3} e^{\omega_S} \sum_{e\in S} \int |\nabla_e F|^2 p_S(s \mid r)\, ds. \end{aligned} \tag{10.9}$$

The minimum characterization of variance is valid for complex functions as well, with the complex mean as minimizer. This proof records the entire density ratio; it is the bounded-density perturbation argument, not a replacement of either measure.

Let $\mathscr{H}_\rho = L^2(\rho\, dU)$, and define the exact isometry

$$J_S : L^2(\rho_{S^c}\, dr) \longrightarrow \mathscr{H}_\rho, \qquad (J_S F)(s,r) = F(r).$$

Its adjoint is

$$(J_S^* G)(r) = \int G(s,r) p_S(s \mid r)\, ds.$$

Fubini verifies the inner-product identity for the adjoint and $J_S^* J_S = I$. Thus $P_S^\rho = J_S J_S^*$ and $Q_S^\rho = I - P_S^\rho$ are orthogonal projections; $\mathscr{K}_S^\rho = \ker J_S^*$ is the conditional-mean-zero space. The multiplication map $U_\psi : G \mapsto \psi G$ is a unitary $\mathscr{H}_\rho \to \mathcal{H}_L$. The exact ground-state transform and its closed form are

$$\mathscr{L}_\rho = U_\psi^{-1}(H-\mathcal{E})U_\psi = -\kappa(\Delta + \langle \nabla \log\rho, \nabla \cdot \rangle), \qquad q_\rho[G] = \kappa \sum_e \int |\nabla_e G|^2 \rho\, dU. \tag{10.10}$$

At each fixed box the density and its reciprocal are smooth and bounded on a compact space. Differentiating the conditional integral shows that P_S^ρ, Q_S^ρ preserve H^1: a retained derivative gives the conditional integral of the input derivative and a bounded smooth derivative of the conditional kernel; an eliminated derivative of the output is zero. The constants in this domain-preservation statement need not be uniform, and are not used for the coercivity bound below. Applying Q_S^ρ to smooth H^1 approximants proves that $H^1 \cap \mathscr{K}_S^\rho$ is the form closure of its smooth elements and dense in $\mathscr{K}_S^\rho$.

Define C_S^ρ as the self-adjoint operator of the restricted closed form $q_\rho|_{H^1\cap\mathscr{K}_S^\rho}$. Explicitly its domain consists of those G in that form space for which there is $h \in \mathscr{K}_S^\rho$ satisfying $q_\rho(F,G) = \langle F, h\rangle_\rho$ for every F in the form space, and $C_S^\rho G = h$. This definition makes no unproved claim about the domain of a formal operator product.

Theorem 10.2. For every positive coupling and every finite box,

$$C_S^\rho \geq \frac{3\kappa}{4} e^{-\omega_S} I, \qquad \|(C_S^\rho)^{-1}\| \leq \frac{4}{3\kappa} e^{\omega_S}. \tag{10.11}$$

In particular for a single eliminated link e,

$$C_{\{e\}}^\rho \geq \frac{3\kappa}{4} e^{-8\pi r_e \xi} I \geq \frac{3\kappa}{4} e^{-32\pi\xi} I. \tag{10.12}$$

These single-link bounds are independent of all exterior volume and valid for the actual interacting conditional density.

Proof. A vector $G \in \mathscr{K}_S^\rho$ has zero conditional mean for almost every r. Apply (10.9) to its conditional slices and multiply by $\rho_{S^c}(r)$; integrate in r. Fubini gives

$$\|G\|_\rho^2 \leq \frac{4}{3} e^{\omega_S} \sum_{e \in S} \int |\nabla_e G|^2 \rho \, dU \leq \frac{4}{3\kappa} e^{\omega_S} q_\rho[G].$$

Smooth approximation extends this inequality to the entire restricted form domain. The form representation and spectral theorem yield (10.11); $r_e \leq 4$ gives (10.12). The unitary U_ψ carries this form onto the full physical excitation form restricted to $U_\psi \mathscr{K}_S^\rho$, so the constant has not been transferred between different operators without a domain map. Gauge transformations preserve the density and intertwine conditional integration by endpoint Haar invariance. Consequently these projections, forms, and inverses preserve the physical gauge-invariant subspaces as well. □

For comparison with the original parameters, the last bound in (10.12) is

$$\frac{3 g_{\mathrm{YM}}^2}{2a} \exp\left(-\frac{8\pi}{g_{\mathrm{YM}}^4} \right).$$

For larger eliminated sets the bound is also explicit, but its dependence on $\sum_{e \in S} r_e$ is retained. No volume-independent extensive-block bound is inferred by deleting that sum. Nor does a uniform conditional one-link gap alone tensorize into a uniform gap for the correlated full density. The proof has established an inverse for the actual eliminated operator, not a separated-correlation estimate or the Millennium spectral conclusion. The full finite-box spectral bound proved in Section 12 uses an independently convergent cluster construction, not an assumed tensorization of these conditional inequalities.

The indexed Villani PDF has stable unit ID PUBUNIT-3B56736161C0B49850EA7433, SHA-256 3AE2DB61 CBFFADFEEF0DE738742F4F3CE81B21874DACB1AD2991A7000434675D. Its contents and the complete printed pp.387–389, including equation (14.28), were read directly; the book is cited, not copied into this repository. The page reference is to that exact local edition rather than an assumed page number in every edition.

11. An explicit convergent cluster coordinate construction for the full vacuum

The source method used here is the commuting creation-coordinate construction in D. A. Yarotsky, *Quasi-particles in weak perturbations of non-interacting quantum lattice systems*, arXiv:math-ph/0411042v1, Section 2, equations labelled mmo, c1, vsum, thl and sumj in the retained source TeX (primary source). That section explicitly permits infinite-dimensional single-site spaces and unbounded on-site operators. Its smallness constants are not numerical. The following proof derives its own constants for the complete four-link Wilson interaction. The method is not claimed new. No finite-spin truncation is made, and no rescaling of the original Hamiltonian or vacuum is made.

Write

$$H_0 = \kappa \sum_e E_e, \qquad V_p = -b W_p, \qquad H = H_0 + 2bM + \sum_p V_p, \qquad \gamma = \frac{3\kappa}{4}. \tag{11.1}$$

Thus $\|V_p\| \le J := 2b$, each face has $s = 4$ distinct links, and each link belongs to at most $d = 4$ faces. The scalar $2bM$ remains in (11.1) and in the eigenvalue equation below. It commutes with every coordinate transformation and has zero nonconstant component; these facts, not deletion of its value, account for its absence from the nonconstant fixed-point equation.

Let $\Omega = 1$ in the original product Haar probability space. For $I \subset \mathsf{E}_L$, put

$$\mathcal{H}'_I = \bigotimes_{e\in I} \big(L^2(SU(2)) \ominus \mathbb{C}1\big), \qquad \mathcal{H}'_\varnothing = \mathbb{C}.$$

The canonical orthogonal tensor decomposition gives a unitary map

$$\mathcal{T} : \bigoplus_{I\subset \mathsf{E}_L} \mathcal{H}'_I \longrightarrow \mathcal{H}_L, \qquad \mathcal{T}(z) = \sum_I z_I \otimes 1_{I^c}. \tag{11.2}$$

Its inverse component Π_I is obtained by Haar integration over I^c and application of the nonconstant projection in every link of I. Product projections prove both inverse identities in (11.2). For nonempty I, set $H_I = \kappa \sum_{e\in I} E_e$ on $\mathcal{H}'_I$. The separate nonzero spins prove

$$H_I \ge \gamma|I|, \qquad \|H_I^{-1}\| \le (\gamma|I|)^{-1}. \tag{11.3}$$

For $z_I \in \mathcal{H}'_I$, define the bounded operator, on the I factor and tensored with the exterior identity, by

$$\widehat{z}_I f = z_I \langle 1_I, f\rangle_{L^2(SU(2)^I)}. \tag{11.4}$$

The inner product is the one fixed in Section 1, so (11.4) is linear in f. Its norm is $\|z_I\|$. Integration of any one common nonconstant link proves

$$\widehat{z}_I \widehat{y}_K = \begin{cases} 0, & I \cap K \ne \varnothing, \\ \widehat{z_I \otimes y_K}, & I \cap K = \varnothing. \end{cases} \tag{11.5}$$

For entangled vectors the same calculation follows by approximation by finite elementary tensors in $\mathcal{H}'_I, \mathcal{H}'_K$; operator norms converge by (11.4). In particular all these operators commute. A product of more than $N = |\mathsf{E}_L|$ nonempty creation operators is zero.

Use the graph whose vertices are physical links and whose distinct vertices are adjacent precisely when they share a face. For nonempty I, let $\ell(I)$ be the smallest number of edges of a connected subgraph containing I; extra vertices are allowed. The finite full-box graph is connected. Define

$$\omega(I) = 2^{\ell(I)+1}.$$

Monotonicity under inclusion holds. For a face support $p = \partial p$ the induced graph is a four-vertex clique, so $\ell(p) = 3$ and $\omega(p) = 16$. If each I_j meets p, connected minimizing subgraphs and a tree on p have connected union. Therefore

$$\omega\left(p \cup \bigcup_{j=1}^n I_j\right) \le 16 \prod_{j=1}^n \omega(I_j). \tag{11.6}$$

No replacement of physical coordinates is involved in this graph.

Take the Banach space of collections $z = (z_I)_{I\ne\varnothing}$, $z_I \in \mathrm{Dom}(H_I)$, with norm

$$\|z\|_* = \max_x \sum_{I\ni x} \omega(I) \frac{\|H_I z_I\|}{\gamma}. \tag{11.7}$$

There are finitely many I, and each energy norm is complete by the bounded inverse in (11.3), so (11.7) is complete. In particular, writing $a_I = \omega(I)\|z_I\|$, one has

$$\sup_x \sum_{I\ni x} |I| a_I \le \|z\|_*, \qquad \sum_{I: I\cap p\ne\varnothing} a_I \le s\|z\|_*. \tag{11.8}$$

Let $C(z) = \sum_{I \neq \varnothing} \widehat{z}_I$. It is bounded in each finite box. Its exponential and inverse are the finite polynomials $e^{C(z)}$ and $e^{-C(z)}$ by (11.5). On the full domain of H_0,

$$[H_0, \widehat{z}_I] = \widehat{H_I z_I}, \qquad e^{-C(z)} H_0 e^{C(z)} = H_0 + \sum_{I \neq \varnothing} \widehat{H_I z_I}. \tag{11.9}$$

To check domains, the exterior part of H_0 commutes with (11.4); the local part annihilates the Haar bra, and its action on the ket is $H_I z_I$. This first proves the commutator on finite spectral tensors. Its bounded right-hand side extends it to the full graph domain by approximation. Thus $C(z)$, its powers, and both exponentials preserve that domain. The second commutator in (11.9) vanishes by (11.5), proving the exact conjugation formula.

The fixed-point map is

$$\mathcal{F}(z)_I = -H_I^{-1} \Pi_I e^{-C(z)} \left(\sum_p V_p \right) e^{C(z)} \Omega, \qquad I \neq \varnothing. \tag{11.10}$$

The projection output is an arbitrary vector of $\mathcal{H}'_I$; (11.3) places $\mathcal{F}(z)_I$ in the correct operator domain. We now bound (11.10), including all clusters, rather than assuming the convergence estimate quoted in the source.

For a fixed face p, expand

$$e^{-C(z)} V_p e^{C(z)} = \sum_{n \geq 0} \frac{1}{n!} [\cdots [[V_p, C(z)], C(z)], \ldots, C(z)]. \tag{11.11}$$

Only indices I_j meeting p contribute: a creation operator disjoint from p commutes with V_p and with all other creation operators. Every n-fold nested commutator expands into 2^n ordered words. A nonzero word has disjoint creation supports on each side of V_p. Since they all meet the four-link support of V_p, there are at most four on either side. Thus (11.11) actually ends at $n = 8$. Infinite exponential sums used below are upper bounds for this finite exact polynomial.

There is a dimension-independent bound on the tensor-sector sum of each word acting on Ω. Outside p, V_p acts as identity. A common outside link of two creation supports annihilates the word, by integrating a nonconstant factor. Otherwise the outside excitation set is exactly $(\bigcup_j I_j) \setminus p$. Only the choice of excited links inside p varies in the output. There are at most $2^s = 16$ such orthogonal sectors. Cauchy–Schwarz and (11.4) give

$$\sum_K \|\Pi_K(\text{word})\Omega\| \leq 2^{s/2} J \prod_{j=1}^{n} \|z_{I_j}\| = 4J \prod_{j=1}^{n} \|z_{I_j}\|. \tag{11.12}$$

The argument is unchanged for entangled creation vectors by the elementary-tensor approximation already used in (11.5).

Each nonempty output sector K is contained in $p \cup \bigcup_j I_j$. To bound the anchored sum for $x \in K$, cover its possibilities by $x \in p$ and $x \in I_j$. For the first possibility there are at most d choices of p; (11.8) bounds each remaining index sum by sR, where $R = \|z\|_*$. For the second, choose j, then $I_j \ni x$. At most $d|I_j|$ faces meet that set; the first inequality in (11.8) absorbs this full factor $|I_j|$. Thus the respective anchored bounds for the product sums are

$$d(sR)^n, \qquad ndR(sR)^{n-1}. \tag{11.13}$$

The second quantity is zero for $n = 0$. Combining (11.6), (11.10)–(11.13), and the exact cancellation of H_I against its inverse yields

$$\|\mathcal{F}(z)\|_* \leq \frac{4Jd\,16}{\gamma} \sum_{n \geq 0} \frac{2^n}{n!} \left[(sR)^n + nR(sR)^{n-1} \right] = \frac{2048}{3} \xi (1 + 2R) e^{8R}. \tag{11.14}$$

This anchor counting, rather than the norm of the full perturbation, removes the factor M.

For two collections in a ball of radius R, telescope each multilinear word in (11.11), inserting their difference in one position at a time. The counting (11.13) holds with the norm of that difference in its position, whether

the anchor is there or elsewhere. For the homogeneous degree-n bound this multiplies its coefficient by nR^{n-1} instead of R^n. Differentiating the positive majorant series in (11.14) proves

$$\|\mathcal{F}(z) - \mathcal{F}(y)\|_* \le \frac{2048}{3}\xi(10 + 16R)e^{8R}\|z - y\|_*. \tag{11.15}$$

This is a bound on the exact finite commutator map, not on a truncated approximation to the vacuum.

Set $R_* = 1/16$. Since $\sum_{n\ge0}(1/2)^n/n! < \sum_{n\ge0}(1/2)^n = 2$, (11.14)–(11.15) give, on the explicit nonempty interval

$$0 < \xi \le \frac{1}{49152}, \tag{11.16}$$

$$\|\mathcal{F}(z)\|_* \le 1536\xi \le \frac{1}{32} < R_*, \qquad \mathrm{Lip}(\mathcal{F}) \le \frac{45056}{3}\xi \le \frac{11}{36} < 1. \tag{11.17}$$

Iteration from zero therefore converges in the complete ball to a unique fixed point z. More explicitly, successive increments are bounded by a geometric series of ratio $11/36$; the limit solves (11.10) by continuity, and applying (11.15) to two fixed points proves uniqueness. Complex conjugation preserves the real Wilson potential, the Haar projections and the inverses H_I^{-1}. It commutes with the iteration, so z, $e^{C(z)}\Omega$, and the eigenvalue below are real.

Equations (11.9)–(11.10) show that

$$\phi = e^{C(z)}\Omega, \qquad H\phi = \varepsilon\phi, \qquad \varepsilon = 2bM + \Pi_\varnothing e^{-C(z)}\left(\sum_p V_p\right)e^{C(z)}\Omega. \tag{11.18}$$

Every nonempty creation product has zero Haar integral, hence $\langle\Omega,\phi\rangle = 1$, so $\phi \ne 0$. The next section proves that $\varepsilon = \mathcal{E}$, and identifies the exact scalar relating ϕ to the unchanged physical unit vacuum. No equality of these differently specified vectors is presumed.

12. Uniform full spectral estimate and the extensive electric covariance

In this section (11.16) holds. Fix its constructed z and abbreviate $C = C(z)$. The bounded invertible similarity

$$\widetilde{H} = e^{-C}(H - \varepsilon)e^{C} = H_0 + \mathcal{K}, \qquad \widetilde{H}\Omega = 0 \tag{12.1}$$

has the full domain of H_0, by (11.9). It is not asserted unitary. Both similarities and the exact physical Hamiltonian, including its ground energy, remain explicit.

Give the finite orthogonal sector decomposition the additional norm

$$\|f\|_{\oplus,1} = \sum_I \|\Pi_I f\|, \qquad \|f\| \le \|f\|_{\oplus,1} \le 2^{N/2}\|f\|. \tag{12.2}$$

The second inequality is Cauchy–Schwarz over all 2^N sectors. Thus the norm has exactly the same vectors and topology in each fixed box. The equivalence constant depends on volume; no uniform claim about that equivalence is used.

For $u_I \in \mathrm{Dom}(H_I)$, (11.5), (11.9), (12.1) and $\widetilde{H}\Omega = 0$ give exactly

$$\mathcal{K}\,\widehat{u}_I\Omega = e^{-C}\left[\sum_p V_p, \widehat{u}_I\right]e^{C}\Omega. \tag{12.3}$$

In detail, commute $\widetilde{H}$ with $\widehat{u}_I$; the H_0 commutator supplies $H_I u_I$, while the creation sum in (11.9) commutes with $\widehat{u}_I$. For the remaining term, e^C commutes with $\widehat{u}_I$, proving (12.3). Faces disjoint from I have zero commutator, so at most $d|I|$ faces remain.

Expand each remaining conjugation as in (11.11). All additional creation indices still have to meet p: the derivation identity and commutation with $\widehat{u}_I$ allow the nested commutators to be written as the commutator with $\widehat{u}_I$ of the corresponding conjugated V_p. The output outside p now has the fixed excited support $(I \cup \bigcup_j I_j) \setminus p$, or the word vanishes. Thus (11.12) applies with an extra factor $\|u_I\|$. The commutator with $\widehat{u}_I$ contributes two words. Using $\sum_{K:K\cap p\neq\varnothing} \|z_K\| \le sR_*$ gives

$$\|\mathcal{K}\widehat{u}_I\Omega\|_{\oplus,1} \le 8Jd\,|I|\,e^{8R_*}\|u_I\|. \tag{12.4}$$

The bound also applies to the empty component with value zero. Summing (12.4) and using (11.3) proves the relative operator bound

$$\|\mathcal{K}f\|_{\oplus,1} \le c_\xi \|H_0 f\|_{\oplus,1}, \qquad c_\xi = \frac{512}{3}\xi \le \frac{1}{288} < 1, \quad f \in \mathrm{Dom}(H_0). \tag{12.5}$$

Indeed $8Jd/\gamma = 256\xi/3$, and $e^{8R_*} = e^{1/2} < 2$. The bound on finite spectral tensors extends in the graph norm; all factors in (12.3) preserve the required domain as proved above.

Here is the resolvent argument with no appeal to a missing uniform perturbation theorem. On every nonempty sector I, $\mathrm{Spec}(H_I) \subset [\gamma|I|, \infty)$. For real $t < 0$, functional calculus gives

$$\|\mathcal{K}(H_0 - t)^{-1}\|_{\oplus,1} \le c_\xi \sup_{\lambda\ge\gamma} \frac{\lambda}{\lambda - t} \le c_\xi < 1.$$

For $0 < t < (1 - c_\xi)\gamma$, it gives instead

$$\|\mathcal{K}(H_0 - t)^{-1}\|_{\oplus,1} \le \frac{c_\xi\gamma}{\gamma - t} < 1. \tag{12.6}$$

The empty sector causes no term because $\mathcal{K}\Omega = 0$. The geometric Neumann series for $(I + \mathcal{K}(H_0 - t)^{-1})^{-1}$, followed by $(H_0 - t)^{-1}$, is a bounded inverse of $\widetilde{H} - t$. The domain is verified by the resolvent's graph-domain image. Norm equivalence (12.2) makes it also a bounded inverse in the original Hilbert space, without changing the numerical spectral exclusion (12.6). Similarity (12.1) transfers this inverse to $H - \varepsilon - t$.

The original H is self-adjoint, ε is its real eigenvalue, and no spectrum lies below it by the $t < 0$ argument. Thus $\varepsilon = \mathcal{E}$. Its simple positive ground vector was proved in Section 1. The exact identification is therefore

$$\psi = \alpha e^C\Omega, \qquad \alpha = \langle\Omega, \psi\rangle > 0, \qquad \alpha^2\|e^C\Omega\|^2 = 1. \tag{12.7}$$

This records the physical scalar and preserves the original unit vacuum rather than replacing it with a coefficient-normalized vector. The spectrum above it obeys the explicit volume-independent estimate

$$(H - \mathcal{E})\big|_{\psi^\perp} \ge \gamma(1 - c_\xi)I = \frac{3\kappa}{4}\left(1 - \frac{512}{3}\xi\right)I \ge \frac{287\kappa}{384}I. \tag{12.8}$$

This follows from (12.6) and the spectral theorem, since the ground eigenvalue is simple. Restricting the same inequality to the gauge-invariant subspace retains it, because that closed subspace reduces H and contains ψ. Section 15 proves that the cluster similarity itself intertwines this constraint. Its restricted free threshold is exactly 3κ, so (15.13) improves the physical restriction of (12.8), not the full-space inequality or the numerical fixed-point constants. In original parameters its range and last lower bound are

$$g_{\mathrm{YM}}^4 \ge 12288, \qquad \frac{287g_{\mathrm{YM}}^2}{192a}. \tag{12.9}$$

This is a strong-coupling, fixed-spatial-regulator theorem for the full interacting finite boxes. No spatial continuum limit or Millennium conclusion follows by deleting its coupling range.

The spectral estimate has an immediate quantitative consequence for the previously unresolved extensive covariance upper bound. Every v_w in Section 7 is perpendicular to ψ, so the spectral theorem and (12.8) imply

$$\|v_w\|^2 \le \frac{4}{3\kappa(1 - c_\xi)}\,\mathfrak{q}_{H-\mathcal{E}}[v_w].$$

Combining this with the already proved all-weight estimate (7.2) gives, for every real weight vector and every box,

$$0 \le w^T C w \le \frac{\xi^2 \|\mathsf{I} w\|_2^2/4 + 468000\xi^3 \|w\|_2^2}{1 - 512\xi/3}. \tag{12.10}$$

The norm of I is at most 4: on each face, $(\sum_{e\in\partial p} w_e)^2 \le 4\sum_{e\in\partial p} w_e^2$; summing faces and retaining $r_e \le 4$ proves $\|\mathsf{I} w\|_2^2 \le 16\|w\|_2^2$. Consequently the fully extensive matrix upper bound is

$$0 \le C \le \frac{4\xi^2 + 468000\xi^3}{1 - 512\xi/3}\, I_{\mathbb{R}^{\mathsf{E}_L}}. \tag{12.11}$$

This removes the ℓ^1-squared volume loss from an upper bound on the exact covariance. This argument alone does not assert the entrywise absolute row summability of C, or an ℓ^2 remainder of order ξ^3 around $\xi^2 \mathsf{I}^T \mathsf{I}/16$. Positive-semidefinite order and absolute-value entry sums are not interchanged. Sections 13–14 prove the separate absolute-row and two-sided claims by an explicit locality and spectral-filter calculation.

For the original native weights the exact substitution is

$$\|v_{w^{\mathrm{nat}}}\|^2 \le \frac{\xi^2 \mathcal{A}_L + 468000\xi^3 S_{2,L}}{1 - 512\xi/3}. \tag{12.12}$$

Using $S_{2,L} \le 2\mathcal{A}_L$, proved in Section 8, yields

$$\|v_{w^{\mathrm{nat}}}\|^2 \le \xi^2 \mathcal{A}_L \frac{1 + 936000\xi}{1 - 512\xi/3}. \tag{12.13}$$

Unlike (8.1), this upper control has no L^3 relative loss. Equation (8.1) remains a valid two-sided remainder with its original constants; (12.13) supplies a different, stronger extensive upper control on the nonempty range (11.16). On the entire leading kernel $\mathsf{I} w = 0$, (12.10) also yields

$$\|v_w\|^2 \le \frac{468000\xi^3}{1 - 512\xi/3} \|w\|_2^2. \tag{12.14}$$

This complements, and does not discard, the fourth-order ℓ^1-weighted bound (7.3) and the parent's exact fourth-order calculation. The upper bounds (12.10)–(12.14) alone do not give a two-sided estimate for the minimum over all signed actual states. The uniform minimum, as opposed to every individual signed quotient, is obtained in (15.21) by the gauge-restricted spectral argument. Equations (15.15)–(15.17) also strengthen these PSD upper bounds by a factor of four. The original fourth-order kernel coefficients are retained; (16.21) supplies their uniform covariance and energy remainders.

Finally the same result improves exact extensive elimination, not only the electric variational family. For every nonempty eliminated set S, a vector $G \in \ker J_S^*$ satisfies

$$\int G\rho \, dU = \int (J_S^* G)(r) \rho_{S^c}(r)\, dr = 0.$$

Thus $U_\psi G = \psi G$ is perpendicular to the actual ground vector. Apply (12.8) to its form and use (10.10), first on smooth vectors and then by the proved form closure. This yields

$$C_S^\rho \ge \frac{3\kappa}{4}(1 - 512\xi/3) I \ge \frac{287\kappa}{384} I \quad \left(0 < \xi \le \frac{1}{49152}\right), \tag{12.15}$$

$$\|(C_S^\rho)^{-1}\| \le \frac{4}{3\kappa(1 - 512\xi/3)} \le \frac{384}{287\kappa}. \tag{12.16}$$

Every constant in (12.15)–(12.16) is independent of the size, shape and location of S, as well as the exterior volume. The operator is precisely the restricted full form C_S^ρ defined in Section 10, containing derivatives in all link directions. It is not replaced by a conditional-fibre differential operator containing only eliminated derivatives. The all-positive-coupling bound (10.11) remains available outside (11.16). On their common range both lower bounds hold, so the larger of the two constants may be used. The unitary and gauge maps remain exactly those proved in Section 10.

For the reducing gauge-invariant conditional-complement space, (15.22)–(15.23) prove the sharper inverse constant $96/(287\kappa)$. The projection and unitary supplying that restriction are explicit.

13. Absolute decay of the unbounded electric covariance

Throughout Sections 13–14, $0 < \xi \le 1/49152$. Retain H, ψ and E_e exactly as in Section 1, and set

$$\Delta_* = \frac{287\kappa}{384}, \qquad Q_\psi = I - |\psi\rangle\langle\psi|, \qquad \alpha_t(O) = e^{itH} O e^{-itH}.$$

Equation (12.8), not an additional assumption, gives $A|_{\psi^\perp} \ge \Delta_*$, where $A = H - \mathcal{E}$. Let $d(e, f)$ be distance in the shared-face graph of Section 11. We will prove

$$|C_{ef}| \le 1200\xi e^{-d(e,f)/4} \qquad (e \ne f). \tag{13.1}$$

This is a bound on the original electric operators, which are unbounded. No finite-spin Hamiltonian or approximate ground state is introduced.

The two primary methods are B. Nachtergaele and R. Sims, *On the dynamics of lattice systems with unbounded on-site terms in the Hamiltonian*, arXiv:1410.8174v1, Sections 2–3, especially Proposition 2.1, Lemma 2.2 and the proof of Theorem 3.1 (source), and the same authors, *Lieb-Robinson Bounds and the Exponential Clustering Theorem*, Commun. Math. Phys. 265 (2006), 119–130, arXiv:math-ph/0506030v3, Theorem 2, Section 3.2 and Lemma 1 (source). The former explicitly repairs domain and strong-continuity issues in earlier unbounded-on-site proofs. Both source TeX files are retained. We prove the numerical bounds and the electric-operator passage below; neither is obtained by assigning constants to an existential statement.

13.1. A second electric moment and an exact cutoff comparison

For $S = \{e\}$, joint spectral calculus in (2.7) gives, for $f \in \mathrm{Dom}(E_e)$,

$$\|E_e^2 R_{\{e\}} f\| \le \kappa^{-1} \|E_e f\|.$$

Indeed, on a nonconstant link sector the multiplier satisfies $\lambda_e^2/(\kappa\lambda_e + k) \le \lambda_e/\kappa$; it vanishes on the constant link sector. Applying this to the smooth vector $W_e^\star \psi$ in (2.8), and noting $E_e^2 P_{\{e\}} \psi = 0$, proves

$$\|E_e^2 \psi\| \le \xi \|E_e(W_e^\star \psi)\| \le \frac{3r_e}{2}\xi + 8r_e^2\xi^2 \le 6\xi + 128\xi^2 \le 7\xi. \tag{13.2}$$

Here the full product rule is

$$E_e(W_e^\star \psi) = \tfrac{3}{4} W_e^\star \psi + W_e^\star E_e \psi - 2D_e\psi.$$

The three terms have bounds $3r_e/2$, $4r_e^2\xi$, and $4r_e^2\xi$, respectively: the last follows from $\|D_e\psi\| \le r_e\sqrt{\gamma_e} \le 2r_e^2\xi$, using (10.5). Thus the mixed derivative has been estimated, not omitted.

For any $\Lambda > 0$, define the bounded one-link observable

$$B_e^\Lambda = E_e \mathbf{1}_{[0,\Lambda]}(E_e), \qquad \|B_e^\Lambda\| \le \Lambda.$$

It is a functional calculus of the original E_e; it changes neither H nor ψ. For its spectral tail, $\lambda \mathbf{1}_{\lambda>\Lambda} \le \lambda^2/\Lambda$, hence

$$\|(E_e - B_e^\Lambda)\psi\| \le 7\xi/\Lambda, \qquad \|Q_\psi B_e^\Lambda \psi\| \le \|B_e^\Lambda \psi\| \le \|E_e\psi\| \le 8\xi. \tag{13.3}$$

Define $\mathrm{Cov}_\psi(O, P) = \langle Q_\psi O\psi, Q_\psi P\psi\rangle$ for self-adjoint observables with ψ in their domains. Subtracting the two centered inner products, retaining both difference terms, gives

$$|C_{ef} - \mathrm{Cov}_\psi(B_e^\Lambda, B_f^\Lambda)| \le 112\xi^2/\Lambda. \tag{13.4}$$

Both original link Casimirs and their spectral projections commute with endpoint gauge transformations; every cutoff observable is physical.

13.2. Locality for the complete dynamics

For bounded observables O_e, O_f on distinct links, we claim

$$\|[\alpha_t(O_e), O_f]\| \le 2\|O_e\|\|O_f\|e^{-d(e,f)}(e^{v|t|}-1), \qquad v = 192b. \tag{13.5}$$

This intermediate statement holds at every positive coupling. Here are the domain construction and the full counting proof.

For a finite link set X, introduce the auxiliary self-adjoint operator

$$H_X^0 = H_0 + 2bM + \sum_{\partial p \subset X} V_p, \qquad V_p = -bW_p.$$

It has the same H^2 domain as H, and its dynamics preserves the algebra of bounded operators supported on X. Its difference from H is bounded. The propagator

$$W_X(t,s) = e^{itH_X^0}e^{-i(t-s)H}e^{-isH_X^0}$$

has inverse $W_X(s,t)$, and on the common domain differentiating gives

$$\partial_t W_X(t,s) = -iK_X(t)W_X(t,s), \qquad K_X(t) = e^{itH_X^0}(H - H_X^0)e^{-itH_X^0}.$$

This bounded self-adjoint generator is strongly continuous. Its vectorwise Dyson integrals exist, and on each compact time interval their norms are bounded by the exponential series with parameter $\sup_t \|K_X(t)\|$. Iteration of the integral equation gives the series; iteration for the difference of two solutions gives zero, since its n-th bound has a factorial denominator. Consequently the domain derivative agrees with that strong solution on the whole Hilbert space. This justifies the differentiation below for bounded observables without assuming operator-norm continuity of the unbounded on-site evolution.

Let $\beta_{X,t}(O) = W_X(0,t)OW_X(t,0)$, and let $\partial_{\mathrm{P}}X$ be the faces meeting both X and X^c. For O supported on X, the strong commutator derivative and Jacobi identity give

$$\begin{aligned} F(t) &= [\beta_{X,t}(O), O_f], \\ F'(t) &= i[L_X(t), F(t)] + i[\beta_{X,t}(O), [O_f, L_X(t)]], \\ L_X(t) &= \sum_{p \in \partial_{\mathrm{P}}X} \alpha_t(V_p). \end{aligned}$$

Terms outside X commute with O, and terms inside X are already in H_X^0; the crossing faces are therefore exactly the stated ones. Conjugate this equation by the unitary strong propagator generated by $-L_X(t)$. Integrating its remaining inhomogeneous term and taking norms proves

$$\|F(t)\| \le \|[O, O_f]\| + 2\|O\| \sum_{p \in \partial_{\mathrm{P}}X} \int_0^{|t|} \|[\alpha_{\mathrm{sgn}(t)s}(V_p), O_f]\|\, ds.$$

At any fixed final t, insert $O = e^{itH_X^0}O_X e^{-itH_X^0}$, which still has support X and norm $\|O_X\|$. This is a pointwise use of the already proved inequality, not differentiation of a time-dependent initial observable. It yields the recurrence

$$\begin{aligned} \|[\alpha_t(O_X), O_f]\| \le{}& 2\|O_X\|\|O_f\|\mathbf{1}_{f \in X} \\ &+ 2\|O_X\| \sum_{p \in \partial_{\mathrm{P}}X} \int_0^{|t|} \|[\alpha_{\mathrm{sgn}(t)s}(V_p), O_f]\|\, ds. \end{aligned} \tag{13.6}$$

The Hilbert spaces here are separable. Every strongly continuous commutator has a measurable norm, as the supremum over a countable dense set of unit vectors; these scalar integrals are well-defined.

Iterate (13.6) from $X = \{e\}$. A term reaching f in n interactions is a chain of n faces, the first containing e, consecutive faces intersecting, and the last containing f. Such a chain gives a shared-face link path of at most n steps; therefore $n \geq d(e, f)$. There are at most 4 choices of the first face and 16 choices of every subsequent face, since a face contains four links and each link meets at most four faces. Each interaction has norm at most $2b$. The remainder after n iterations is bounded by a constant times $(64b|t|)^n/n!$ and tends to zero. Dropping only counted positive terms gives

$$\|[\alpha_t(O_e), O_f]\| \leq 2\|O_e\|\|O_f\| \sum_{n \geq d(e,f)} \frac{(64b|t|)^n}{n!} \leq 2\|O_e\|\|O_f\| e^{-d(e,f)}(e^{64\exp(1)b|t|} - 1).$$

The last inequality uses $\exp(n - d) \geq 1$ for $n \geq d$. The series for $\exp(1)$, with $n! \geq 2^{n-1}$ for $n \geq 2$ and a strict inequality for $n \geq 3$, gives $\exp(1) < 3$. This proves (13.5), including its explicit $192b$. The constant $2bM$ has commuted through every displayed conjugation; the original Hamiltonian and its energy origin remain unchanged.

13.3. Spectral filtering with the actual vacuum moments

Take bounded self-adjoint O_e, O_f, set $q_e = Q_\psi O_e \psi,\ q_f = Q_\psi O_f \psi$, and put $s_0 = \|q_e\|\|q_f\|$. For

$$h(t) = \langle \psi, [O_e, \alpha_t(O_f)]\psi \rangle = \langle q_e, e^{itA} q_f \rangle - \langle q_f, e^{-itA} q_e \rangle$$

one has $|h(t)| \leq 2s_0$. For $\alpha > 0$, introduce the scalar integral

$$I_\alpha = \lim_{\epsilon \downarrow 0} \frac{1}{2\pi} \int_{\mathbb{R}} \frac{e^{-\alpha t^2} h(t)}{\epsilon + it}\, dt.$$

The limit is absolutely dominated: (13.5) makes $h(t) = O(|t|)$ at zero, while the Gaussian controls infinity. For $\epsilon > 0$, use $(\epsilon + it)^{-1} = \int_0^\infty e^{-\epsilon u - itu}\, du$ and integrate the Gaussian in t. Fubini is justified by its integrable absolute majorant. Letting $\epsilon \downarrow 0$ gives the bounded spectral multiplier

$$F_\alpha(\lambda) = \frac{1}{\sqrt{4\pi\alpha}} \int_0^\infty e^{-(u-\lambda)^2/(4\alpha)}\, du$$

and the exact identity

$$I_\alpha = \langle q_e, F_\alpha(A) q_f \rangle - \langle q_f, F_\alpha(-A) q_e \rangle.$$

The Gaussian transform itself follows by differentiating its integral with respect to its Fourier variable, integrating by parts, and fixing the constant at zero by the squared Gaussian integral. For $\lambda \geq \Delta_*$, substitution followed by $(x + y)^2 \geq x^2 + y^2$ gives

$$0 \leq 1 - F_\alpha(\lambda) = F_\alpha(-\lambda) \leq \tfrac{1}{2} e^{-\Delta_*^2/(4\alpha)}.$$

The total variation of a cross spectral measure is at most $\|q_e\|\|q_f\|$: on a disjoint measurable partition apply Cauchy–Schwarz to the two sequences of spectral-projection norms. Thus every spectral interchange above is controlled, and

$$|\operatorname{Cov}_\psi(O_e, O_f) - I_\alpha| \leq s_0 e^{-\Delta_*^2/(4\alpha)}.$$

Splitting its time integral at $T > 0$, (13.5) controls the part $|t| \leq T$, while $2s_0$ controls the rest. The explicit estimates

$$\frac{e^{vt} - 1}{t} \leq v e^{vt}, \qquad \int_T^\infty \frac{e^{-\alpha t^2}}{t}\, dt \leq \frac{e^{-\alpha T^2}}{2\alpha T^2}$$

follow respectively by integrating the exponential derivative and by using $1/t \leq t/T^2$. Choosing $\alpha = \Delta_*/(2T)$ proves

$$\begin{aligned} |\operatorname{Cov}_\psi(O_e, O_f)| \leq{}& \frac{2\|O_e\|\|O_f\|}{\pi} v T e^{-d(e,f)+vT} \\ &+ s_0 \left(1 + \frac{2}{\pi \Delta_* T}\right) e^{-\Delta_* T/2}. \end{aligned} \tag{13.7}$$

The Gaussian is only an integral filter in this proof. The dynamics, vacuum and interaction have never been replaced by Gaussian ones.

For $d = d(e, f) \geq 1$, now use

$$O_e = B_e^{\Lambda}, \quad O_f = B_f^{\Lambda}, \quad \Lambda = e^{d/4}, \qquad T = \frac{d}{\Delta_* + 2v}.$$

The original parameters give exactly

$$\frac{v}{\Delta_*} = \frac{73728}{287}\xi \leq \frac{3}{574}, \qquad vT \leq \frac{3d}{580} < \frac{d}{100}, \qquad \Delta_* T \geq \frac{287d}{290}. \tag{13.8}$$

Consequently the first term in (13.7) is at most

$$\frac{73728}{287}\xi d\, e^{-49d/100} \leq 1100\xi e^{-d/4}.$$

We used $2/\pi < 1$, and $de^{-6d/25} \leq 25/6$, which follows from $e^x \geq x$; the exact remaining coefficient is $307200/287 < 1100$. By (13.3), $s_0 \leq 64\xi^2$. The second term in (13.7) is at most $192\xi^2 e^{-d/4}$: its parenthesis is less than 3 for $d \geq 1$ and $\Delta_* T/2 \geq 287d/580 > d/4$. Finally (13.4) contributes at most $112\xi^2 e^{-d/4}$. Their sum is bounded by $(1100\xi + 304\xi^2)e^{-d/4} \leq 1200\xi e^{-d/4}$. This proves (13.1) for the original unbounded covariance.

14. A volume-uniform two-sided remainder and the original native quotient

Define the real symmetric remainder

$$\mathscr{R} = C - \frac{\xi^2}{16}\mathsf{I}^{\mathsf{T}}\mathsf{I}.$$

For every entry (5.3) gives $|\mathscr{R}_{ef}| \leq 42500\xi^3$. For $d(e, f) \geq 2$ there is no shared face, so $\mathscr{R}_{ef} = C_{ef}$ exactly and (13.1) applies. These are two estimates of the same exact entries; no correlation is set to zero.

The injective indexing map $x(n, i) = 2n + \mathbf{e}_i$ and its inverse were proved in Section 9. Any two links on one face have indexing ℓ^1 distance at most 2, as inspection of its four midpoints shows. A graph ball of integer radius D therefore lies inside the integer cube of coordinate side $4D + 1$, and contains at most $(4D + 1)^3$ links, including near the original box boundary. For a shell of radius n we use the valid upper bound $(4n + 1)^3$, not a difference of two bounding cube volumes. For every integer $D \geq 1$, this proves

$$\sup_e \sum_f |\mathscr{R}_{ef}| \leq 42500\xi^3(4D + 1)^3 + 1200\xi \sum_{n>D} (4n + 1)^3 e^{-n/4}. \tag{14.1}$$

Here is a closed, explicit bound with no remaining infinite sum. For $0 < q < 1$ put

$$G(A, q) = \frac{A^3}{1 - q} + \frac{12A^2 q}{(1 - q)^2} + \frac{48Aq(1 + q)}{(1 - q)^3} + \frac{64q(1 + 4q + q^2)}{(1 - q)^4}.$$

The geometric series and its first three applications of $q\, d/dq$ give respectively

$$\sum_{j\geq 0} q^j = \frac{1}{1 - q}, \quad \sum_{j\geq 0} jq^j = \frac{q}{(1 - q)^2}, \quad \sum_{j\geq 0} j^2 q^j = \frac{q(1 + q)}{(1 - q)^3}, \quad \sum_{j\geq 0} j^3 q^j = \frac{q(1 + 4q + q^2)}{(1 - q)^4}.$$

Termwise differentiation is justified by uniform convergence on each compact subinterval of $(0, 1)$. Expanding $(A + 4j)^3$ and using these four identities proves

$$\sum_{n>D} (4n + 1)^3 q^n = q^{D+1} G(4D + 5, q). \tag{14.2}$$

Write $z = \log(1/\xi) > 0$, set $D = \lceil 8z \rceil$, and use $q = e^{-1/4} < 4/5$, the latter following from $e^{1/4} > 1 + 1/4$. Then $q^D \le \xi^2$, $4D + 1 \le 32z + 5$, and $4D + 5 \le 32z + 9$. All coefficients of G are positive. Define the explicit cubic polynomial

$$\mathcal{P}(z) = 42500(32z + 5)^3 + 1200G(32z + 9, 4/5), \qquad \epsilon_\xi = \xi\mathcal{P}(\log(1/\xi)). \tag{14.3}$$

Equations (14.1)–(14.2) now prove

$$\sup_e \sum_f |\mathscr{R}_{ef}| \le \xi^3\mathcal{P}(\log(1/\xi)) = \xi^2\epsilon_\xi. \tag{14.4}$$

The same bound holds for column sums by symmetry. For completeness, Cauchy–Schwarz in the nonnegative weights $|\mathscr{R}_{ef}|$ gives

$$\sum_e |(\mathscr{R}w)_e|^2 \le \left(\sup_e \sum_f |\mathscr{R}_{ef}|\right) \sum_f |w_f|^2 \sum_e |\mathscr{R}_{ef}|.$$

Thus (14.4) bounds its operator norm on $\ell^2(\mathsf{E}_L)$ and yields the two-sided inequality

$$\left| w^{\mathsf{T}}Cw - \frac{\xi^2}{16}\|\mathsf{I}w\|_2^2 \right| \le \xi^2\epsilon_\xi\|w\|_2^2 \quad \text{for every real } w. \tag{14.5}$$

In particular the absolute rows of C are uniformly summable: $\sum_f t_{ef} = 4r_e \le 16$, so $\sup_e \sum_f |C_{ef}| \le \xi^2(1+\epsilon_\xi)$. The remainder has order $\xi^3(1 + \log(1/\xi))^3$, uniformly in the full box. No order-ξ^4 assertion is substituted for that proved order.

There is an entirely explicit interval on which the relative estimate is nonvacuous:

$$0 < \xi \le 10^{-16} \quad \Longrightarrow \quad \epsilon_\xi < 1/32. \tag{14.6}$$

To verify every constant, $\mathcal{P}$ has positive coefficients and degree three. For $z \ge 3$, every $e^{-z}z^k$, $0 \le k \le 3$, is nonincreasing, by differentiation. Hence $e^{-z}\mathcal{P}(z)$ is nonincreasing there. The elementary bounds $e < 3 < 10 < e^3$, proved by the exponential series, give $16 < 16\log 10 < 48$. At $z_0 = 16\log 10$,

$$e^{-z_0}\mathcal{P}(z_0) \le 10^{-16}\mathcal{P}(48) = \frac{356710369517}{20000000000000} < \frac{1}{32}.$$

The exact integer evaluation is $\mathcal{P}(48) = 178355184758500$. Monotonicity proves (14.6), and the same exponential-series bounds show $\epsilon_\xi \to 0$ as $\xi \downarrow 0$. The smaller interval is used for the explicit positive lower denominators below; (14.5) holds on all of (11.16).

For every nonzero nonnegative weight vector, summing the facewise inequalities $(\sum w_e)^2 \ge \sum w_e^2$ gives $\|\mathsf{I}w\|_2^2 \ge 2\|w\|_2^2$. Equations (14.5)–(14.6) imply

$$\|v_w\|^2 = w^{\mathsf{T}}Cw \ge \frac{3\xi^2}{32}\|w\|_2^2 > 0, \qquad \left| \frac{\mathfrak{q}_A[v_w]}{\|v_w\|^2} - 3\kappa \right| \le \kappa(3744000\xi + 32\epsilon_\xi). \tag{14.7}$$

For the second inequality subtract 3κ times (14.5) from (7.2): the numerator error is at most $\kappa\xi^2(351000\xi + 3\epsilon_\xi)\|w\|_2^2$. Divide by the positive lower bound just proved. The conclusion is uniform over all boxes and all such weights, not merely over a prescribed finite support.

For the unchanged native weights of (1.4), (14.5) reads

$$\left| \|v_{w^{\mathrm{nat}}}\|^2 - \frac{\xi^2}{4}\mathcal{A}_L \right| \le \xi^2\epsilon_\xi S_{2,L} \le 2\xi^2\epsilon_\xi\mathcal{A}_L. \tag{14.8}$$

Thus its relative denominator error is at most $8\epsilon_\xi$, with no L^3 loss. On (14.6),

$$\frac{3\xi^2}{16}\mathcal{A}_L \le \|v_{w^{\mathrm{nat}}}\|^2 \le \frac{5\xi^2}{16}\mathcal{A}_L, \tag{14.9}$$

and the exact physical native electric-state quotient satisfies (14.7). No original n_2^2 weight or original scale has been changed. The physical substitution remains $\xi = 1/(4g_{\mathrm{YM}}^4)$, $\kappa = 2g_{\mathrm{YM}}^2/a$; the explicit interval (14.6) is $g_{\mathrm{YM}}^4 \geq 2500000000000000$. This conservative numerical interval is not claimed optimal.

For signed weights (14.5) is still valid, but the nonnegative-weight lower bound is not asserted. On the full leading kernel it gives

$$0 \leq \|v_w\|^2 \leq \xi^2 \epsilon_\xi \|w\|_2^2 \qquad (\mathsf{I}w = 0). \tag{14.10}$$

The fourth-order forms already proved in the parent remain intact. Equations (7.3) and (12.14) remain additional upper bounds, and the smallest of these proved bounds may be used. A uniform fourth-order kernel remainder and the minimum over all signed electric weights have not been obtained by this argument. Section 15 below obtains a uniform bound on that minimum by proving the gauge-equivariance of the full cluster similarity and using the exact physical free threshold. It does not identify the fourth-order kernel quotient by dividing (14.10). Section 16 instead constructs the exact two-face correction function, bounds its full-vacuum residual, and proves that kernel quotient with an explicit uniform remainder in (16.24). Nor does (14.7) remove the fixed-regulator restriction on the small-angle remainder in (1.4), construct a spatial continuum theory, or determine the Millennium mass gap.

15. The gauge-restricted threshold and the full signed electric minimum

The physical constraint improves the full-space estimate (12.8) by a factor of four, without changing the interaction or the interval (11.16). We prove the gauge maps, the free threshold, and the restricted resolvent before using the improvement for the actual interacting states.

The literature source for the finite-graph gauge Hilbert space and its vertex-invariant representation is John C. Baez, *Spin Network States in Gauge Theory*, arXiv:gr-qc/9411007v1, the section *Gauge Theory on a Graph*, Lemmas 1, 2, and the lemma labelled lem2.5 in the original TeX (primary source; *Advances in Mathematics* 117 (1996), 253–272, DOI 10.1006/aima.1996.0012). The retained source is literature/Baez-spin-network-states.tex, lines 174–321. The source constructs the invariant Hilbert space from edge representations and vertex intertwiners. The leaf argument below proves the particular free spectral bound directly, including the original open boundary and every endpoint gauge transformation.

15.1. Exact gauge projection, source convention, and support sectors

Retain the entire vertex set and define

$$\mathcal{G}_L = SU(2)^{\mathsf{V}_L}, \qquad (\alpha_g U)_e = g_{s(e)} U_e g_{t(e)}^{-1}, \qquad (\mathsf{T}_g f)(U) = f(\alpha_{g^{-1}} U). \tag{15.1}$$

There is no restriction of g_v to the identity at boundary vertices and no external charged representation at a vertex. These are precisely the gauge transformations of Section 1. Group multiplication gives $\alpha_g \alpha_h = \alpha_{gh}$, hence $\mathsf{T}_g \mathsf{T}_h = \mathsf{T}_{gh}$. Left and right Haar invariance on each link prove $\|\mathsf{T}_g f\| = \|f\|$ and $\mathsf{T}_g^* = \mathsf{T}_{g^{-1}}$. Translations of continuous functions are continuous in g, and density extends this to strong continuity on $\mathcal{H}_L$. Thus the strong Haar integral

$$\mathsf{P}_{\mathcal{G}} = \int_{\mathcal{G}_L} \mathsf{T}_g \, dg, \qquad \mathcal{H}_{\mathrm{phys}} = \{f : \mathsf{T}_g f = f \text{ for every } g\} = \operatorname{Ran} \mathsf{P}_{\mathcal{G}} \tag{15.2}$$

exists and is bounded. Haar invariance gives $\mathsf{T}_h \mathsf{P}_{\mathcal{G}} = \mathsf{P}_{\mathcal{G}}$; inversion invariance gives $\mathsf{P}_{\mathcal{G}}^* = \mathsf{P}_{\mathcal{G}}$. Integrating the first identity proves $\mathsf{P}_{\mathcal{G}}^2 = \mathsf{P}_{\mathcal{G}}$. Conversely the integral fixes every invariant vector, proving both range identities in (15.2).

Baez uses the coordinate action $A_e \mapsto g_{t(e)} A_e g_{s(e)}^{-1}$. The exact dictionary, on the same directed graph, is

$$\jmath : SU(2)^{\mathsf{E}_L} \longrightarrow SU(2)^{\mathsf{E}_L}, \qquad A_e = \jmath(U)_e = U_e^{-1}, \qquad \jmath^{-1} = \jmath. \tag{15.3}$$

Indeed $(g_s U_e g_t^{-1})^{-1} = g_t U_e^{-1} g_s^{-1}$. The pullback $\mathsf{J} f = f \circ \jmath$, from the source's Haar Hilbert space to ours, is therefore a unitary intertwiner with inverse the same pullback. Haar measure is invariant under inversion. Inversion is an isometry for the bi-invariant metric with the T_a frame of (1.2), so it preserves the Sobolev domains and intertwines every E_e. This proves the gauge-space dictionary; it does not silently reverse a Wilson traversal in (1.1).

Each T_g is a product of link isometries, so it preserves H^1, H^2 and commutes with E_e on its operator domain. In (1.1) the endpoint factors cancel at the four consecutive vertices, leaving conjugation by g_n inside the trace. Thus every W_p, and consequently H, H_0, V_p, is gauge invariant on its stated domain. The gauge projection also preserves these domains, by integrating in the corresponding graph norm.

Let P_e be Haar averaging in the one link e, followed by the constant lift. Endpoint Haar invariance gives $P_e \mathsf{T}_g = \mathsf{T}_g P_e$. In Section 11 notation, the embedded sector projection is exactly

$$\mathsf{Q}_I = \prod_{e \in I} (I - P_e) \prod_{f \notin I} P_f, \qquad \mathsf{Q}_I f = (\Pi_I f) \otimes 1_{I^c}. \tag{15.4}$$

Products commute, so Q_I commutes with the gauge representation and with $\mathsf{P}_{\mathcal{G}}$. The finite sum of all Q_I is the identity, by expanding $\prod_e [(I - P_e) + P_e]$. Restricting the unitary (11.2) consequently gives the exact orthogonal decomposition

$$\mathcal{H}_{\text{phys}} = \bigoplus_{I \subset \mathsf{E}_L} \left(\mathcal{H}'_I \otimes 1_{I^c} \right)^{\mathcal{G}_L}. \tag{15.5}$$

The inverse map remains the collection of Haar projections (15.4). The action on the I factor is the restriction of the product link representation, not a gauge group with its boundary removed.

15.2. Leaf annihilation and the exact free physical threshold

For a nonempty set I of links, consider the undirected graph with these edges and their endpoints. Suppose a vertex v has exactly one incident edge e in I. For

$$f \in \left(\mathcal{H}'_I \otimes 1_{I^c} \right)^{\mathcal{G}_L},$$

vary only g_v. On the I factors this translates only the variable U_e, on the left or the right according to its original orientation. On the exterior factors it acts on constants. Since f is invariant, integration over this single $SU(2)$ gauge variable gives

$$f = \int_{SU(2)} \mathsf{T}_{g_v} f \, dg_v = P_e f = 0. \tag{15.6}$$

The second equality is exactly left or right Haar averaging; the last equality follows from the $I - P_e$ factor in (15.4). For arbitrary, possibly entangled L^2 vectors the identity follows first on elementary tensors, then by density and the boundedness of the two averages. Thus no pointwise representative or unstated smoothness is used in the leaf argument.

It follows that every nonempty support of a nonzero invariant sector has degree at least two at every one of its incident vertices. Such a finite graph contains a cycle: choose a simple path of maximal length; at its last vertex there is a neighbor other than the preceding vertex. Maximality places that neighbor earlier in the path, and the resulting segment closes a simple cycle. The original cubic graph has no loops or multiple edges. The coloring $n \mapsto (-1)^{n_1+n_2+n_3}$ changes sign along every edge, so every cycle has even length. Its length is therefore at least four. In particular,

$$\left(\mathcal{H}'_I \otimes 1_{I^c} \right)^{\mathcal{G}_L} \neq \{0\}, \quad I \neq \varnothing \quad \Longrightarrow \quad |I| \geq 4. \tag{15.7}$$

This proof includes vertices on faces, edges, and corners of the open box. Omitting a boundary gauge transformation would invalidate the corresponding step (15.6); no such omission has been made.

On a nonempty support (11.3) gives $H_I \geq (3\kappa/4)|I|$, on its full domain and hence on its invariant subspace. Combining (15.5)–(15.7) and summing the orthogonal nonempty sectors yields the form inequality

$$H_0 \big|_{\mathcal{H}_{\text{phys}} \cap \Omega^\perp} \geq 3\kappa I. \tag{15.8}$$

The constant is attained. For any elementary face p, its unchanged W_p is gauge invariant and is a fundamental matrix coefficient in each of its four distinct link variables, including the two inverse traversals. Bi-invariance of the Casimir gives

$$E_e W_p = \tfrac{3}{4} W_p \quad (e \in \partial p), \qquad E_e W_p = 0 \quad (e \notin \partial p).$$

Integration in one participating link, with all others held fixed, gives $\int W_p \, dU = 0$ and $\int |W_p|^2 dU = 1$. For the latter, the product inside the trace is Haar distributed under that integral, also for an inverse occurrence, and the fundamental matrix-coefficient identity $\int U_{ij} \overline{U_{k\ell}} \, dU = \delta_{ik}\delta_{j\ell}/2$ gives the unit norm of its trace. Consequently

$$H_0 W_p = 3\kappa W_p, \qquad W_p \in \mathcal{H}_{\mathrm{phys}} \cap \Omega^{\perp}, \qquad \|W_p\| = 1. \tag{15.9}$$

This proves the exact free physical threshold 3κ, rather than merely a lower estimate. No free vector is substituted for the interacting vacuum in the argument that follows.

15.3. Gauge-equivariance of the interacting cluster similarity

The fixed-point construction of Section 11 may be performed within the invariant sector collections. To prove that statement, assume each $z_I \otimes 1_{I^c}$ is gauge invariant. The gauge operator factorizes as $\mathsf{T}_{g,I} \otimes \mathsf{T}_{g,I^c}$. Both z_I and 1_I are fixed by the first factor. Hence the rank-one creation operator (11.4) satisfies

$$\mathsf{T}_g \widehat{z}_I \mathsf{T}_g^{-1} = |\mathsf{T}_{g,I} z_I\rangle\langle \mathsf{T}_{g,I} 1_I| \otimes I_{I^c} = \widehat{z}_I. \tag{15.10}$$

The projections Π_I intertwine the full representation with its sector action by (15.4). The operator H_I^{-1} commutes with this action by the reducing property and functional calculus. Since V_p and Ω are invariant, (11.10) therefore sends invariant collections to invariant collections.

Iteration starts at zero. Every iterate is invariant, and the limit in the energy norm (11.7) is invariant because fixed subspaces of bounded representations are closed. Thus the actual fixed point has the claimed invariance. Its creation sum $C(z)$ and the two finite polynomial exponentials commute with T_g and $\mathsf{P}_{\mathcal{G}}$. The domain preservation proved in (11.9) also holds upon restriction. In particular the exact relation remains

$$\psi = \alpha e^{C(z)} \Omega, \qquad \alpha = \langle \Omega, \psi \rangle > 0,$$

with the scalar and unit physical vacuum specified in (12.7).

Write $\widetilde{H} = H_0 + \mathcal{K}$ as in (12.1). All three operators preserve the physical subspace; the bounded invertible similarity and its inverse do as well. The restriction of the sector norm (12.2) is still equivalent to the Hilbert norm, with its same explicitly volume-dependent upper comparison. The relative estimate is unchanged:

$$\|\mathcal{K} f\|_{\oplus,1} \le c_\xi \|H_0 f\|_{\oplus,1}, \qquad c_\xi = \frac{512}{3}\xi \le \frac{1}{288}, \quad f \in \mathrm{Dom}(H_0) \cap \mathcal{H}_{\mathrm{phys}}. \tag{15.11}$$

We do not replace the per-link bound used to prove this estimate by a larger value or alter its numerical contraction.

On every nonempty invariant sector, the spectral parameter λ of H_I satisfies $\lambda \ge 3\kappa$ by (15.7). Therefore, for $0 < t < 3\kappa(1 - c_\xi)$,

$$\|\mathcal{K}(H_0 - t)^{-1}\|_{\oplus,1;\mathrm{phys}} \le c_\xi \sup_{\lambda \ge 3\kappa} \frac{\lambda}{\lambda - t} = \frac{3\kappa c_\xi}{3\kappa - t} < 1. \tag{15.12}$$

The constant sector contributes zero because $\mathcal{K}\Omega = 0$, exactly as in Section 12. The Neumann series therefore gives a bounded inverse for $\widetilde{H} - t$ on the physical subspace. Its image is in the restricted H_0 domain: factor the inverse as $(H_0 - t)^{-1}[I + \mathcal{K}(H_0 - t)^{-1}]^{-1}$. Norm equivalence and the invariant similarities transfer that inverse to $H - \mathcal{E} - t$ on the original physical Hilbert space. No form positivity is claimed for the nonunitary similarity itself. Applying the spectral theorem to the original self-adjoint restriction, with its simple ground eigenvalue, proves

$$(H - \mathcal{E})\big|_{\mathcal{H}_{\mathrm{phys}} \cap \psi^{\perp}} \ge 3\kappa \left(1 - \frac{512}{3}\xi\right) I \ge \frac{287\kappa}{96} I, \qquad 0 < \xi \le \frac{1}{49152}. \tag{15.13}$$

In original parameters the last constant is $287 g_{\mathrm{YM}}^2/(48a)$ and the range is $g_{\mathrm{YM}}^4 \ge 12288$. The full, not necessarily invariant Hilbert-space bound remains (12.8), with constant $287\kappa/384$. The fourfold improvement is asserted exactly on the physical restriction just proved.

15.4. Covariance order, signed states, and an attained uniform minimum

Every E_e commutes with the gauge representation, so each centered vector v_e is in $\mathcal{H}_{\mathrm{phys}} \cap \psi^{\perp}$. There is no sign restriction on the coefficients in a real linear combination. Applying (15.13) to every v_w, including a zero vector, gives the matrix inequality

$$\mathcal{N} \;\geq\; 3\kappa(1-c_\xi)C \;\geq\; 0 \quad \text{on } \mathbb{R}^{\mathsf{E}_L}. \tag{15.14}$$

Combining it with the upper side of (7.2) yields

$$0 \leq w^{\mathsf{T}} C w \leq \frac{\xi^2 \|\mathsf{I} w\|_2^2/16 + 117000\xi^3 \|w\|_2^2}{1 - 512\xi/3}. \tag{15.15}$$

Here $351000/3 = 117000$; the leading coefficient is $1/16$. The exact facewise estimate $\|\mathsf{I} w\|_2^2 \leq 16\|w\|_2^2$ from Section 12 gives

$$0 \leq C \leq \frac{\xi^2 + 117000\xi^3}{1 - 512\xi/3} I_{\mathbb{R}^{\mathsf{E}_L}}. \tag{15.16}$$

On $\ker \mathsf{I}$ the numerator of (15.15) is exactly $117000\xi^3\|w\|_2^2$. This is an upper bound, not a fourth-order identification or a discarded null direction. For the unchanged native weights,

$$\|v_{w^{\mathrm{nat}}}\|^2 \leq \frac{\xi^2 \mathcal{A}_L/4 + 117000\xi^3 S_{2,L}}{1 - 512\xi/3} \leq \frac{\xi^2 \mathcal{A}_L}{4} \frac{1 + 936000\xi}{1 - 512\xi/3}. \tag{15.17}$$

The last step uses $S_{2,L} \leq 2\mathcal{A}_L$, without changing the coefficients or coordinates in the original state.

To define the signed minimum without dividing by an unproved denominator, let $\mathcal{H}_{\mathrm{phys},\mathbb{R}}$ denote the real-valued invariant functions and define

$$\mathsf{V}_\xi : \mathbb{R}^{\mathsf{E}_L} \longrightarrow \mathcal{H}_{\mathrm{phys},\mathbb{R}} \cap \psi^{\perp}, \qquad \mathsf{V}_\xi w = v_w, \qquad \mathscr{W}_\xi = \operatorname{Ran} \mathsf{V}_\xi. \tag{15.18}$$

Its Gram identity is $\langle \mathsf{V}_\xi w, \mathsf{V}_\xi z \rangle = w^{\mathsf{T}} C z$. Thus $\ker \mathsf{V}_\xi = \ker C$: $Cw = 0$ implies zero norm of $\mathsf{V}_\xi w$; conversely a zero vector has zero inner product with every $\mathsf{V}_\xi z$. The induced map

$$\overline{\mathsf{V}}_\xi : \big(\mathbb{R}^{\mathsf{E}_L}/\ker C,\, \langle [w],[z] \rangle_C = w^{\mathsf{T}} C z\big) \longrightarrow \mathscr{W}_\xi \tag{15.19}$$

is an isometric linear bijection. The bilinear form is well-defined because C annihilates its kernel, and it is positive on every nonzero equivalence class by the Gram identity. The inverse sends a vector v_w to its class $[w]$; two preimages differ by $\ker C$, which proves independence of choice. The same identity for the energy form shows that $\mathcal{N}$ annihilates $\ker C$ and descends to the quotient. This records the exact information loss. No equality $\ker C = \ker \mathsf{I}$, or $\ker C = \{0\}$, is assumed.

On $0 < \xi \leq 1/680000$, (7.5)–(7.6) give a nonzero v_e for every link, so $\mathscr{W}_\xi \neq \{0\}$. It is finite dimensional and consists of smooth vectors. The energy form is a continuous quadratic form on this finite dimensional space. Its restriction to the unit sphere attains a minimum by compactness. Through (15.19) this number is exactly

$$\mathfrak{m}_L(\xi) = \min_{\substack{w \in \mathbb{R}^{\mathsf{E}_L} \\ w^{\mathsf{T}} C w > 0}} \frac{w^{\mathsf{T}} \mathcal{N} w}{w^{\mathsf{T}} C w}. \tag{15.20}$$

Write $\Delta_{\mathrm{phys},L}(\xi)$ for the first positive spectral value of $H - \mathcal{E}$ on $\mathcal{H}_{\mathrm{phys}}$. Its existence follows from compact resolvent, the simple ground eigenvalue, and the nonzero physical excited subspace; this is the same excitation operator as in (15.13). The spectral variational principle, (15.14), and the actual single-link trial bound (7.6) prove the complete chain

$$\begin{aligned} 3\kappa - 512\kappa\xi &= 3\kappa(1-c_\xi) \\ &\leq \Delta_{\mathrm{phys},L}(\xi) \leq \mathfrak{m}_L(\xi) \leq 3\kappa + 2472000\kappa\xi, \qquad 0 < \xi \leq \frac{1}{680000}. \end{aligned} \tag{15.21}$$

Every $L \geq 2$ is covered by these same constants and coupling interval. In particular the estimate permits weights that depend on L, ξ, have arbitrary signs, or lie in $\ker \mathsf{I}$, whenever they represent a nonzero actual

state. Such a state is included in the minimization, not excluded by a leading-coefficient test. The quotient construction retains all zero actual states as the explicitly identified kernel, where an energy quotient has no defined value.

Equation (15.21) proves both uniform statements $\Delta_{\mathrm{phys},L}(\xi) = 3\kappa + O(\kappa\xi)$ and $\mathfrak{m}_L(\xi) = 3\kappa + O(\kappa\xi)$, with the displayed two-sided constants as their meaning. It does not assert that every signed state's quotient is near 3κ. In particular it does not replace the parent's fixed-box fourth-order kernel quotient or prove its remainder uniform. The separate calculation (16.17)–(16.24) supplies that uniform remainder on the entire original incidence kernel. The earlier two-sided nonnegative/native estimate (14.7) also remains valid with its original range. In physical parameters the range of (15.21) is $g_{\mathrm{YM}}^4 \geq 170000$ and its central scale is $3\kappa = 6g_{\mathrm{YM}}^2/a$. No lattice-spacing limit or coupling continuation is taken.

15.5. The physical conditional-complement operator

Retain $J_S, Q_S^\rho, \mathscr{K}_S^\rho, U_\psi, q_\rho$ and their domains from Section 10. Let the gauge group act on $\mathscr{H}_\rho$ by the same pullback (15.1). It is unitary because $\rho = \psi^2$ is invariant. The retained-variable space has the induced endpoint action on its retained links. Under a simultaneous transformation of (s, r), invariance of $\rho(s, r)$ and Haar measure in s proves invariance of the marginal $\rho_{S^c}(r)$. Changing variables in the conditional integral then gives

$$J_S^* \mathsf{T}_g = \mathsf{T}_{g,S^c} J_S^*, \qquad \mathsf{T}_g J_S = J_S \mathsf{T}_{g,S^c}. \tag{15.22}$$

It follows that Q_S^ρ and the gauge projection commute. They preserve the form domain as proved in Section 10 and by the isometric gauge action. Averaging smooth approximants over the compact gauge group and then applying Q_S^ρ proves form density of the smooth invariant conditional-mean-zero vectors in

$$\mathscr{K}_{S,\mathrm{phys}}^\rho = \mathscr{K}_S^\rho \cap \mathscr{H}_\rho^{\mathcal{G}_L}.$$

The closed form restricted to $H^1 \cap \mathscr{K}_{S,\mathrm{phys}}^\rho$ therefore defines a self-adjoint operator $C_{S,\mathrm{phys}}^\rho$ by exactly the variational domain rule given for C_S^ρ in Section 10. Equations (15.22) and invariance of q_ρ also show that this is the reducing physical restriction of C_S^ρ.

The exact unitary $U_\psi G = \psi G$ intertwines the gauge representations. For a conditional-mean-zero G, Fubini gives $\int G\rho\, dU = 0$; consequently $U_\psi G$ is physical and perpendicular to ψ when G is invariant. Apply (15.13) under this unitary and the form identity (10.10):

$$C_{S,\mathrm{phys}}^\rho \geq 3\kappa(1 - 512\xi/3) I \geq \frac{287\kappa}{96} I, \qquad \|(C_{S,\mathrm{phys}}^\rho)^{-1}\| \leq \frac{1}{3\kappa(1 - 512\xi/3)} \leq \frac{96}{287\kappa}. \tag{15.23}$$

This holds for every nonempty eliminated set S, independently of its size and the exterior volume, on (11.16). It concerns the full form with derivatives in all link directions, not a differential operator containing only the eliminated derivatives. For noninvariant conditional-mean-zero vectors the previously proved constants (12.15)–(12.16), not (15.23), remain the stated bounds. The mathematical distinction is implemented by the proved reducing projection and unitary, not by declaring the two constructions unrelated.

16. Uniform fourth-order forms on the entire signed incidence kernel

Retain the full Hamiltonian (1.2), its actual positive unit vacuum ψ, $A = H - \mathcal{E}$, the gauge constraint (15.1), and every original real edge weight. Throughout this section $0 < \xi \leq 1/49152$, unless an explicitly smaller interval is displayed. Write $\mathscr{Z}_L = \ker \mathsf{I}$. We prove volume-uniform remainders for the first surviving covariance and energy forms on all of $\mathscr{Z}_L$, not merely on one alternating weight.

The exact two-face coefficients are the existing calculation in the parent source wilson_variation/ELECTRIC_COVARIANCE_UNSIGNED_INCIDENCE_20260908.md, Sections 4–5, equations (4.1)–(5.8). The source's electric operator denoted H_0 there is $\sum_e E_e$; our H_0 in (11.1) is $\kappa \sum_e E_e$. Accordingly define here $\mathfrak{h}_0 = \sum_e E_e$, so $H_0 = \kappa\mathfrak{h}_0$, on the same domain. Every occurrence of a physical energy below retains $\kappa = 2g_{\mathrm{YM}}^2/a$. No change of Hamiltonian or vacuum is made. We first record the two-face functions and their exact derivative dictionary needed for the new remainder calculation.

16.1. The retained two-face modes and a local correction function

Let $\mathscr{P}_2$ be the unordered pairs of distinct elementary faces sharing a link. Their unique shared link is $e(p,q)$; their other six links form the set $O(p,q)$. Write

$$R_{pq} = \operatorname{tr}(A_{pq}B_{pq}), \qquad D_{pq} = W_pW_q - \tfrac{1}{2}R_{pq}, \qquad G_{pq} = \nabla_{e(p,q)}W_p \cdot \nabla_{e(p,q)}W_q.$$

Here the original face words, cyclically based at the shared link, are U_eA_{pq} and $U_e^{-1}B_{pq}$, reversing a full face traversal when necessary. The trace is unchanged under that full reversal in $SU(2)$. The words A_{pq}, B_{pq} contain the six distinct outer links with the original inherited inverse occurrences. No individual inverse occurrence is dropped.

The Pauli contraction with $T_a = -i\sigma_a/2$ is

$$\sum_a \operatorname{tr}(T_aC)\operatorname{tr}(T_aD) = -\tfrac{1}{2}\operatorname{tr}(CD) + \tfrac{1}{4}\operatorname{tr}C\operatorname{tr}D.$$

Inserting the minus sign from differentiation of the inverse shared link proves the first identity below. The product rule for E_e then proves the shared-link label 2 of D_{pq}:

$$\begin{aligned} G_{pq} &= \tfrac{3}{8}R_{pq} - \tfrac{1}{4}D_{pq}, \\ \mathfrak{h}_0R_{pq} &= \tfrac{9}{2}R_{pq}, \qquad \mathfrak{h}_0D_{pq} = \tfrac{13}{2}D_{pq}, \\ \|R_{pq}\|_0^2 &= 1, \qquad \|D_{pq}\|_0^2 = \tfrac{3}{4}, \qquad \langle R_{pq}, D_{pq}\rangle_0 = 0. \end{aligned} \tag{16.1}$$

The subscript 0 denotes the original full product Haar measure, used to compute coefficients, not substituted for the physical density. Indeed each outer link has Casimir $3/4$ on both functions; on the shared link R_{pq} has Casimir zero. The identity $E_e(W_pW_q) = 2D_{pq}$ follows from the displayed Pauli contraction, and gives the remaining Casimir label.

For completeness, the norms in (16.1) can be checked by conditioning on the shared link. The two disjoint outer words are independent Haar matrices. Specifically take $P = U_eA_{pq}$ and $Q = B_{pq}U_e^{-1}$; then $\operatorname{tr}(PQ) = R_{pq}$ and $\operatorname{tr}P\operatorname{tr}Q = W_pW_q$. For these independent holonomies $P = p_0I + ip_a\sigma_a$, $Q = q_0I + iq_a\sigma_a$, put $x = p_0q_0$, $y = \sum_{a=1}^3 p_aq_a$. Their uniform unit-sphere second moments give $\mathbb{E}x^2 = 1/16$, $\mathbb{E}y^2 = 3/16$, $\mathbb{E}xy = 0$. Now $R = 2(x-y)$ and $D = 3x + y$; multiplying these expressions proves all three Haar products in (16.1). Their individual Haar means are zero, by integration in an outer fundamental link.

The functions R_{pq}, D_{pq}, for all unordered pairs, are mutually orthogonal except for their own squared norms. Here is the geometric check retained from the parent. A coplanar pair has a 2-by-1 rectangular outer boundary, which determines its two unit faces. A perpendicular pair has a unit-cube bounding box; the two vertices absent from its outer six-cycle form the edge opposite the shared edge. They determine that shared edge and the two faces. The two cases have respectively two and three nonzero coordinate spans. Thus different pairs have different outer supports. Two R's then have a $3/4$-versus-zero link label somewhere. An R and a D differ at the latter's label-2 link. Two D's differ either at that label-2 link or at an outer link. Self-adjointness of the differing link Casimir proves the claimed orthogonality in every case.

For a physical link e and pair $\{p,q\}$, define the exact real coefficient

$$\begin{aligned} c_{e;pq} &= -\mathbf{1}_{\{e=e(p,q)\}} + \tfrac{1}{6}\mathbf{1}_{\{e\in O(p,q)\}}, \\ F_e &= \sum_{\{p,q\}\in\mathscr{P}_2} c_{e;pq}G_{pq}, \\ \mathcal{B}_e &= \sum_{\{p,q\}\in\mathscr{P}_2} c_{e;pq}\left(\tfrac{1}{12}R_{pq} - \tfrac{1}{26}D_{pq}\right). \end{aligned} \tag{16.2}$$

Equation (16.1) proves $\mathfrak{h}_0\mathcal{B}_e = F_e$ and $\int \mathcal{B}_e\,dU = 0$. These functions are bounded, real, smooth and gauge invariant. They are not truncated Hamiltonians.

Every contributing pair has e in at least one of its faces. There are at most four such faces and at most twelve adjacent faces for each, giving at most 48 pairs. Their link union lies in the shared-face graph ball of radius

two about e. Each radius-one ball has at most 13 links, so a radius-two ball is contained in a union of 13 such balls and has at most 169 links. The pointwise bounds are

$$\|\mathcal{B}_e\|_\infty \le 24, \qquad \sum_f \|\nabla_f \mathcal{B}_e\|_\infty \le 168. \tag{16.3}$$

To verify the constants, $|R| \le 2$, $|D| \le 5$. On an outer link $|\nabla R| \le 1$ and $|\nabla D| \le 2 + 1/2 = 5/2$, using (6.2) and the product rule. On the shared link $\nabla R = 0$, $|\nabla D| \le 4$. The absolute value of a shared-coefficient summand in (16.2) is at most $2/12 + 5/26 < 1/2$. Its derivative at any one of its seven links is at most $\max\{1/12 + (5/2)/26, 4/26\} < 1/2$. An outer-coefficient summand is a further factor $1/6$ smaller. Summing at most 48 summands and seven derivative locations proves (16.3).

16.2. A full-vacuum third-order state residual

Set $h_e = W_e^\star/4$, which is a function, and put $d_e = E_e\psi - \xi h_e\psi$. Section 4 gives $\|d_e\| \le 4200\xi^2$. We need its differentiated counterpart, not a derivative of a norm estimate. Introduce $f_e = \psi - (\xi/3)W_e^\star\psi$. The exact product rule gives

$$E_e f_e = d_e - \tfrac{\xi}{3} W_e^\star E_e\psi + \tfrac{2\xi}{3} D_e\psi.$$

Here D_e is the differential contraction of Section 6, not the two-face function D_{pq}. Equations (3.1), (10.5) and (6.2) give $\|E_e\psi\| \le 8\xi$, $\|\nabla_e\psi\| \le 8\xi$, $\|D_e\psi\| \le 32\xi$. Thus $\|E_e f_e\| \le (4200 + 128/3)\xi^2 = (12728/3)\xi^2$. The exact one-link spectral inequality $\|\nabla_e f\| \le (2/\sqrt{3})\|E_e f\|$ holds also when f has a constant component. Apply it to f_e and retain the product-rule remainder:

$$\begin{aligned} \left\|\nabla_e\psi - \tfrac{\xi}{3}(\nabla_e W_e^\star)\psi\right\| &\le \frac{2}{\sqrt{3}}\frac{12728}{3}\xi^2 + \frac{64}{3}\xi^2 \\ &< 5000\xi^2. \end{aligned} \tag{16.4}$$

For the last numerical inequality use $2/\sqrt{3} < 7/6$; the resulting coefficient is $44740/9 < 5000$. All differentiations are on the actual smooth vacuum.

From (6.1) and the full ground-state product rule,

$$Ad_e = \kappa\xi\left[-2D_e\psi + \tfrac{1}{2}\sum_{p\ni e}\sum_{f\in\partial p}\nabla_f W_p\cdot\nabla_f\psi\right]. \tag{16.5}$$

Insert (16.4) for each differentiated vacuum. The error from the first term has norm at most $2\cdot 4\cdot 5000\kappa\xi^3$. The error from the second has norm at most $(1/2)\cdot 4\cdot 4\cdot 5000\kappa\xi^3$. The full leading function is exactly F_e of (16.2). To check every cancellation, a repeated face contributes $-2/3$ from the first sum and $4/6$ from the second, hence zero. An unordered distinct pair with shared link e contributes $-4/3 + 2/6 = -1$; a pair with e on its outer boundary contributes $1/6$. These are precisely $c_{e;pq}$. Consequently

$$\|Ad_e - \kappa\xi^2 F_e\psi\| \le 80000\kappa\xi^3. \tag{16.6}$$

Define the original local differential observable and its centered residual state by

$$\mathscr{D}_e = E_e - \xi h_e - \xi^2\mathcal{B}_e, \qquad s_e = Q_\psi\mathscr{D}_e\psi, \qquad Q_\psi = I - |\psi\rangle\langle\psi|. \tag{16.7}$$

It is a bounded real multiplication perturbation of E_e, with the same one-link operator domain tensored with the exterior space. It is self-adjoint on that domain; every product below is first evaluated on smooth ψ. The vector is gauge invariant, real, smooth and perpendicular to the same ψ. Since

$$A(\mathcal{B}_e\psi) = \kappa F_e\psi - 2\kappa\sum_f \nabla_f\mathcal{B}_e\cdot\nabla_f\psi,$$

the last term has norm at most $2\kappa\cdot 168\cdot 8\xi = 2688\kappa\xi$. Equations (16.6)–(16.7), including the exact centering killed by A, imply

$$\|As_e\| \le 82688\kappa\xi^3 < 83000\kappa\xi^3, \qquad \|s_e\| \le 28000\xi^3. \tag{16.8}$$

The second assertion is the proved physical inverse bound $96/(287\kappa)$ from (15.13), applied to s_e: $96\cdot 83000/287 < 28000$. It is not obtained from an assumed locality of the full vacuum.

16.3. Exact finite-range residual energy and its extensive consequence

The support of $\mathscr{D}_e$ is inside the radius-two link ball about e. The commutator $[H, \mathscr{D}_f]$ has support inside the radius-two ball about f as well. Indeed $[H, E_f]$ uses only the faces at f, while the commutator of H with either multiplication function in (16.7) contains only that function's derivatives on its own link support. All multiplication potentials commute with those functions. No distant coefficient is generated by these exact commutators.

Let $\mathcal{S}_{ef} = \langle s_e, A s_f \rangle$. Expanding the double commutator on the smooth vacuum gives

$$\mathcal{S}_{ef} = \tfrac{1}{2}\langle \psi, [\mathscr{D}_e, [H, \mathscr{D}_f]]\psi \rangle, \qquad \mathcal{S}_{ef} = 0 \quad \text{when } d(e,f) > 4. \tag{16.9}$$

For the first identity no commutation of $\mathscr{D}_e, \mathscr{D}_f$ is assumed. The expansion equals $\langle \mathscr{D}_e\psi, A\mathscr{D}_f\psi\rangle + \langle \mathscr{D}_f\psi, A\mathscr{D}_e\psi\rangle$; the two terms are equal because all functions and differential operators are real and A is self-adjoint. Centering changes neither term. For the second identity the two indicated supports are disjoint, so their differential operators commute on the smooth domain.

The radius-four link ball has at most $17^3 = 4913$ vertices by the unchanged coordinate map of Section 14. Equation (16.8) gives $|\mathcal{S}_{ef}| \le 28000 \cdot 83000\kappa\xi^6$. The symmetric row-sum estimate used in (6.6) now gives, for every real weight w, with $s_w = \sum_e w_e s_e$,

$$0 \le \mathfrak{q}_A[s_w] \le 10^{14}\kappa\xi^6 \|w\|_2^2, \qquad \|s_w\| \le 6 \cdot 10^6 \xi^3 \|w\|_2. \tag{16.10}$$

The exact row coefficient is $4913 \cdot 28000 \cdot 83000 = 11417812000000 < 10^{14}$. Applying (15.13) once more bounds the squared norm by $(96/287)10^{14}\xi^6\|w\|_2^2$; $(96/287)10^{14} < (6 \cdot 10^6)^2$. This is the step which removes the number of nonzero weights from the residual estimate. The physical s_w and its full energy form remain unchanged.

16.4. Uniform covariance and energy estimates for the correction functions

Put

$$u_e = Q_\psi \mathcal{B}_e \psi, \qquad \mathcal{C}^B_{ef} = \langle u_e, u_f \rangle, \qquad \mathcal{C}^{B,0}_{ef} = \langle \mathcal{B}_e, \mathcal{B}_f \rangle_0.$$

Each $\mathcal{B}_e$ uses at most 169 links, and a product uses at most 338. Equations (3.2)–(3.3), with $32\sqrt{169}/3 < 139$ and $32\sqrt{338}/3 < 197$, give

$$|\langle \mathcal{B}_e \rangle_\psi| \le 7000\xi, \qquad |\mathcal{C}^B_{ef} - \mathcal{C}^{B,0}_{ef}| \le 230000\xi. \tag{16.11}$$

In detail, the first coefficient is at most $48 \cdot 139(1 + 139/49152) < 7000$. For the second use the product bound $24^2 = 576$, and retain the product of the two means. The coefficient is at most

$$2 \cdot 576 \cdot 197 + \frac{2 \cdot 576 \cdot 197^2 + 7000^2}{49152} = \frac{175757179}{768} < 230000.$$

This controls every pair locally; it does not yet sum all pairs.

We also require the large-distance bound

$$|\mathcal{C}^B_{ef}| \le 10000 e^{-d(e,f)/4} \quad \text{when } d(e,f) \ge 5. \tag{16.12}$$

Here is the support-dependent version of the method in Section 13. For bounded O_X, O_Y on disjoint link sets, the exact recurrence (13.6) has the initial term $2\|O_X\|\|O_Y\|\mathbf{1}_{X\cap Y \ne \varnothing}$. The first face in an iterated chain has at most $4|X|$ choices; each later face has at most 16 choices. Reaching Y requires at least $d(X,Y)$ faces. The same strong evolution and factorial-remainder proof therefore gives

$$\|[\alpha_t(O_X), O_Y]\| \le 2|X|\|O_X\|\|O_Y\| e^{-d(X,Y)}(e^{192b|t|} - 1). \tag{16.13}$$

The extra factor is the explicitly counted $|X|$, not the volume of the full box or an assumed single-link estimate for a multi-link observable. This is the crossing-set construction of Nachtergaele–Sims, arXiv:1410.8174v1, Section 3, equations fbd, iterandum and the complete Step 4; the original proof has been read directly in the retained TeX.

For $\mathcal{B}_e, \mathcal{B}_f$, take X, Y their actual supports and set $s = d(e,f) - 4 \geq 1$, so $d(X,Y) \geq s$. Repeat the explicitly evaluated Gaussian filter (13.7), now with the $|X|$ in (16.13), and take $T = s/(\Delta_* + 2v)$, $\Delta_* = 287\kappa/384$, $v = 192b$. The spectral-integral proof is unchanged because its only observable hypotheses are boundedness and self-adjointness. This is the method of Nachtergaele–Sims, arXiv:math-ph/0506030v3, Section 3.2; its spectral and Gaussian integral proof is read and retained, not inferred from a title. The numerical estimates (13.8) give $vT < s/100$ and $\Delta_* T \geq 287s/290$. Since $|X| \leq 169$ and $\|u_e\|\|u_f\| \leq 576$, the two terms of the filter are at most

$$\frac{576 \cdot 169}{100} s\, e^{-99s/100} + 1728 e^{-s/4} \leq \left(\frac{576 \cdot 169}{74} + 1728 \right) e^{-s/4}.$$

We used $se^{-74s/100} \leq 100/74$, proved by $\exp(x) \geq x$, and $2/\pi < 1$. Finally $e^{-s/4} = \exp(1) e^{-d(e,f)/4} < 3e^{-d(e,f)/4}$; the resulting coefficient $337824/37 < 10000$ proves (16.12).

For $d(e,f) > 4$, the two correction functions have disjoint supports and zero individual Haar means, so $\mathcal{C}^{B,0}_{ef} = 0$ by Fubini. Use (16.11) up to integer distance D and (16.12) outside it. With the generating function G already explicitly evaluated in (14.2), define

$$\begin{aligned} \mathcal{P}_B(z) &= 230000(16z+5)^3 + 10000 G(16z+9, 4/5), \\ r_B(\xi) &= \xi \mathcal{P}_B(\log(1/\xi)). \end{aligned} \tag{16.14}$$

Take $D = \lceil 4 \log(1/\xi) \rceil$, which is at least four on the present interval. The same coordinate shell count gives

$$\begin{aligned} \sup_e \sum_f |\mathcal{C}^B_{ef} - \mathcal{C}^{B,0}_{ef}| &\leq 230000\xi(4D+1)^3 + 10000 \sum_{n>D} (4n+1)^3 e^{-n/4} \\ &\leq r_B(\xi). \end{aligned} \tag{16.15}$$

Indeed $e^{-D/4} \leq \xi$, $4D + 1 \leq 16 \log(1/\xi) + 5$, and $4D + 5 \leq 16 \log(1/\xi) + 9$; (14.2) evaluates the full tail. The symmetric row and column bound also bounds the ℓ^2 operator norm, by the weighted Cauchy–Schwarz proof following (14.4). Thus (16.15) applies to arbitrary extensive real weights without an ℓ^1 loss.

For energy, retain the exact multiplication-state form

$$\mathcal{E}^B_{ef} = \mathfrak{q}_A[u_e, u_f] = \kappa \int \rho \sum_h \nabla_h \mathcal{B}_e \cdot \nabla_h \mathcal{B}_f \, dU.$$

Its Haar counterpart is $\mathcal{E}^{B,0}_{ef} = \kappa \langle \mathcal{B}_e, \mathfrak{h}_0 \mathcal{B}_f \rangle_0$, by integration by parts. Both matrices vanish when $d(e,f) > 4$. The multiplier in the integral has sup norm at most $168^2 = 28224$ and support size at most 338. Formula (3.3) bounds the coefficient of its expectation difference by

$$2 \cdot 28224 \cdot 197(1 + 197/49152) = \frac{1429097691}{128} < 12000000.$$

The same radius-four row count proves, for all real w,

$$|w^{\mathsf{T}}(\mathcal{E}^B - \mathcal{E}^{B,0})w| \leq 6 \cdot 10^{10} \kappa \xi \|w\|_2^2. \tag{16.16}$$

The exact row coefficient $4913 \cdot 12000000 = 58956000000$ is smaller than the stated constant.

16.5. The full kernel identity and its uniform fourth-order remainder

For $w \in \mathscr{Z}_L$, the two face sums vanish separately. Thus the sum of its six outer weights at each adjacent pair is $-2w_{e(p,q)}$. Substituting in (16.2) gives the exact function identity

$$\mathcal{B}_w := \sum_e w_e \mathcal{B}_e = \sum_{\{p,q\} \in \mathscr{P}_2} w_{e(p,q)} \left(-\tfrac{1}{9} R_{pq} + \tfrac{2}{39} D_{pq} \right) = S_w. \tag{16.17}$$

Indeed the summed $c_{e;pq}$ is $-w_{e(p,q)} + (1/6)(-2w_{e(p,q)}) = -4w_{e(p,q)}/3$. This is precisely the parent's surviving two-face function; the present calculation has not chosen a different weight family or discarded one of its Casimir components.

Define

$$\mathcal{K}_L(w,z)=\sum_e \binom{r_e}{2} w_e z_e, \qquad d_0=\frac{196}{13689}, \qquad e_0=\frac{8}{117}.$$

Since $2\le r_e\le 4$, $\|w\|_2^2\le \mathcal{K}_L(w,w)\le 6\|w\|_2^2$. The orthogonality in (16.1) gives exactly

$$\|\mathcal{B}_w\|_0^2=d_0\mathcal{K}_L(w,w), \qquad \kappa\langle \mathcal{B}_w,\mathfrak{h}_0\mathcal{B}_w\rangle_0=\kappa e_0\mathcal{K}_L(w,w). \tag{16.18}$$

The retained component sums are $d_0=1/81+(3/4)(4/1521)$ and $e_0=(9/2)/81+(13/2)(3/4)(4/1521)$. Each pair is counted once through its unique shared link.

The first-order term in (16.7) cancels exactly on $\mathscr{Z}_L$, because $\sum_e w_e h_e=(1/4)\sum_p (\mathsf{l}w)_p W_p=0$. Consequently the actual centered state satisfies

$$v_w=\xi^2 u_w+s_w, \qquad u_w=Q_\psi \mathcal{B}_w\psi, \qquad s_w=\sum_e w_e s_e, \qquad w\in\mathscr{Z}_L. \tag{16.19}$$

Both terms on the right are physical vectors of the full interacting Hilbert space, on the domains used above.

Set the explicit positive errors

$$\begin{aligned}\varepsilon_C(\xi)&=r_B(\xi)+12\cdot 10^6\xi\sqrt{6d_0+r_B(\xi)}+36\cdot 10^{12}\xi^2,\\ \varepsilon_N(\xi)&=6\cdot 10^{10}\xi+2\cdot 10^7\xi\sqrt{6e_0+6\cdot 10^{10}\xi}+10^{14}\xi^2.\end{aligned} \tag{16.20}$$

Then, simultaneously for every original box and every real $w\in\mathscr{Z}_L$,

$$\begin{aligned}\big|\|v_w\|^2-d_0\xi^4\mathcal{K}_L(w,w)\big|&\le \xi^4\varepsilon_C(\xi)\|w\|_2^2,\\ \big|\mathfrak{q}_A[v_w]-\kappa e_0\xi^4\mathcal{K}_L(w,w)\big|&\le \kappa\xi^4\varepsilon_N(\xi)\|w\|_2^2.\end{aligned} \tag{16.21}$$

Here is the complete error calculation. Equations (16.15),(16.18) give

$$\big|\|u_w\|^2-d_0\mathcal{K}_L(w,w)\big|\le r_B(\xi)\|w\|_2^2, \qquad \|u_w\|\le\sqrt{6d_0+r_B(\xi)}\,\|w\|_2.$$

Expand the squared norm of (16.19). Its cross term is at most $2\xi^2\|u_w\|\|s_w\|$, and its residual square is $\|s_w\|^2$. Substitution of (16.10) yields exactly the first line of (16.20), with no hidden weight-dependent constant.

For energy, (16.16),(16.18) give

$$|\mathfrak{q}_A[u_w]-\kappa e_0\mathcal{K}_L(w,w)|\le 6\cdot 10^{10}\kappa\xi\|w\|_2^2, \qquad \mathfrak{q}_A[u_w]\le\kappa(6e_0+6\cdot 10^{10}\xi)\|w\|_2^2.$$

The nonnegative form satisfies Cauchy–Schwarz: apply ordinary Cauchy–Schwarz to $A^{1/2}u_w, A^{1/2}s_w$. Thus the energy cross term in (16.19) is at most $2\xi^2\sqrt{\mathfrak{q}_A[u_w]\mathfrak{q}_A[s_w]}$. Use (16.10), whose square-root coefficient is 10^7. The residual energy contributes $10^{14}\kappa\xi^6\|w\|_2^2$. These are exactly the three terms in ε_N, proving the second line of (16.21).

Both errors tend to zero as $\xi\downarrow 0$. Indeed $\mathcal{P}_B$ is a fixed cubic with positive coefficients, so $\xi\mathcal{P}_B(\log(1/\xi))\to 0$; each remaining displayed term then tends to zero. More explicitly, (16.21) gives a covariance remainder $O(\xi^5(1+\log(1/\xi))^3)\|w\|_2^2$ and an energy remainder $O(\kappa\xi^5)\|w\|_2^2$, with constants independent of L,w. These real-coupling estimates do not assert that the fixed-box even Taylor expansion has a nonzero fifth-order term. The parent's $O_L(\xi^6)$ expansion remains true on its stated fixed-box interval; its constant is not silently promoted to a uniform one.

16.6. Nonzero kernel states and the uniform two-face energy quotient

The explicit interval

$$0<\xi\le 10^{-24} \tag{16.22}$$

has $\varepsilon_C(\xi)<d_0/2$. To check this without a numerical logarithm, put $z=\log(1/\xi)$. The positive-coefficient cubic $\mathcal{P}_B$ has $e^{-z}\mathcal{P}_B(z)$ nonincreasing for $z\ge 3$; differentiate each monomial $e^{-z}z^k$, $0\le k\le 3$. At the largest allowed ξ, $24<24\log 10<72$, using the exponential-series bounds $e<10<e^3$. The exact polynomial value is

$$\mathcal{P}_B(72)=437811569040000.$$

It follows for all of (16.22) that

$$r_B(\xi) \le \frac{437811569040000}{10^{24}} < 10^{-9}, \qquad 6d_0 + r_B(\xi) < 1.$$

The second and third terms of ε_C are therefore bounded by $12 \cdot 10^6/10^{24}$ and $36 \cdot 10^{12}/10^{48}$. Their sum with the displayed rational bound is less than $10^{-9} < d_0/2$; every number in this comparison is rational. This conservative interval is explicit, not asserted optimal.

For every nonzero real $w \in \mathscr{Z}_L$, (16.21) now proves

$$\|v_w\|^2 \ge \xi^4(d_0 - \varepsilon_C(\xi))\|w\|_2^2 \ge \tfrac{1}{2}d_0\xi^4\|w\|_2^2 > 0. \tag{16.23}$$

The exact state map (15.18) is therefore injective on the entire incidence kernel throughout the same volume-independent interval. Its restriction is a bijection onto its image, and (16.23) bounds its inverse, in the original edge norm and physical state norm, by $\xi^{-2}\sqrt{2/d_0}$. Neither the weights nor the state has been rescaled in that assertion.

Finally $e_0/d_0 = 234/49$. Subtract $(234/49)\kappa$ times the first line of (16.21) from the second before dividing by the strictly positive bound (16.23). The leading $\mathcal{K}_L(w,w)$ terms cancel with their exact coefficients. The result is

$$\left|\frac{\mathfrak{q}_A[v_w]}{\|v_w\|^2} - \frac{234}{49}\kappa\right| \le \kappa\,\frac{\varepsilon_N(\xi) + (234/49)\varepsilon_C(\xi)}{d_0 - \varepsilon_C(\xi)} \quad \left(0 \ne w \in \mathscr{Z}_L,\ 0 < \xi \le 10^{-24}\right). \tag{16.24}$$

Its right-hand coefficient tends to zero independently of the box and of every signed, volume-dependent or coupling-dependent kernel weight. This proves the uniform first surviving two-face quotient of the actual states, retaining both free Casimir components only as coefficients of the full-vacuum calculation.

The minimum over all signed actual states in (15.21) and the kernel quotients in (16.24) describe the same state map on different explicitly specified domains. Their central energies are 3κ and $(234/49)\kappa$; the difference $(234/49 - 3)\kappa = (87/49)\kappa > 0$ is explicit. No implication of unrelatedness is drawn: (16.17)–(16.19) prove the exact cancellation map and the surviving physical vectors. The present injectivity assertion is on $\mathscr{Z}_L$; it is not a proof that $\ker C = \{0\}$ on the entire edge space for a uniform coupling interval. Section 17 proves that stronger statement by controlling the mixed one-face and two-face coefficient covariance; it does not infer it from the kernel restriction. The physical parameters remain $\xi = 1/(4g_{\mathrm{YM}}^4)$, $\kappa = 2g_{\mathrm{YM}}^2/a$. Interval (16.22) is $g_{\mathrm{YM}}^4 \ge 25000000000000000000000000$. No spatial continuum construction, interchange of limits, or Millennium conclusion is asserted by these finite-regulator strong-coupling estimates.

17. Mixed face coordinates and uniform injectivity on the full edge space

We retain the same actual vacuum and all real edge weights, without restricting to $\ker \mathsf{I}$. The local residual construction of Section 16 already applies to these weights. Its remaining leading function contains both the one-face and two-face coordinates. We now control their complete mixed covariance. Throughout $0 < \xi \le 1/49152$; the smaller interval used for a positive lower bound will be stated separately.

17.1. Exact coefficient map, image and inverse

For each unordered adjacent pair $\alpha = \{p, q\}$, put

$$f_p = W_p, \qquad f_\alpha = \tfrac{1}{12}R_{pq} - \tfrac{1}{26}D_{pq}, \qquad b_0 = \tfrac{1}{144} + \tfrac{3}{4\cdot 676} = \frac{49}{6084}.$$

The index set is the disjoint union $\mathscr{J}_L = \mathsf{P}_L \sqcup \mathscr{P}_2$. These are real, gauge-invariant smooth functions with zero Haar means, supported on at most seven original links and of absolute value at most two. The pair functions in fact have absolute value less than $1/2$, by the constants in (16.3). For different faces the Wilson functions are orthogonal by integration in a link appearing in only one face. Each has squared norm one. Every one-face function is orthogonal to every R_{pq}: their four- and six-link fundamental supports differ. It is

orthogonal to every D_{pq}, whose shared link has Casimir two rather than zero or $3/4$. Together with (16.1), these facts prove the entire Haar Gram matrix, including its mixed blocks:

$$\langle f_j, f_k\rangle_0 = \begin{cases} 1, & j = k \in \mathsf{P}_L, \\ b_0, & j = k \in \mathscr{P}_2, \\ 0, & j \neq k. \end{cases} \tag{17.1}$$

For the original weight $w \in \mathbb{R}^{\mathsf{E}_L}$, define

$$\begin{aligned} z_p &= (\mathsf{I}w)_p, \\ c_\alpha &= \tfrac{1}{6}(z_p + z_q) - \tfrac{4}{3} w_{e(p,q)}, \\ \mathsf{T}_\xi w = a, \qquad a_p &= \tfrac{\xi}{4} z_p, \quad a_\alpha = \xi^2 c_\alpha, \\ F_w &= \sum_{j \in \mathscr{J}_L} a_j f_j = \xi h_w + \xi^2 \mathcal{B}_w. \end{aligned} \tag{17.2}$$

The formula for c_α follows by summing the coefficients in (16.2): the six outer weights sum to $z_p + z_q - 2w_e$. Thus no term of $\mathcal{B}_w$ is lost. The exact state identity for all weights is

$$v_w = Q_\psi F_w \psi + s_w. \tag{17.3}$$

Every link has at least two incident faces. Fix the lexicographically first pair $\alpha_e = \{p_e, q_e\}$ of them, using the original integer face labels. On the whole coefficient space define

$$\begin{aligned} \mathsf{L}_\xi &: \mathbb{R}^{\mathscr{J}_L} \longrightarrow \mathbb{R}^{\mathsf{E}_L}, \\ (\mathsf{L}_\xi a)_e &= \frac{a_{p_e} + a_{q_e}}{2\xi} - \frac{3 a_{\alpha_e}}{4\xi^2}. \end{aligned} \tag{17.4}$$

Substitution of (17.2) gives $(z_{p_e} + z_{q_e})/8 - 3c_{\alpha_e}/4 = w_e$. Consequently $\mathsf{L}_\xi \mathsf{T}_\xi = I$ exactly, $\ker \mathsf{T}_\xi = \{0\}$, and $\mathsf{T}_\xi \mathsf{L}_\xi$ is an idempotent with image $\operatorname{Ran} \mathsf{T}_\xi$. A coefficient array a lies in this image exactly when $a = \mathsf{T}_\xi \mathsf{L}_\xi a$. This is an explicit system of linear equations, with inverse (17.4) on that image. The powers of ξ belong to this recorded coefficient map; the original physical state and its norm have not been changed.

Let $(\mathsf{J}z)_\alpha = z_p + z_q$. Each face has at most twelve adjacent faces, since its four links each meet at most three other faces. Therefore

$$\|\mathsf{J}z\|_2^2 \le 2 \sum_{\{p,q\} \in \mathscr{P}_2} (z_p^2 + z_q^2) \le 24 \|z\|_2^2.$$

Using $w_{e(p,q)} = (\mathsf{J}z)_\alpha/8 - 3c_\alpha/4$ at every pair, not only the selected pair, gives

$$\|w\|_2^2 \le \mathcal{K}_L(w,w) \le \tfrac{3}{4}\|z\|_2^2 + \tfrac{9}{8}\|c\|_2^2. \tag{17.5}$$

Indeed $|x+y|^2 \le 2|x|^2 + 2|y|^2$ gives the coefficients $24/32 = 3/4$ and $18/16 = 9/8$. Write $d_0 = 196/13689$ as before. The exact Haar value is

$$\begin{aligned} \mathcal{H}_\xi(w) := \|F_w\|_0^2 &= \tfrac{\xi^2}{16}\|\mathsf{I}w\|_2^2 + b_0 \xi^4 \|c\|_2^2, \\ \mathcal{H}_\xi(w) &\ge \tfrac{d_0}{2}\xi^4 \|w\|_2^2. \end{aligned} \tag{17.6}$$

For the second inequality multiply (17.5) by $(d_0/2)\xi^4$. The pair coefficients agree because $9d_0/16 = b_0$, and the face coefficient is bounded by $\xi^2/16$, since $\xi \le 1$ and $3d_0/8 < 1/16$. No inverse singular value of I was used.

17.2. The full actual-vacuum Gram matrix

Anchor each face $(n;i,j)$, $i<j$, at link (n,i); anchor each pair at its unique shared link. At most two face indices and at most six pair indices have the same anchor. All supports lie in the radius-one link ball of their anchor. Hence the number of indices whose anchors lie within distance D of a fixed anchor is at most $8(4D+1)^3$.

Put

$$\mathsf{G}^{\psi}_{jk}=\langle Q_\psi f_j\psi, Q_\psi f_k\psi\rangle, \qquad \mathsf{G}^{0}_{jk}=\langle f_j,f_k\rangle_0.$$

Each support has at most seven links, and two supports together have at most fourteen. In (3.2)–(3.3) use $(32/3)\sqrt{7}<29$, $(32/3)\sqrt{14}<40$. The means satisfy

$$|\langle f_j\rangle_\psi|\le 116\xi(1+29\xi)<117\xi.$$

The product has absolute value at most four. Keeping the product of the means, the entrywise difference is bounded by

$$|\mathsf{G}^{\psi}_{jk}-\mathsf{G}^{0}_{jk}|\le \big[320+(12800+13689)/49152\big]\xi<400\xi. \tag{17.7}$$

For anchor distance $d\ge 3$, the supports are disjoint and their distance is at least $s=d-2\ge 1$. Apply the full-dynamics estimate (16.13) with $|X|\le 7$ and $\|f_j\|\|f_k\|\le 4$, and the exact Gaussian integral proof (13.7), retaining $T=s/(\Delta_*+2v)$. Here $s_0\le 4$. The two resulting terms are bounded by

$$\frac{28}{100}se^{-99s/100}+12e^{-s/4}\le\left(\frac{28}{74}+12\right)e^{-s/4}.$$

Since $e^{1/2}<2$, this proves

$$|\mathsf{G}^{\psi}_{jk}|\le 32e^{-d/4} \qquad (d\ge 3). \tag{17.8}$$

The constants follow from the same strong on-site evolution and spectral integrals as in Section16.4, using the original Nachtergaele–Sims source proofs cited there. No infinite-volume Hamiltonian or tensorized correlated vacuum is assumed.

For $d\ge 3$, $\mathsf{G}^{0}_{jk}=0$ also follows from disjoint supports and zero Haar means. Define, with G from (14.2),

$$\begin{aligned}\mathcal{P}_{\mathrm{mix}}(z)&=8\left[400(16z+5)^3+32G(16z+9,4/5)\right],\\ \rho_{\mathrm{mix}}(\xi)&=\xi\mathcal{P}_{\mathrm{mix}}(\log(1/\xi)).\end{aligned} \tag{17.9}$$

Taking $D=\lceil 4\log(1/\xi)\rceil\ge 2$, summing (17.7) inside and (17.8) outside, with the anchor multiplicity eight, gives

$$\sup_j\sum_k|\mathsf{G}^{\psi}_{jk}-\mathsf{G}^{0}_{jk}|\le\rho_{\mathrm{mix}}(\xi), \qquad \|\mathsf{G}^{\psi}-\mathsf{G}^{0}\|_{\ell^2\to\ell^2}\le\rho_{\mathrm{mix}}(\xi). \tag{17.10}$$

The summation is exactly $8[400\xi(4D+1)^3+32\sum_{n>D}(4n+1)^3e^{-n/4}]$. Equation (14.2) evaluates its complete tail and $e^{-D/4}\le\xi$ proves (17.10). The symmetric row/column argument of (14.4) proves the operator-norm statement. Thus the mixed blocks have been bounded, not set to zero in the actual density.

For $a=\mathsf{T}_\xi w$, put

$$\mathcal{A}_\xi(w)=\|a\|_2^2=\tfrac{\xi^2}{16}\|z\|_2^2+\xi^4\|c\|_2^2.$$

Equations (17.1) and (17.10) imply

$$\big|\|Q_\psi F_w\psi\|^2-\mathcal{H}_\xi(w)\big|\le\rho_{\mathrm{mix}}(\xi)\mathcal{A}_\xi(w), \qquad b_0\mathcal{A}_\xi(w)\le\mathcal{H}_\xi(w). \tag{17.11}$$

In particular the following complete all-weight covariance error keeps both coefficient arrays:

$$\begin{aligned}\big|\|v_w\|^2-\mathcal{H}_\xi(w)\big|&\le\rho_{\mathrm{mix}}(\xi)\mathcal{A}_\xi(w)\\ &\quad+12\cdot10^6\xi^3\|w\|_2\sqrt{\mathcal{H}_\xi(w)+\rho_{\mathrm{mix}}(\xi)\mathcal{A}_\xi(w)}\\ &\quad+36\cdot10^{12}\xi^6\|w\|_2^2.\end{aligned} \tag{17.12}$$

This is the squared norm expansion of (17.3), the exact bound (17.11), and the original residual norm (16.10). It applies to signed, volume-dependent and coupling-dependent weights, including mixtures approaching $\ker\mathsf{I}$.

17.3. A positive uniform interval for every raw weight

On $0 < \xi \le 10^{-24}$, the positive-coefficient cubic in (17.9) gives $\rho_{\mathrm{mix}}(\xi) \le 10^{-24}\mathcal{P}_{\mathrm{mix}}(72)$, by the monomial monotonicity and $24 < 24\log 10 < 72$ used in Section16.6. Direct rational evaluation of (17.9) gives $\mathcal{P}_{\mathrm{mix}}(72) = 7044756359040 < 10^{13}$, so

$$\rho_{\mathrm{mix}}(\xi) < 10^{-11} < b_0/4, \qquad (24 \cdot 10^6 \xi)^2 < d_0/2. \tag{17.13}$$

Both numerical comparisons are rational and hold on the whole stated interval. Equations (17.11), (17.13) show $\|Q_\psi F_w \psi\| \ge (3/4)\sqrt{\mathcal{H}_\xi(w)}$, since $1 - \rho_{\mathrm{mix}}/b_0 > 3/4 > (3/4)^2$. The second inequality in (17.13) and (16.10) give $\|s_w\| \le (1/4)\sqrt{d_0/2}\xi^2\|w\|_2$. Use (17.6) in the reverse triangle inequality for (17.3). For every real w the result is

$$\boxed{\|v_w\|^2 \ge \frac{d_0}{8}\xi^4\|w\|_2^2, \qquad C(\xi) \ge \frac{d_0}{8}\xi^4 I_{\mathbb{R}^{\mathsf{E}_L}} \quad (0 < \xi \le 10^{-24}).} \tag{17.14}$$

This is on the entire original edge space, uniformly in L. It strengthens, rather than substitutes for, the sharper kernel lower bound (16.23).

The physical state map $V_\xi : w \mapsto v_w$ of (15.18) therefore has zero kernel on this interval and is bijective onto its actual image, with inverse norm at most $\xi^{-2}\sqrt{8/d_0}$. An explicit inverse on that image is

$$w = C(\xi)^{-1}\big(\langle v_e, v_w\rangle\big)_{e\in\mathsf{E}_L}. \tag{17.15}$$

Indeed the vector on the right before applying C^{-1} is precisely Cw. Inequality (17.14) proves existence of this finite-matrix inverse, and the physical inverse-norm bound follows directly by taking square roots in (17.14). No identification of these vectors as individual eigenvectors is made. The original signed minimum (15.21), kernel quotient (16.24), and full mixed covariance (17.12) all refer to this same injective state map on their stated intervals. The original spacing $a > 0$, coupling g_{YM} and every Hamiltonian term remain unchanged. This calculation does not take a spatial continuum limit or continue beyond the displayed strong-coupling interval.

18. Complete mixed energy and the original electric-state Ritz spectrum

Write $A = H - \mathcal{E}$, with exactly the Hamiltonian, domains and actual vacuum of Section 1. The coefficients $z, c, a = \mathsf{T}_\xi w$, the function F_w, and the positive quadratic form $\mathcal{H}_\xi(w)$ are those of Section 17. We first work on $0 < \xi \le 1/49152$, and then give an explicit smaller interval for all quotients and ordered Ritz values. All weights remain the original real link arrays; none is replaced by a unit vector or a different choice of physical coefficients.

18.1. The complete coefficient energy matrix

For $j, k \in \mathscr{J}_L$ define

$$\mathsf{K}^{\psi}_{jk} = \kappa \int \psi^2 \sum_e \nabla_e f_j \cdot \nabla_e f_k \, dU, \qquad \mathsf{K}^{0}_{jk} = \kappa \int \sum_e \nabla_e f_j \cdot \nabla_e f_k \, dU. \tag{18.1}$$

Equation (1.5), polarized over real smooth functions, proves that K^{ψ} is exactly the energy Gram matrix of $Q_\psi f_j \psi$. Centering contributes no derivative. Every function and state here is smooth on the full compact link manifold, so both the form and operator domains are available.

Haar integration by parts gives $\mathsf{K}^0_{jk} = \kappa\langle f_j, \mathfrak{h}_0 f_k\rangle_0$. Each W_p has free energy 3. For a pair, retaining the two distinct eigenvalues in (16.1) gives

$$\langle f_\alpha, \mathfrak{h}_0 f_\alpha\rangle_0 = \frac{9/2}{144} + \frac{(13/2)(3/4)}{676} = \frac{1}{26}.$$

The link-Casimir orthogonality proved in Sections16.1 and17.1 annuls every other Haar entry, including the face/pair entries. Thus the entire coefficient matrix and its original-weight form are

$$\begin{aligned} \mathsf{K}^0_{jk} &= \begin{cases} 3\kappa, & j=k\in \mathsf{P}_L,\\ \kappa/26, & j=k\in \mathscr{P}_2,\\ 0, & j\neq k, \end{cases}\\ \mathcal{J}_\xi(w) &:= \kappa^{-1}a^t\mathsf{K}^0 a = \frac{3\xi^2}{16}\|z\|_2^2+\frac{\xi^4}{26}\|c\|_2^2,\\ 3\mathcal{H}_\xi(w) &\le \mathcal{J}_\xi(w)\le q_*\mathcal{H}_\xi(w), \qquad q_*:=\frac{234}{49}. \end{aligned} \tag{18.2}$$

Indeed $(1/26)/b_0 = 234/49$, with the unchanged $b_0 = 49/6084$. These are Haar coefficient calculations, not a replacement of the actual matrix K^ψ.

For a face, $\sum_e \|\nabla_e f_p\|_\infty \le 4$, by (6.2). For a pair, the seven derivative bounds preceding (16.3) each are less than $1/2$; their sum is less than $7/2 < 4$. Consequently the real multiplier $M_{jk} = \sum_e \nabla_e f_j \cdot \nabla_e f_k$ has $\|M_{jk}\|_\infty \le 16$. Its support has at most fourteen links. Apply the actual local marginal estimate (3.3) with $q_S < 40\xi$. It yields

$$|\mathsf{K}^\psi_{jk} - \mathsf{K}^0_{jk}| \le 1280\kappa\xi(1+40\xi) < 1300\kappa\xi. \tag{18.3}$$

If the anchors of j,k have link-graph distance greater than two, their link supports are disjoint. Each summand in M_{jk} then vanishes pointwise. Both energy entries, not merely their difference, are zero. There are at most $8(4\cdot 2+1)^3 = 8\cdot 9^3$ indices at anchor distance at most two from any one anchor. The exact row coefficient is $1300\cdot 8\cdot 9^3 = 7581600 < 8000000$. Put

$$\chi_E(\xi) = 8000000\xi.$$

The symmetric row/column estimate therefore proves

$$\begin{aligned} \sup_j \sum_k |\mathsf{K}^\psi_{jk} - \mathsf{K}^0_{jk}| &\le \kappa\chi_E(\xi),\\ \|\mathsf{K}^\psi - \mathsf{K}^0\|_{\ell^2\to\ell^2} &\le \kappa\chi_E(\xi),\\ |\mathfrak{q}_A[Q_\psi F_w\psi] - \kappa\mathcal{J}_\xi(w)| &\le \kappa\chi_E(\xi)\mathcal{A}_\xi(w). \end{aligned} \tag{18.4}$$

All actual mixed entries are retained in the left sides of (18.3)–(18.4); their smallness was calculated from the vacuum.

Now use the exact identity (17.3), not an identification with its leading part. Positivity of A and Cauchy–Schwarz for its form give the complete energy error

$$\begin{aligned} |w^t\mathcal{N}w - \kappa\mathcal{J}_\xi(w)| \le{}& \kappa\chi_E(\xi)\mathcal{A}_\xi(w)\\ &+ 2\cdot 10^7\kappa\xi^3\|w\|_2\sqrt{\mathcal{J}_\xi(w)+\chi_E(\xi)\mathcal{A}_\xi(w)}\\ &+ 10^{14}\kappa\xi^6\|w\|_2^2. \end{aligned} \tag{18.5}$$

The cross term is bounded by $2\sqrt{\mathfrak{q}_A[Q_\psi F_w\psi]\mathfrak{q}_A[s_w]}$; (18.4) bounds its first factor and (16.10) its second. This proves (18.5) for every weight, including any dependence of the weight on the box and coupling.

18.2. Uniform relative errors and every electric-state quotient

Keep $d_0 = 196/13689$ and ρ_{mix} from (17.9). Define the following explicit functions of the original ξ:

$$\begin{aligned} r(\xi) &= \rho_{\mathrm{mix}}(\xi)/b_0, \qquad t_C(\xi) = 6\cdot 10^6\sqrt{2/d_0}\,\xi,\\ t_N(\xi) &= 10^7\sqrt{2/d_0}\,\xi,\\ \beta_C(\xi) &= r(\xi) + 2t_C(\xi)\sqrt{1+r(\xi)} + t_C(\xi)^2,\\ \beta_N(\xi) &= \frac{\chi_E(\xi)}{b_0} + 2t_N(\xi)\sqrt{q_* + \frac{\chi_E(\xi)}{b_0}} + t_N(\xi)^2. \end{aligned} \tag{18.6}$$

By (17.6) and (17.11), $\|w\|_2 \le \sqrt{2/d_0}\xi^{-2}\sqrt{\mathcal{H}_\xi(w)}$ and $\mathcal{A}_\xi(w) \le \mathcal{H}_\xi(w)/b_0$. Substitution into every term of (17.12) and (18.5), followed by the two inequalities in (18.2), proves

$$\begin{aligned} |w^t C w - \mathcal{H}_\xi(w)| &\le \beta_C(\xi)\mathcal{H}_\xi(w), \\ |w^t \mathcal{N} w - \kappa\mathcal{J}_\xi(w)| &\le \kappa\beta_N(\xi)\mathcal{H}_\xi(w). \end{aligned} \tag{18.7}$$

For example the covariance cross term, divided by $\mathcal{H}_\xi(w)$, is at most $12\cdot 10^6\sqrt{2/d_0}\xi\sqrt{1+r}$, exactly the middle term of β_C. The energy calculation has the same retained two factors and yields its displayed β_N. All these functions tend to zero as $\xi \downarrow 0$: ρ_{mix} is ξ times a fixed cubic in $\log(1/\xi)$, and all other factors are displayed.

On the explicit nonempty interval $0 < \xi \le 10^{-24}$, set $x_0 = 10^{-24}$ only for recording numerical bounds. Equation (17.13), $2/d_0 = 13689/98 < 144$, and rational comparisons give

$$\begin{aligned} \beta_C(\xi) < B_C &:= \frac{10^{-11}}{b_0} + 4(72\cdot 10^6 x_0) + (72\cdot 10^6 x_0)^2 < 10^{-8}, \\ \beta_N(\xi) < B_N &:= \frac{8000000 x_0}{b_0} + 6(120\cdot 10^6 x_0) + (120\cdot 10^6 x_0)^2 < 2\cdot 10^{-15}. \end{aligned} \tag{18.8}$$

Here $1 + 10^{-11}/b_0 < 4$ and $q_* + 8000000x_0/b_0 < 9$ justify respectively the square-root bounds two and three used above. The polynomial monotonicity proved in Section17.3 makes these bounds valid on the whole interval, not just at its upper endpoint. Define

$$\delta(\xi) = \frac{\beta_N(\xi) + q_*\beta_C(\xi)}{1 - \beta_C(\xi)}.$$

Its denominator is positive and $\delta(\xi) < (B_N + q_* B_C)/(1 - B_C) < 10^{-7}$. For every nonzero original weight, (17.14) permits division by the actual norm. Subtract the two quotients with their denominators retained:

$$\begin{aligned} \frac{w^t\mathcal{N}w}{w^tCw} - \frac{\kappa\mathcal{J}_\xi(w)}{\mathcal{H}_\xi(w)} &= \frac{w^t\mathcal{N}w - \kappa\mathcal{J}_\xi(w)}{w^tCw} \\ &\quad - \frac{\kappa\mathcal{J}_\xi(w)}{\mathcal{H}_\xi(w)}\frac{w^tCw - \mathcal{H}_\xi(w)}{w^tCw}. \end{aligned}$$

Since $w^tCw \ge (1-\beta_C)\mathcal{H}_\xi(w)$, this proves

$$\boxed{\left|\frac{w^t\mathcal{N}w}{w^tCw} - \frac{\kappa\mathcal{J}_\xi(w)}{\mathcal{H}_\xi(w)}\right| \le \kappa\delta(\xi), \quad 0 < \xi \le 10^{-24}, \quad w \ne 0.} \tag{18.9}$$

The bound is uniform in every original box and every such weight. It controls all quotients, not only the minimum or the incidence kernel. In particular they all lie between $\kappa(3 - \delta(\xi))$ and $\kappa(q_* + \delta(\xi))$. The independent stronger physical lower bound $3\kappa(1 - 512\xi/3)$ from (15.13) still applies.

For a precise record of the interpolation, put

$$\eta_\xi(w) = \frac{b_0\xi^4\|c\|_2^2}{\mathcal{H}_\xi(w)}.$$

Then $0 \le \eta_\xi(w) \le 1$, and the coefficient quotient satisfies the exact identity

$$\frac{\mathcal{J}_\xi(w)}{\mathcal{H}_\xi(w)} = 3 + \frac{87}{49}\eta_\xi(w). \tag{18.10}$$

This records both contributions with their original powers of coupling. It is not a change of state or its norm.

18.3. Exact compression, inverse and retained leakage

Let $\mathcal{H}_{\mathrm{phys},\mathbb{R}}$ be the real-valued part of the gauge-invariant Hilbert space. The real operator A preserves it. Set $\mathscr{S}_\xi = \operatorname{Ran} V_\xi \subset \mathcal{H}_{\mathrm{phys},\mathbb{R}} \cap \psi^\perp$, where $V_\xi w = v_w$. On the interval above, (17.14) proves $\dim \mathscr{S}_\xi = N$. Every vector of this space is smooth. The adjoint and the orthogonal projection are

$$\begin{aligned} V_\xi^* f &= (\langle v_e, f\rangle)_e, \qquad V_\xi^* V_\xi = C, \\ P_{\mathscr{S}_\xi} &= V_\xi C^{-1} V_\xi^*. \end{aligned} \tag{18.11}$$

The last operator is self-adjoint because C is positive and symmetric. Its square is itself by $V_\xi^* V_\xi = C$. Its range is contained in $\mathscr{S}_\xi$, and it fixes $V_\xi w$ for every w; thus it is the stated projection. The inverse of V_ξ on its image is $C^{-1}V_\xi^*$, which is the exact formula (17.15).

Define the self-adjoint finite-dimensional compression $R_\xi = P_{\mathscr{S}_\xi} A|_{\mathscr{S}_\xi}$. For $u, v \in \mathscr{S}_\xi$, $\langle u, R_\xi v\rangle = \langle u, Av\rangle = \langle Au, v\rangle$. Its coordinate intertwiner and its full uncompressed action are exactly

$$\begin{aligned} R_\xi V_\xi &= V_\xi C^{-1}\mathcal{N}, \\ AV_\xi &= V_\xi C^{-1}\mathcal{N} + \mathcal{L}_\xi, \qquad \mathcal{L}_\xi := (I - P_{\mathscr{S}_\xi})AV_\xi, \\ V_\xi^* \mathcal{L}_\xi &= 0. \end{aligned} \tag{18.12}$$

These follow by inserting (18.11) and $V_\xi^* A V_\xi = \mathcal{N}$. The domain of $\mathcal{L}_\xi$ is the entire finite edge space; its codomain is $\mathscr{S}_\xi^\perp \cap \mathcal{H}_{\mathrm{phys},\mathbb{R}} \cap \psi^\perp$. Thus $C^{-1}\mathcal{N}$ is self-adjoint in the exact inner product $\langle w, z\rangle_C = w^t C z$. This records its metric rather than replacing it by the Euclidean one. The map V_ξ is an isometry from this metric space onto $\mathscr{S}_\xi$. Its Rayleigh quotient is exactly the first quotient in (18.9). No vanishing of $\mathcal{L}_\xi$, or invariance of the state image under A, is used or asserted.

18.4. Ordered Ritz values and the exact top coefficient eigenspace

Let B_0 be the diagonal coefficient matrix with entries one on faces and b_0 on pairs, and let B_1 have entries three on faces and $1/26$ on pairs. On the original edge space define

$$\mathsf{H}_\xi = \mathsf{T}_\xi^t B_0 \mathsf{T}_\xi, \qquad \mathsf{J}_\xi = \mathsf{T}_\xi^t B_1 \mathsf{T}_\xi, \qquad K_\xi^{\mathrm{coeff}} = \mathsf{H}_\xi^{-1}\mathsf{J}_\xi. \tag{18.13}$$

Their quadratic forms are respectively $\mathcal{H}_\xi$ and $\mathcal{J}_\xi$, and (17.6) proves the inverse exists. The operator K_ξ^{coeff} is self-adjoint in the positive metric H_ξ. Let its ordered eigenvalues be $\mu_1 \le \cdots \le \mu_N$, retaining their dependence on L, ξ, and let $\lambda_1 \le \cdots \le \lambda_N$ be those of R_ξ, with physical energy units retained.

Here is the full variational comparison argument. For any self-adjoint operator B in a finite-dimensional positive inner product g, take a g-orthonormal eigenbasis with ordered eigenvalues ν_i. A j-dimensional subspace intersects the g-orthogonal complement of the first $j-1$ eigenvectors in a nonzero vector: the projection onto their span has rank at most $j-1$. Such a vector has Rayleigh quotient at least ν_j, by its eigenbasis expansion. The span of the first j eigenvectors has maximum quotient exactly ν_j. Hence

$$\nu_j = \min_{\substack{W \text{ linear} \\ \dim W = j}} \max_{0 \neq w \in W} \frac{g(w, Bw)}{g(w, w)}. \tag{18.14}$$

The eigenbasis used to prove this identity does not change the original physical vectors or either retained Gram matrix. Both problems in (18.13) and (18.12) use precisely the same collection of linear subspaces of the raw edge space. Apply (18.9) to every nonzero vector of every one of those subspaces, then take the maximum and minimum in (18.14). This proves

$$\boxed{|\lambda_j/\kappa - \mu_j| \le \delta(\xi) \quad (1 \le j \le N), \qquad 3 \le \mu_j \le \frac{234}{49}.} \tag{18.15}$$

No volume-dependent matrix perturbation constant enters.

The top eigenspace of the coefficient problem is computable exactly, without solving a large characteristic polynomial:

$$q_*\mathsf{H}_\xi - \mathsf{J}_\xi = \frac{87}{49}\frac{\xi^2}{16}\mathsf{I}^t\mathsf{I}, \qquad \ker(K_\xi^{\mathrm{coeff}} - q_* I) = \ker \mathsf{I}. \tag{18.16}$$

The first identity follows by retaining both diagonal blocks in (18.13); the pair block vanishes because $q_* b_0 = 1/26$. Its right side is positive, with zero quadratic form exactly when $\mathsf{I}w = 0$. Invertibility of H_ξ then proves the stated equality of kernels.

The exact original-box incidence calculation in the parent, *Electric covariance and unsigned incidence*, Section3, equations(3.2)–(3.9), gives $\dim\ker \mathsf{I} = 3m(m+1) = N - M$, $m = 2L$. For explicitness, its coordinate proof writes $w_i(n) = (-1)^{n_1+n_2+n_3} a_i(n)$ and turns the kernel equations into $D_j a_i + D_i a_j = 0$. Commuting differences in three distinct directions gives $D_k D_j a_i = 0$. The complete inverse parameters are the longitudinal boundary values $f_i(n_i)$ and anchored potentials $\varphi_{ij}(x,y) = \sum_{r=-L}^{x-1} u_{ij}(r,y)$, where $u_{ij}(x,y) = a_i(x\mathbf{e}_i + y\mathbf{e}_j - L\mathbf{e}_k) - f_i(x)$. They obey $\varphi_{ij}(-L,y) = \varphi_{ij}(x,-L) = 0$, and reconstruct

$$a_i(n) = f_i(n_i) + \sum_{j>i} D_i\varphi_{ij}(n_i,n_j) - \sum_{j<i} D_i\varphi_{ji}(n_j,n_i).$$

The vanishing transverse mixed difference proves the additive decomposition giving these parameters; summing $D_j u_{ij} + D_i u_{ji} = 0$ proves both reconstructed cross derivatives. The boundary values and finite sums fix the inverse uniquely. There are $3m$ longitudinal and $3m^2$ anchored potential entries, on the original edge and vertex domains. Thus (18.16), including multiplicities, gives

$$\begin{aligned} \mu_{M+1} = \cdots = \mu_N = \frac{234}{49}, \qquad \mu_j < \frac{234}{49} \quad (1 \le j \le M), \\ \left|\lambda_j - \frac{234}{49}\kappa\right| \le \kappa\delta(\xi) \quad (M+1 \le j \le N). \end{aligned} \tag{18.17}$$

No uniform separation of μ_M from q_* is asserted. The last $N - M$ Ritz values need not be exactly equal; the comparison preserves the actual mixed energy and covariance.

18.5. Consequence for the true physical excitation eigenvalues

The gauge-invariant restriction of A has compact resolvent because its gauge projection commutes with the full resolvent; the restriction is a compression of a compact operator. Remove its simple vacuum. The remaining physical space is infinite dimensional: characters of every integer and half-integer spin of a fixed face holonomy are gauge invariant, and are linearly independent by integration in one face link. Removing one dimension does not change that fact. Let $\epsilon_1 \le \epsilon_2 \le \cdots$ be the true excitation eigenvalues there, counted with multiplicity.

The eigenbasis proof of (18.14) applies also to this positive compact-resolvent operator: the first j eigenvectors give the upper variational inequality, and expansion of the nonzero intersection with their first $j-1$ orthogonal complement gives the lower one. This uses the form domain for the nonnegative convergent spectral sum. Restricting the admissible j-dimensional spaces to $\mathscr{S}_\xi \subset \mathrm{Dom}\, A$ therefore gives

$$\begin{aligned} 3\kappa(1 - 512\xi/3) \le \epsilon_j \le \lambda_j \le \kappa\big(\mu_j + \delta(\xi)\big) \quad (1 \le j \le N), \\ \#\{j : \epsilon_j \le \kappa(q_* + \delta(\xi))\} \ge N. \end{aligned} \tag{18.18}$$

The lower bound is the independent physical gap proof (15.13). The upper inequalities and count are for the original interacting operator, not the free coefficient operator. They do not transfer the exact coefficient multiplicity in (18.17) into a physical eigenvalue multiplicity. Equation (18.12) gives the exact map and retained leakage underlying that distinction. All energies are measured from the actual $\mathcal{E}$; no level below that spectral minimum has been introduced.

For the original nonnegative native weights in (1.4), the sharper targeted bounds of Section14 remain in force. The present result additionally controls every mixed signed weight and the full ordered electric-state compression. It leaves $a > 0$, $\kappa = 2g_{\mathrm{YM}}^2/a$, $\xi = 1/(4g_{\mathrm{YM}}^4)$, and the geometric input (1.3) unchanged. No spatial continuum limit or Millennium conclusion is inferred from this finite-regulator strong-coupling result.

18.6. Precise literature dependency and an attainment counterexample

The canonical corpus routes the variational method to Mathieu Lewin, *Spectral Theory and Quantum Mechanics*, Universitext, Springer2024, DOI10.1007/978-3-031-66878-4, Section5.6.1, printed pages201–204. Equations(5.42) and(5.44) are the min–max value and restriction inequality used here; (18.14) and(18.18) provide their proofs in the exact present spaces. Two details in that edition must not be imported as extra mathematical claims.

First, the sentence on page202 asserting that the minimizing subspaces are exactly eigenvector-generated low-energy spaces, and the concluding necessity argument on page204, fail for the following explicit example. On $\mathbb{R}^3$ with its Euclidean inner product take

$$B = \operatorname{diag}(0, 1, 2), \qquad W = \operatorname{span}(e_2, e_1 + e_3).$$

For all real s, t,

$$\langle se_2 + t(e_1 + e_3), B[se_2 + t(e_1 + e_3)]\rangle = s^2 + 2t^2 = \|se_2 + t(e_1 + e_3)\|^2. \tag{18.19}$$

Every nonzero vector of W has quotient one, the second eigenvalue of B. The dimension-intersection proof of (18.14) proves this is the minimizing value. Yet W is not contained in $\operatorname{span}(e_1, e_2)$, and $W \cap \operatorname{span}(e_3) = \{0\}$. Failure of containment does not imply the nonzero intersection used in the source's necessity argument. The min–max value itself is unaffected; only that characterization of all attaining spaces is refuted.

Second, the matrix following (5.44) represents the stated ordinary eigenvalues with an orthonormal basis. For a general basis v_i, the exact coordinate problem is $\mathcal{N}w = \lambda Cw$, with both Gram matrices retained, as proved in (18.11)–(18.12). Already on $\mathbb{R}$, the identity operator and basis vector $v_1 = 2$ give $\mathcal{N} = (4)$, $C = (4)$, and eigenvalue one, not four. These independent finite-dimensional checks explain exactly which source assertions are used and which are corrected. They assert nothing about unread editions or other parts of the book. No protected bulk text is reproduced in this note.

19. Original negative period value, collapse, and negative magnetic eigenvalues

This section corrects the previous Section19: its auxiliary two-coordinate form was not the actual Euclidean metric, and its negative-direction witness was false. Neither inference may be used to exclude the original negative period value. The following are calculations in the retained period matrix and the existing full link Hilbert space.

19.1. The original expression and the complete metric map

Retain the expression, including its denominator and its domain,

$$\Delta_{\tau_T,\beta_T,\mu_T} := \operatorname{Im}\tau_T \left(\operatorname{Im}\beta_T - \frac{6(\operatorname{Im}\mu_T)^2}{\operatorname{Im}\tau_T}\right), \qquad \operatorname{Im}\tau_T \neq 0. \tag{19.1}$$

Use the original dictionary $L_T = \operatorname{Im}\tau_T$, $q_T = -\operatorname{Im}\beta_T$, $m_T = \operatorname{Im}\mu_T$, $D = L_T q_T + 6m_T^2$. Substitution into the displayed expression, without replacing it by a different condition, gives the exact relation

$$\Delta_{\tau_T,\beta_T,\mu_T} = -D. \tag{19.2}$$

In particular, the original domain $D > 0$ already has $\Delta_{\tau_T,\beta_T,\mu_T} < 0$. A claim that this sign alone lies outside the existing construction is false.

Write $M_{\mathrm{cov}} \in \mathbb{N}_{>0}$ for the cover degree parameter called M in the parent cusp note, to retain its identity

without confusing it with this note's number of faces. Keep all real and imaginary entries:

$$A_T = \begin{pmatrix} 6\,\mathrm{Re}\,\mu_T & \mathrm{Re}\,\tau_T \\ \mathrm{Re}\,\beta_T & \mathrm{Re}\,\mu_T \end{pmatrix}, \qquad B_T = \begin{pmatrix} 6m_T & L_T \\ -q_T & m_T \end{pmatrix},$$
$$P_{M_{\mathrm{cov}}} = \begin{pmatrix} 6\,\mathrm{Re}\,\mu_T & \mathrm{Re}\,\tau_T & M_{\mathrm{cov}} & 0 \\ 6m_T & L_T & 0 & 0 \\ \mathrm{Re}\,\beta_T & \mathrm{Re}\,\mu_T & 0 & M_{\mathrm{cov}} \\ -q_T & m_T & 0 & 0 \end{pmatrix}. \tag{19.3}$$

The original physical coordinate order is

$$(x_1, x_2, x_3, x_4) = (\mathrm{Re}\,z_1, \mathrm{Im}\,z_1, \mathrm{Re}\,z_2, \mathrm{Im}\,z_2).$$

The recorded permutation $R(x_1, x_2, x_3, x_4) = (x_1, x_3, x_2, x_4)$ has $R^tR = I$, $\det R = -1$, and

$$Q := RP_{M_{\mathrm{cov}}} = \begin{pmatrix} A_T & M_{\mathrm{cov}} I_2 \\ B_T & 0 \end{pmatrix},$$
$$Q^{-1} = \begin{pmatrix} 0 & B_T^{-1} \\ M_{\mathrm{cov}}^{-1} I_2 & -M_{\mathrm{cov}}^{-1} A_T B_T^{-1} \end{pmatrix}, \quad B_T^{-1} = \frac{1}{D}\begin{pmatrix} m_T & -L_T \\ q_T & 6m_T \end{pmatrix}. \tag{19.4}$$

Both products in (19.4) are the identity: their off-diagonal blocks are respectively $-A_TB_T^{-1} + A_TB_T^{-1}$ and $M_{\mathrm{cov}}^{-1}A_T - M_{\mathrm{cov}}^{-1}A_T$, and their diagonal blocks are I_2. This proves the inverse for either sign of $D \neq 0$.

For $G = P_{M_{\mathrm{cov}}}^t P_{M_{\mathrm{cov}}}$ the exact quadratic form is

$$G = \begin{pmatrix} A_T^tA_T + B_T^tB_T & M_{\mathrm{cov}}A_T^t \\ M_{\mathrm{cov}}A_T & M_{\mathrm{cov}}^2 I_2 \end{pmatrix},$$
$$(r,z)^tG(r,z) = |A_Tr + M_{\mathrm{cov}}z|^2 + |B_Tr|^2, \tag{19.5}$$
$$\det P_{M_{\mathrm{cov}}} = -M_{\mathrm{cov}}^2 D, \qquad \det G = M_{\mathrm{cov}}^4 D^2, \qquad \sqrt{\det G} = M_{\mathrm{cov}}^2|D|.$$

Here $r, z \in \mathbb{R}^2$. Zero quadratic value forces $B_Tr = 0$, then $r = 0$ by (19.4), then $z = 0$. Thus G is positive definite for both signs of D. To compute the determinant of Q, exchange its two column blocks. This uses four transpositions and gives the block triangular matrix with diagonal blocks $M_{\mathrm{cov}}I_2, B_T$. Multiplication by R gives the sign in (19.5).

The map $\phi : [y] \mapsto [P_{M_{\mathrm{cov}}}y]$ from $\mathbb{R}^4/\mathbb{Z}^4$ to $\mathbb{R}^4/P_{M_{\mathrm{cov}}}\mathbb{Z}^4$ is a bijection with inverse $[x] \mapsto [P_{M_{\mathrm{cov}}}^{-1}x]$. Its lift has zero kernel; its composition with the target quotient has kernel $\mathbb{Z}^4$. Equation (19.5) proves that ϕ is an isometry from the first torus with metric G to the second with the Euclidean metric. Its oriented Jacobian is $-M_{\mathrm{cov}}^2 D$; its density Jacobian is $M_{\mathrm{cov}}^2|D|$. These are two explicitly related quantities, not an argument for discarding the sign. On the original $D > 0$ domain the latter is exactly the parent's $M_{\mathrm{cov}}^2 D$.

19.2. Exact relation to the auxiliary form and the collapsing direction

The form introduced in the previous version of this section was

$$G_\eta = \begin{pmatrix} 2\,\mathrm{Im}\,\tau_T & 2\,\mathrm{Im}\,\mu_T \\ 2\,\mathrm{Im}\,\mu_T & \mathrm{Im}\,\beta_T/3 \end{pmatrix} = \begin{pmatrix} 2L_T & 2m_T \\ 2m_T & -q_T/3 \end{pmatrix}. \tag{19.6}$$

Its relation to the imaginary period block is an invertible linear coordinate map, not an identification of their metrics:

$$\mathfrak{F} : \left\{ \begin{pmatrix} 6m & L \\ -q & m \end{pmatrix} : L, m, q \in \mathbb{R} \right\} \longrightarrow \mathrm{Sym}_2(\mathbb{R}),$$
$$B \longmapsto \begin{pmatrix} 2B_{12} & B_{11}/3 \\ B_{11}/3 & B_{21}/3 \end{pmatrix}, \tag{19.7}$$
$$\mathfrak{F}^{-1}\begin{pmatrix} u & v \\ v & w \end{pmatrix} = \begin{pmatrix} 3v & u/2 \\ 3w & v/2 \end{pmatrix}, \qquad \det G_\eta = -\frac{2}{3}D = \frac{2}{3}\Delta_{\tau_T,\beta_T,\mu_T}.$$

Entry-by-entry composition proves both inverse identities. The kernel is zero and every fibre is a singleton. This map retains all three imaginary parameters; it does not encode A_T or M_{cov}.

On the original $D > 0$ domain, the following two vectors give the signature without a guessed sign-reversal argument:

$$(1,0)G_\eta(1,0)^t = 2L_T, \qquad (-m_T/L_T, 1)G_\eta(-m_T/L_T, 1)^t = -\frac{D}{3L_T}. \tag{19.8}$$

They have opposite signs for every $L_T \neq 0$. Nevertheless the actual metric in (19.5) is positive definite. An indefinite nondegenerate form also has an absolute volume density $\sqrt{|\det G_\eta|}$; the former claim that indefiniteness prevents any positive volume density was false.

There is an explicit collapse at $D = 0$, rather than a new region added to avoid it. In the retained coefficient coordinates put

$$k_T = (L_T, -6m_T)^t, \qquad v_T = (k_T, -M_{\mathrm{cov}}^{-1}A_T k_T)^t.$$

Direct multiplication, with all parameters unchanged, gives

$$B_T k_T = (0, -D)^t, \qquad Q v_T = (0, 0, 0, -D)^t, \qquad v_T^t G v_T = D^2. \tag{19.9}$$

At $D = 0$, $L_T \neq 0$ makes B_T rank one; (19.4)'s inverse ceases to exist. The kernel of $P_{M_{\mathrm{cov}}}$ is exactly $\mathbb{R}v_T$: solve $B_T r = 0$, then $z = -M_{\mathrm{cov}}^{-1}A_T r$. Thus the extended real matrix has rank three. For fixed M_{cov} and bounded period parameters approaching that locus with $|L_T| \geq \ell > 0$, the least metric eigenvalue satisfies

$$0 \leq \lambda_{\min}(G) \leq \frac{D^2}{|k_T|^2 + M_{\mathrm{cov}}^{-2}|A_T k_T|^2} \leq \frac{D^2}{\ell^2}. \tag{19.10}$$

The variational inequality follows by testing the unit direction $v_T/|v_T|$; it is not a change of the original period parameters. The oriented determinant and density in (19.5) both approach zero. This is a statement about the displayed period family; no assertion that every such coefficient path occurs in a completed S6 threefold has been inserted.

For completeness, the same collapse has a calculated classical action. The full inverse metric is obtained by multiplying $Q^{-1}Q^{-t}$. Its leading two-by-two block is $B_T^{-1}B_T^{-t}$; its determinant is D^{-2}. The induced two-form Gram entry is therefore $\mathcal{K}_{12,12} = D^{-2}$. Pullback of the original connection preserves every commutator: $\widetilde{F}_{ij} = \sum_{\alpha,\beta} P_{\alpha i}P_{\beta j}F_{\alpha\beta} \circ P$, by the chain rule and bilinearity. Consequently, for $\widetilde{F}_{12} = 2\pi H_c$, other components zero, $c(H_c, H_c) = -\operatorname{tr}(H_c^2)/2 = 1$, the unchanged action is

$$S = \frac{M_{\mathrm{cov}}^2|D|}{2g_{\mathrm{YM}}^2}(2\pi)^2 D^{-2} = \frac{2\pi^2 M_{\mathrm{cov}}^2}{g_{\mathrm{YM}}^2|D|}. \tag{19.11}$$

For fixed $M_{\mathrm{cov}}, g_{\mathrm{YM}} > 0$ it diverges as $D \to 0$. The absolute Jacobian here is required by the proved Euclidean pullback, not an unexplained deletion of an unfavorable sign. On $D > 0$ this is exactly the original action. Collapse and this action divergence are not yet a quantum spectral limit.

19.3. The sign map inside the original magnetic operator

Return to the full spatial box and its original $a, g_{\mathrm{YM}} > 0$. The finite, nonwrapping patch has $h_1(n) = e^{-\theta n_2 H_c}$, $h_2(n) = h_3(n) = I$, $H_c = i\sigma_3 = -2T_3$, $\theta = 2\pi a^2/D$. The parent derivation retains

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \frac{1}{D}\begin{pmatrix} m_T & -L_T \\ q_T & 6m_T \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \quad du_1 \wedge du_2 = D^{-1}dy_1 \wedge dy_2, \tag{19.12}$$

and the original connection

$$\begin{aligned} A_{\mathrm{orig}} &= \frac{2\pi H_c}{D^2}(m_T y_1 - L_T y_2)(q_T dy_1 + 6m_T dy_2), \\ f(y) &= \frac{2\pi}{D^2}\left(\frac{m_T q_T}{2}y_1^2 - L_T q_T y_1 y_2 - 3m_T L_T y_2^2\right), \\ A_{\mathrm{orig}} &= (2\pi/D)H_c y_1 dy_2 + H_c df. \end{aligned} \tag{19.13}$$

For the dy_1 component both sides equal $2\pi H_c(m_T q_T y_1 - L_T q_T y_2)/D^2$. For the dy_2 component, the right side's coefficient is $2\pi H_c[(D - L_T q_T)y_1 - 6m_T L_T y_2]/D^2$, equal to the left because $D - L_T q_T = 6m_T^2$. The two gauge maps $e^{-H_c f(y)}$ and $e^{-(2\pi/D)H_c y_1 y_2}$ produce $-(2\pi/D)H_c y_2 dy_1$. Integrating with the parent's inverse-parallel-transport convention gives the stated links. This proves the local dictionary for either sign of $D \neq 0$; it does not remove periodic transition edges or assert a completed threefold construction.

Define $T_h F(U) = F((h_e^{-1}U_e)_e)$ and retain the original gauge projection Π. Conjugation by a vertex gauge map conjugates h_e at its source vertex. Exactly one possibly nonidentity h_e has any given source. The independent vertex integrals therefore give, on invariant inputs,

$$\Pi T_h \Pi = K_\theta \Pi, \qquad K_\theta = \prod_{e=(n,1)} C_{e,\theta n_2}, \qquad C_{e,s}F(U) = \int F(\dots, g e^{sH_c} g^{-1} U_e, \dots)\, dg. \tag{19.14}$$

Each factor is a Haar average of unitary link translations, hence a contraction and a map preserving pointwise nonnegative functions. The matrix $J = i\sigma_1$ obeys $JH_cJ^{-1} = -H_c$. Replacing g by gJ proves $C_{e,-s} = C_{e,s} = C_{e,s}^*$. Thus, exactly at fixed a, L, g_{YM},

$$K_{-\theta} = K_\theta, \qquad K_{\theta+2\pi} = K_\theta, \qquad K_{2\pi k} = I \quad (k \in \mathbb{Z}). \tag{19.15}$$

The periodic identities use the original integer n_2 and $e^{2\pi H_c} = I$, without changing the angle variable. For the actual vacuum define $c_\theta = \langle \psi, K_\theta \psi\rangle$, $\chi_\theta = (K_\theta - c_\theta)\psi$, $d_\theta = \|\chi_\theta\|^2$, and $n_\theta = \langle K_\theta\psi, (H - \mathcal{E})K_\theta\psi\rangle$. All four are identical at θ and $-\theta$. In particular the map $D \mapsto K_{2\pi a^2/D}$ loses the sign of D. At $D = a^2/k$, $k \neq 0$, $\chi_\theta = 0$ exactly. It is then a zero vector, and its Rayleigh quotient is undefined; no zero-energy eigenstate is manufactured by dividing by its norm. These identities neither identify different $|D|$ nor control a spatial continuum limit.

19.4. A negative eigenvalue in the existing gauge-invariant space

Preservation of pointwise nonnegative functions does not imply $\langle F, K_\theta F\rangle \geq 0$. Here is an explicit violation of that proposed inference, within the existing operator.

Use the spin-one representation $\rho_1 = \mathrm{Sym}^2 \mathbb{C}^2$. In its three weight coordinates, $\rho_1(e^{sH_c})$ has diagonal entries $e^{2is}, 1, e^{-2is}$. In the basis $v_0 = e_+ \otimes e_+$, $v_1 = e_+ \otimes e_- + e_- \otimes e_+$, $v_2 = e_- \otimes e_-$, use the complexified generators $E = iT_1 - T_2$, $F = iT_1 + T_2$, acting on both tensor factors. Their full actions are

$$Ev_0 = 0, \quad Ev_1 = 2v_0, \quad Ev_2 = v_1, \qquad Fv_0 = v_1, \quad Fv_1 = 2v_2, \quad Fv_2 = 0.$$

This representation is irreducible: the diagonal circle splits it into three distinct weights, and these generators connect adjacent weights. An invariant subspace for the unitary group has an invariant orthogonal complement and hence a projection commuting with that circle; it is a sum of weight spaces. The connecting generators force either none or all three of them.

Averaging $\rho_1(g e^{sH_c} g^{-1})$ commutes with the representation, so its scalar is fixed by its trace:

$$c_1(s) = \frac{e^{2is} + 1 + e^{-2is}}{3} = \frac{1 + 2\cos(2s)}{3}.$$

Choose $\theta_* = \arccos(1/4) \in (0, \pi)$. Then, exactly,

$$\cos(2\theta_*) = -\frac{7}{8}, \qquad c_1(\theta_*) = -\frac{1}{4}, \qquad e^{\theta_* H_c} = \frac{1}{4}I + \frac{\sqrt{15}}{4}H_c, \qquad D_* = \frac{2\pi a^2}{\theta_*} > 0. \tag{19.16}$$

The factor $1/3$ comes from the trace on the existing three-dimensional representation, not from a substituted model.

Let $\mathcal{P}_0$ consist of the 12-faces based at $n = (r, 0, t)$, $-L \leq r < L$, $-L \leq t \leq L$. Its cardinality is $d_L = (2L)(2L+1)$. For each such face let U_p be its full ordered holonomy from (1.1) and put

$$F_p(U) = \mathrm{tr}\, \rho_1(U_p) = W_p(U)^2 - 1. \tag{19.17}$$

To prove the last equality, diagonalize an arbitrary $SU(2)$ matrix with eigenvalues e^{is}, e^{-is}. Its fundamental trace squared minus one is $e^{2is} + 1 + e^{-2is}$, the symmetric-square trace. Both sides are conjugation invariant, so the equality holds on all $SU(2)$. Gauge transformations conjugate U_p at its base vertex; therefore F_p is gauge invariant and smooth. It is not zero: $F_p(I, \dots, I) = 3$.

The two direction-one edges of p have heights $n_2 = 0, 1$. Their class averages act by $c_1(0) = 1, c_1(\theta)$; inverse traversal has the dual representation, with the same scalar because c_1 is real and even. All other translated edges carry the trivial representation in F_p. Thus

$$K_{\theta_*} F_p = -\tfrac{1}{4} F_p, \qquad \langle F_p, F_q \rangle_{L^2(dU)} = \delta_{pq}. \tag{19.18}$$

Here is the norm proof, retaining the coordinate metric and not assuming independence of overlapping faces. The displayed symmetric-tensor basis has Gram matrix $S = \mathrm{diag}(1, 2, 1)$; it is not rescaled. Write $h_0 = 1, h_1 = 2, h_2 = 1$. Its representation matrices obey $\rho(U)^* S \rho(U) = S$, so $\rho(U)^{-1} = S^{-1} \rho(U)^* S$. The averaging map $X \mapsto \int \rho(U) X \rho(U)^{-1} dU$ commutes with the irreducible representation and preserves the trace. Thus it equals $\mathrm{tr}(X) I/3$. Applying this to each matrix unit and retaining S gives

$$\int \overline{\rho_{ij}(U)} \rho_{kl}(U) dU = \frac{h_j}{3 h_i} \delta_{ik} \delta_{jl}.$$

For the diagonal entries in the trace, $j = i$ and $l = k$, so the retained metric factor is one. Summing those entries gives $\int |\operatorname{tr} \rho(U)|^2 dU = 1$. Integrate one edge of p, holding the others fixed; its holonomy is Haar distributed by left/right invariance and inversion invariance. This proves the norm. For $p \neq q$, choose an edge in $p \setminus q$. The average of its nontrivial representation matrix is zero, since the representation has no invariant vector. The same single-edge integral proves orthogonality, including adjacent faces.

The isometry

$$\mathscr{F} : \mathbb{C}^{\mathcal{P}_0} \longrightarrow L^2(dU)^{\mathrm{gauge}}, \qquad w \longmapsto \sum_{p \in \mathcal{P}_0} w_p F_p \tag{19.19}$$

has zero kernel and inverse coordinates $w_p = \langle F_p, \mathscr{F} w \rangle$ on its image. Its image projector is $P_{\mathscr{F}} f = \sum_p F_p \langle F_p, f \rangle$. On this entire d_L-dimensional image, $K_{\theta_*} = -I/4$. Thus K_{θ_*} is genuinely not a positive-semidefinite operator on the physical gauge-invariant space.

Retain the actual interacting vacuum rather than replacing it by the constant function. Put $\alpha_p = \langle \psi, F_p \rangle$, real because both functions are real. The exact vacuum-orthogonal subspace is $\mathscr{F}(\ker \alpha^t)$, of dimension at least $d_L - 1$. For $\alpha \neq 0$, its coefficient projection is $I - \alpha\alpha^t/(\alpha^t \alpha)$; for $\alpha = 0$ it is I. Composition proves these projections, their images and their kernels. Every state in this subspace retains eigenvalue $-1/4$ for K_{θ_*}. Already at $L = 2$ its dimension is at least nineteen, so six linearly independent such vacuum-orthogonal physical states exist without changing the Hilbert space or invoking an indefinite measure.

19.5. Exact full-Hamiltonian image, without a positivity shortcut

The operator just calculated is related to H by its actual action, not by assuming that its eigenvalues are energies. On each of the four edges of p, $-\Delta_e^T F_p = 2F_p$, and it vanishes on the other edges. To verify the constant directly in the original convention, the preceding E, F actions give $J_3 = [E, F] = \mathrm{diag}(2, 0, -2)$ and

$$-\sum_{a=1}^{3} \rho_{1*}(T_a)^2 = \tfrac{1}{2}(EF + FE) + \tfrac{1}{4} J_3^2 = \tfrac{1}{2} \mathrm{diag}(2, 4, 2) + \tfrac{1}{4} \mathrm{diag}(4, 0, 4) = 2 I_3.$$

Here $\rho_{1*}(T_1) = (E + F)/(2i)$, $\rho_{1*}(T_2) = (F - E)/2$, and $\rho_{1*}(T_3) = -iJ_3/2$, as follows already on the two original tensor factors. Hence the full electric sum is $8F_p$, with $T_a = -i\sigma_a/2$ unchanged.

Each $F_p = W_p^2 - 1$ is even under the central sign change $U_e \mapsto -U_e$ on any single link. For every face r, W_r is odd under that change on any one of its four edges. Consequently Haar invariance gives $\langle F_p, W_r F_q \rangle = 0$ for all $p, q \in \mathcal{P}_0$ and every original face r, even when supports intersect. The full potential and every original face

are retained in

$$H\mathscr{F}w = (8\kappa + 2bM)\mathscr{F}w - b\sum_{r\in \mathsf{P}_L} W_r\,\mathscr{F}w,$$
$$\mathscr{F}^* H\mathscr{F} = (8\kappa + 2bM)I, \qquad P_{\mathscr{F}}\left(\sum_r W_r \mathscr{F}w\right) = 0. \tag{19.20}$$

This is a full state map and its explicit component outside the image, not a dropped interaction or an eigenvector assertion for H. Its domain consists of the finite-dimensional smooth image of $\mathscr{F}$, contained in $\operatorname{Dom} H$.

For every nonzero $w \in \ker \alpha^t$ the corresponding actual vacuum-orthogonal state's two quadratic expressions are therefore

$$\frac{\langle \mathscr{F}w, K_{\theta_*}\mathscr{F}w\rangle}{\|\mathscr{F}w\|^2} = -\frac{1}{4}, \qquad \frac{\langle \mathscr{F}w, (H-\mathcal{E})\mathscr{F}w\rangle}{\|\mathscr{F}w\|^2} = 8\kappa + 2bM - \mathcal{E}. \tag{19.21}$$

The trial function 1 has unit norm, electric energy zero and $\int W_r\, dU = 0$, by the same single-edge nontrivial representation integral. The variational definition gives $\mathcal{E} \le 2bM$, so the latter expression is at least 8κ. This calculation does not discard the negative eigenvalue: it proves its exact relation to the original interacting energy, including the omitted-from-compression component in (19.20).

The original signed electric states are also already included. For $V_\xi w = \sum_e w_e v_e$, linearity gives

$$V_\xi(-w) = -V_\xi w, \qquad (-w)^t C(-w) = w^t C w, \qquad (-w)^t \mathcal{N}(-w) = w^t \mathcal{N} w. \tag{19.22}$$

Here $C, \mathcal{N}$ are matrices, not functions $C(w), \mathcal{N}(w)$. This corrects the old Section19 notation. Mixed signs need not cancel: their exact full-coordinate estimate is (17.14), with its stated finite-regulator interval. It does not decide the $D \to 0$ quantum limit, a continuum gap, or a completed S6 identification. Equations (19.9), (19.11), (19.18) and (19.20) provide the actual collapse, action divergence, negative eigenvalue and interacting state map. No replacement state space has been constructed to protect a conclusion.

19.6. Source identities and scope of the correction

The complete retained period matrix and full action map are in the root manuscript *Retained cusp coordinates and nonlinear gauge-field integration*, Section1, file `retained_cusp_nonlinear_bridge.md`, read directly. The exact magnetic gauge maps, class convolution and vacuum definitions are in *Magnetic continuation, Gauss projection, and the interacting vacuum*, Sections1,2 and4, file `magnetic_translation_true_vacuum.md`, equations(1.1)–(1.4), (2.1)–(2.4) and(4.1)–(4.5). They are local research sources, not published literature or certified by another task's report.

For the finite-graph representation framework the human primary source is John C. Baez, *Spin Network States in Gauge Theory*, Advances in Mathematics117 (1996),253–272, doi:10.1006/aima.1996.0012, arXiv:gr-qc/9411007v1, Section *Gauge Theory on a Graph*, retained TeX lines174–321. His endpoint convention is carried to ours by $A_e = U_e^{-1}$: inversion preserves Haar and sends $A_e \mapsto g_t A_e g_s^{-1}$ to $U_e \mapsto g_s U_e g_t^{-1}$. This supplies the exact convention map. The negative eigenvalue example, its multiplicity bound and the full energy calculation above are explicitly proved here; no novelty claim or attribution to a Jacobi/Fable theorem is made.

20. Finite-angle local states, uniform weighted errors, and the original product

All objects in this section remain in the full box of Section1 with its actual unit vacuum ψ. Write $A = H - \mathcal{E}$. The proof first calculates individual class translations and their weighted sum, then gives the exact map and a quantitative remainder relating that sum to the original product K_θ. The sum is not substituted for the product. We retain every angle, every plaquette and the interaction $\xi = b/\kappa = 1/(4g_{\mathrm{YM}}^4)$.

20.1. Exact all-angle multipliers, with the exceptional angles retained

Let $s_e \in \mathbb{R}$ be a specified angle at each link and define

$$t_e = 1 - \cos s_e, \qquad J_e = I - C_{e,s_e}.$$

The original class average is (19.14), on the same full one-link space. On spin j, put $n = 2j \in \mathbb{N}_0$. Its exact coefficient is

$$c_j(s) = \frac{1}{n+1} \sum_{r=0}^{n} e^{i(2r-n)s}.$$

The terms pair under $r \mapsto n - r$, so this coefficient is real. It lies in $[-1, 1]$ as an average of numbers of modulus one. For every integer k,

$$|e^{iks} - 1| \le |k|\, |e^{is} - 1|, \qquad 1 - \cos(ks) \le k^2 (1 - \cos s).$$

For $k > 0$, factor $e^{iks} - 1 = (e^{is} - 1) \sum_{r=0}^{k-1} e^{irs}$; the triangle inequality proves the first inequality. Negative k has the same modulus, and $k = 0$ is exact. Squaring and dividing by two proves the second. The finite sums of r, r^2, proved by induction on n, give

$$\frac{1}{n+1} \sum_{r=0}^{n} (2r - n)^2 = \frac{n(n+2)}{3} = \frac{4}{3} j(j+1).$$

Consequently, in the joint original Peter–Weyl coordinates,

$$0 \le J_e \le 2I, \qquad 0 \le 1 - c_j(s_e) \le \frac{4}{3} t_e j(j+1), \qquad \|J_e F\| \le \frac{4}{3} t_e \|E_e F\| \quad (F \in \operatorname{Dom} E_e). \tag{20.1}$$

The last statement follows by squaring the scalar multiplier inequality and summing the full orthogonal expansion. No representation is discarded. On the fundamental representation $c_{1/2}(s_e) = \cos s_e$, so $J_e = t_e I$ there. At $s_e \in 2\pi\mathbb{Z}$, $J_e = 0$ on every representation, not merely on the fundamental one. No quotient by $t_e = 0$ is used. The negative coefficients of C_{e,s_e}, including Section19's example, remain present: (20.1) concerns $I - C_{e,s_e}$.

20.2. A local first-order state with an angle-uniform remainder

For the original star $S_e = \bigcup_{p \ni e} \partial p$, retain $r_e \le 4$, $m_{S_e} \le 52$, $q_{S_e} < 39$. Use the operators P_S, Q_S, K_S, B_S, R_S from Section2 and abbreviate $\eta = P_S \psi$, $u = Q_S \psi$, $S = S_e$. They are in the same interacting Hilbert space, with $\|u\| \le q_S \xi$, and

$$J_e \psi = b J_e R_S W_S \eta + b J_e R_S W_S u.$$

This follows from (2.8), $J_e P_S = 0$, and commutation of J_e with R_S. On a face containing e, $W_p \eta$ has four fundamental link labels, local electric energy 3, and $J_e W_p \eta = t_e W_p \eta$. For a face not containing e it is constant at e, and $J_e W_p \eta = 0$. The exterior-resolvent intertwiner proved in Section4 therefore gives the full identity

$$\begin{aligned} J_e \psi - \frac{\xi t_e}{3} W_e^\star \psi = & -\frac{\xi t_e}{3} W_e^\star (3\kappa + K_S)^{-1} K_S \eta \\ & + b J_e R_S W_S u - \frac{\xi t_e}{3} W_e^\star u. \end{aligned} \tag{20.2}$$

The joint multiplier bound (20.1) and (2.7) imply $\|J_e R_S\| \le 4t_e/(3\kappa)$. Since $\|K_S \eta\| \le 2b m_S q_S \xi$, each of the three terms in (20.2) is bounded explicitly. For $0 < \xi \le 1/64$,

$$\begin{aligned} \left\| J_e \psi - \frac{\xi t_e}{3} W_e^\star \psi \right\| &\le t_e \left[\frac{(8m_S + 2r_e) q_S}{3} \xi^2 + \frac{4 r_e m_S q_S}{9} \xi^3 \right] \\ &\le t_e \xi^2 \left(5512 + \frac{10816}{3} \xi \right) \le \frac{16705}{3} t_e \xi^2 < 5600 t_e \xi^2. \end{aligned} \tag{20.3}$$

The t_e factor vanishes exactly at the identity angles. The estimate is uniform in s_e, the link position and the exterior volume. In particular it is not a small-angle Taylor remainder.

For the extensive calculation we also require the full excitation operator applied to this residual. Define bounded self-adjoint real operators and centered physical vectors

$$D_e = J_e - \frac{\xi t_e}{3} W_e^\star, \qquad r_e^s = Q_\psi D_e \psi.$$

The superscript distinguishes the vector from the original integer r_e, the number of faces at the link. All these operators preserve smooth functions. Each J_e is central link convolution, so it commutes with the kinetic energy and the full vertex action; $W_e^\star$ is gauge invariant. Hence r_e^s is a smooth gauge-invariant vector in $\psi^\perp$.

Here are the estimates for the commutator, including the derivative terms. Put $P = P_{\{e\}}$ and $u_e = Q_{\{e\}}\psi$. The fundamental coefficient on $W_e^\star P\psi$ gives

$$(J_e - t_e I) W_e^\star \psi = (J_e - t_e I) W_e^\star u_e.$$

The product rule, (3.1)–(3.2) and (10.5) give

$$\begin{aligned} E_e(W_e^\star u_e) &= \tfrac{3}{4} W_e^\star u_e + W_e^\star E_e \psi - 2\nabla_e W_e^\star \cdot \nabla_e \psi, \\ \|E_e(W_e^\star u_e)\| &\le 4r_e^2\xi + 4r_e^2\xi + 4r_e^2\xi = 12 r_e^2 \xi, \\ \|(J_e - t_e I) W_e^\star \psi\| &\le \left(16 + \frac{16}{3}\right) r_e^2 t_e \xi = \frac{64}{3} r_e^2 t_e \xi. \end{aligned} \tag{20.4}$$

In the last line (20.1) was applied to the first term, while the second contributes $t_e\|W_e^\star u_e\| \le 16 r_e^2 t_e \xi/3$.

Since $H_0 W_e^\star = 3W_e^\star$ for $H_0 = \sum_f E_f$, exact commutation with H yields

$$\begin{aligned} A r_e^s = [H, D_e]\psi = b(J_e - t_e I) W_e^\star \psi - b W_e^\star J_e \psi \\ + \frac{2\kappa\xi t_e}{3} \sum_f \nabla_f W_e^\star \cdot \nabla_f \psi. \end{aligned} \tag{20.5}$$

Only the faces at e appear in $[V, J_e]$; no other potential term is deleted from H. The gradient sum has norm at most

$$\sum_{p \ni e} \sum_{f \in \partial p} 2 r_f \xi \le 32 r_e \xi.$$

Also $\|J_e\psi\| \le (8/3) r_e t_e \xi$ by (20.1). Substitution into (20.5) proves

$$\|A r_e^s\| \le \frac{80 r_e^2 + 64 r_e}{3} \kappa t_e \xi^2 \le 512 \kappa t_e \xi^2, \qquad \|r_e^s\| \le 5600 t_e \xi^2. \tag{20.6}$$

Every coefficient of (20.2) and (20.5) has been retained and bounded.

20.3. Exact finite-range residual energy for arbitrary weights

The support of D_e is the radius-one link star S_e. The commutator $[H, D_f]$ has the same radius-one support at f: $[V, J_f]$ is its face-star commutator, and $[H, W_f^\star]$ consists of the function's derivatives on that star. Thus $[D_e, [H, D_f]] = 0$ on smooth functions when $d(e, f) > 2$.

For real s_e, s_f, the wavefunctions and operators are real. Expanding the double commutator and applying $H\psi = \mathcal{E}\psi$ gives the sum of $\langle D_e\psi, A D_f \psi\rangle$ and its real transpose. Self-adjointness and reality make these equal, with no assumption that D_e, D_f commute. Centering changes neither energy term. It follows that

$$\mathcal{R}_{ef} := \langle r_e^s, A r_f^s\rangle = \tfrac{1}{2}\langle \psi, [D_e, [H, D_f]]\psi\rangle, \qquad \mathcal{R}_{ef} = 0 \ (d(e,f) > 2). \tag{20.7}$$

Equation (20.6) gives $|\mathcal{R}_{ef}| \le 5600 \cdot 512\,\kappa\xi^4 t_e t_f$. The radius-two link ball has at most $9^3 = 729$ links by the original coordinate count in Sections14 and16. Using $2|u_e u_f| \le |u_e|^2 + |u_f|^2$ in this symmetric matrix,

$$\begin{aligned} r_w^s &= \sum_e w_e r_e^s, \qquad u_e = w_e t_e, \\ 0 \le \mathfrak{q}_A[r_w^s] &\le 3 \cdot 10^9 \kappa \xi^4 \|u\|_2^2, \\ \|r_w^s\| &\le 32000\xi^2 \|u\|_2 \qquad (0 < \xi \le 1/49152). \end{aligned} \tag{20.8}$$

The row coefficient is $729 \cdot 5600 \cdot 512 = 2090188800 < 3 \cdot 10^9$. For the last line, the independently proved physical gap (15.13) is at least $287\kappa/96$ and $(96/287)3 \cdot 10^9 < 32000^2$. The argument works for real or complex w; the Gram matrix is real symmetric, so its Hermitian quadratic form has the same bound. Neither the number of nonzero weights nor an exterior volume factor appears in these constants.

Let I be the original unsigned face incidence matrix. The full exact weighted state map is

$$\begin{aligned} \mathcal{U}_s(w) &= Q_\psi \sum_e w_e J_e \psi, \qquad u = (t_e w_e)_e, \quad z = \mathsf{I}u, \\ \Phi_z &= \sum_{p \in \mathsf{P}_L} z_p W_p, \\ \mathcal{U}_s(w) &= \frac{\xi}{3} Q_\psi \Phi_z \psi + r_w^s. \end{aligned} \tag{20.9}$$

Indeed $\sum_e w_e t_e W_e^\star = \sum_p (\sum_{e \in \partial p} u_e) W_p$ by the original face incidence, including all boundary faces. The diagonal coefficient map $w \mapsto u$ has kernel $\{w : t_e w_e = 0 \text{ for every } e\}$, and image the coefficient arrays vanishing at the zero-angle links. On that image its inverse on nonzero-angle coordinates is $w_e = u_e/t_e$; the zero-angle coordinates are a freely specified kernel. This describes the exact information loss before incidence or the actual state map is applied. The residual in (20.9) is not set to zero on $\ker \mathsf{I}$.

20.4. A nonzero all-angle family with an explicit physical quotient

Write $\rho_{\mathrm{mix}}(\xi)$ for the full expression (17.9). The face block of (17.10), and the full mixed-energy estimate (18.4), give, with

$$\begin{aligned} h_s(w) &= \frac{\xi^2}{9}\|z\|_2^2, \qquad v_s(w) = \frac{\xi}{3} Q_\psi \Phi_z \psi, \\ \big|\|v_s(w)\|^2 - h_s(w)\big| &\le \rho_{\mathrm{mix}}(\xi) h_s(w), \\ \big|\mathfrak{q}_A[v_s(w)] - 3\kappa h_s(w)\big| &\le 8 \cdot 10^6 \kappa \xi h_s(w). \end{aligned} \tag{20.10}$$

All actual covariance entries, including off-diagonal ones, are in the left sides; the right sides are the proved operator-norm errors. These inequalities hold for complex z as well, because the actual matrices are real symmetric.

For real $u \ge 0$, $u \ne 0$, each face square is at least the sum of its coefficient squares. Every original link has $r_e \ge 2$. Hence

$$\|\mathsf{I}u\|_2^2 = \sum_p \left(\sum_{e \in \partial p} u_e \right)^2 \ge \sum_e r_e u_e^2 \ge 2\|u\|_2^2.$$

Combining this with (20.8)–(20.10), define the explicit scalar bounds

$$\begin{aligned} b_C(\xi) &= \rho_{\mathrm{mix}}(\xi) + 2(68000\xi)\sqrt{1 + \rho_{\mathrm{mix}}(\xi)} + (68000\xi)^2, \\ b_N(\xi) &= 8 \cdot 10^6 \xi + 2(120000\xi)\sqrt{3 + 8 \cdot 10^6 \xi} + (120000\xi)^2. \end{aligned} \tag{20.11}$$

For the norm residual relative to $\sqrt{h_s(w)}$, $3 \cdot 32000/\sqrt{2} < 68000$. For its energy residual relative to $\sqrt{\kappa h_s(w)}$, $3\sqrt{3 \cdot 10^9}/\sqrt{2} < 120000$. Cauchy–Schwarz in the Hilbert norm and in the nonnegative closed form therefore proves

$$\begin{aligned} \big|\|\mathcal{U}_s(w)\|^2 - h_s(w)\big| &\le b_C(\xi) h_s(w), \\ \big|\mathfrak{q}_A[\mathcal{U}_s(w)] - 3\kappa h_s(w)\big| &\le \kappa b_N(\xi) h_s(w). \end{aligned} \tag{20.12}$$

These estimates are now proved for every real angle array and every original nonnegative w for which $u \neq 0$.

Here is a completely numerical, volume-independent interval on which the denominator is positive. The positive cubic polynomial $\mathcal{P}_{\mathrm{mix}}$ in (17.9) has $\mathcal{P}_{\mathrm{mix}}(48) < 10^{13}$, by exact rational substitution. For $0 < \xi \le 10^{-16}$, the function $\xi\mathcal{P}_{\mathrm{mix}}(\log(1/\xi))$ increases with ξ: each monomial has derivative $z^r - rz^{r-1} \ge 0$, $z = \log(1/\xi) \ge 3$, $0 \le r \le 3$. Also $\log(10^{16}) < 48$, since $e^3 > 10$. Consequently $\rho_{\mathrm{mix}}(\xi) < 10^{-3}$. The two square roots in (20.11) are each less than two, so exact rational bounds give

$$b_C(\xi) < \frac{1}{500}, \qquad b_N(\xi) < 10^{-6}, \qquad \frac{b_N(\xi) + 3b_C(\xi)}{1 - b_C(\xi)} < \frac{1}{100} \quad (0 < \xi \le 10^{-16}). \tag{20.13}$$

In particular $\mathcal{U}_s(w) \neq 0$. Subtracting the retained $3\kappa\|\mathcal{U}_s(w)\|^2$ from the energy and applying (20.12) yields the full quantitative statement

$$\left| \frac{\mathfrak{q}_A[\mathcal{U}_s(w)]}{\|\mathcal{U}_s(w)\|^2} - 3\kappa \right| \le \kappa \, \frac{b_N(\xi) + 3b_C(\xi)}{1 - b_C(\xi)}. \tag{20.14}$$

This is an all-angle weighted sum of local class-translation states in the interacting theory. The original product is related to it next. For signed u the exact map and bound (20.8)–(20.10) still hold, but the inequality $\|\mathsf{I}u\|^2 \ge 2\|u\|^2$ is not asserted.

20.5. Every product term and a two-link vacuum moment

For distinct links e, f, the original two-link comparison operator in Section2 gives a useful bound without a full-volume norm. Take $S = \{e, f\}$. In its joint spectral variables $\lambda_e, \lambda_f, k \ge 0$,

$$\frac{\lambda_e \lambda_f}{\kappa(\lambda_e + \lambda_f) + k} \le \frac{\lambda_e + \lambda_f}{4\kappa}.$$

Multiplying $Q_S\psi = bR_SW_S\psi$ by E_eE_f, which annihilates $P_S\psi$, therefore gives

$$\|E_eE_f\psi\| \le \frac{\xi}{4}\|H_SW_S\psi\| \le 3\xi + 96\xi^2 \le 4\xi \quad (0 < \xi \le 1/49152). \tag{20.15}$$

The constants in the middle inequality follow from the full product rule. There are at most eight faces in W_S; $\|W_S\| \le 16$. Its local electric derivative has bound $\|H_SW_S\|_\infty \le (3/2)(r_e + r_f) \le 12$. The product-rule term $W_SH_S\psi$ has norm at most $16 \cdot 16\xi = 256\xi$. The mixed derivatives have total norm at most $4(r_e^2 + r_f^2)\xi \le 128\xi$, by (10.4). Thus $\|H_SW_S\psi\| \le 12 + 384\xi$, proving (20.15) with both interaction and derivative terms included.

Take any finite list of distinct translated links $e_1, \dots, e_m$ in the same box, with their specified angles, and set $C_i = C_{e_i, s_{e_i}}$, $J_i = I - C_i$. These operators commute across different links. Telescoping their product twice gives the exact identity

$$\prod_{i=1}^{m} C_i - I + \sum_{i=1}^{m} J_i = \sum_{i=1}^{m} \sum_{j<i} J_iJ_j \prod_{k<j} C_k. \tag{20.16}$$

An empty product is the identity. Indeed $\prod_i C_i - I = -\sum_i J_i \prod_{k<i} C_k$, and $I - \prod_{k<i} C_k = \sum_{j<i} J_j \prod_{k<j} C_k$. Each C_k is a contraction. Moving its factors to act after J_iJ_j and applying (20.1) twice and (20.15) proves

$$\left\| \left(\prod_i C_i - I + \sum_i J_i \right) \psi \right\| \le \frac{64}{9}\xi \sum_{j<i} t_{e_i} t_{e_j}. \tag{20.17}$$

This proof retains every term of (20.16); no commutator or product is replaced by a sum as an identity.

For the original K_θ, choose all direction-one links and $s_e = \theta n_2$; put $t_e = 0$ on the other directions. With $z = \mathsf{I}t$, define the native pair sum $\mathcal{T}_2 = \sum_{e<f} t_e t_f$, over distinct direction-one links. Its actual centered state satisfies

$$\begin{aligned} \chi_\theta = Q_\psi K_\theta \psi &= -\mathcal{U}_s(\mathbf{1}) + Q_\psi \sum_i \sum_{j<i} J_iJ_j \prod_{k<j} C_k \psi, \\ \left\| \chi_\theta + \frac{\xi}{3} Q_\psi \Phi_z \psi \right\| &\le 32000\xi^2 \|t\|_2 + \frac{64}{9}\xi\mathcal{T}_2. \end{aligned} \tag{20.18}$$

Thus the original product, the additive local family and the Wilson coefficient family have exact maps and explicitly bounded differences. For $t \neq 0$, divide only the error bound by the positive coefficient size $\xi\|z\|/3$ to obtain

$$\varepsilon_{\mathrm{prod}} = 68000\xi + \frac{64}{3}\frac{\mathcal{T}_2}{\|z\|_2}, \qquad \left|\|\chi_\theta\| - \frac{\xi}{3}\|z\|_2\right| \le \frac{\xi}{3}\|z\|_2\,(\rho_{\mathrm{mix}}(\xi) + \varepsilon_{\mathrm{prod}})\,. \tag{20.19}$$

Here $|\sqrt{1+\epsilon} - 1| \le |\epsilon|$ for $-1 < \epsilon$, applied to (20.10), is valid since $\rho_{\mathrm{mix}} < 10^{-3}$ on the displayed interval $0 < \xi \le 10^{-16}$. The original state is never divided by a presumed nonzero norm.

20.6. Exact native angle sums and the limitation on the prescribed path

All boundary counts can be evaluated before taking any limit. Put $t_r = 1 - \cos(\theta r)$, $-L \le r \le L$. There are $2L(2L+1)$ direction-one links at each height r, giving

$$\begin{aligned}
\sum_e t_e &= 2L(2L+1)\sum_{r=-L}^{L} t_r,\\
\sum_e t_e^2 &= 2L(2L+1)\sum_{r=-L}^{L} t_r^2,\\
2\mathcal{T}_2 &= \left(\sum_e t_e\right)^2 - \sum_e t_e^2,\\
\|\mathsf{I}t\|^2 &= 2L(2L+1)\sum_{r=-L}^{L-1}(t_r + t_{r+1})^2 + 4L^2\sum_{r=-L}^{L}(2t_r)^2.
\end{aligned} \tag{20.20}$$

The last line counts respectively every 12-face and every 13-face; the 23-face coefficient is zero because its boundary has no direction-one edge. The pair identity is the exact expansion of the square with all diagonal and off-diagonal terms retained.

These estimates do not close the original geometric sequence in the companion spatial-continuum calculation. This can be established quantitatively, instead of labeling a divergent bound a result about the true state. That sequence keeps

$$L_j = j^2, \quad a_j = \frac{1}{100j}, \quad D_j = L_{j^2}q_{j^2} + 6m_{j^2}^2, \quad \theta_j = \frac{2\pi}{10^4 j^2 D_j}, \qquad D_j = j^4 + O(j^2). \tag{20.21}$$

The last expansion follows from the companion's retained $B_T = TJ + C_T$, bounded C_T, and $T = j^2$; it is not a replacement of D_j by its leading term. In particular $j^{12}\theta_j^2 \to 4\pi^2/10^8$ and $|\theta_j|L_j \to 0$.

For every real x, Taylor's integral remainder gives

$$\left|1 - \cos x - \frac{x^2}{2}\right| \le \frac{x^4}{24}.$$

Uniformly over the original heights, the relative error in $t_r = \theta_j^2 r^2/2$ is at most $\theta_j^2 L_j^2/12$; at $r = 0$ both sides are exactly zero. Let $S_{1,L}, S_{2,L}$ be the unchanged native sums in Section8 and $w_e^{\mathrm{nat}} = n_2^2$ on direction-one links. Thus

$$\begin{aligned}
\mathcal{T}_2 &= \frac{\theta_j^4}{8}(S_{1,L_j}^2 - S_{2,L_j})\big(1 + O(\theta_j^2 L_j^2)\big),\\
\|\mathsf{I}t\| &= \frac{\theta_j^2}{2}\|\mathsf{I}w^{\mathrm{nat}}\|\big(1 + O(\theta_j^2 L_j^2)\big).
\end{aligned}$$

These relative bounds follow termwise because all coefficients in the pair sum and in the incidence squares are nonnegative. The explicit polynomials already proved in Section8 yield $S_{1,L}/L^5 \to 8/3$, $S_{2,L}/L^7 \to 8/5$, and $\|\mathsf{I}w^{\mathrm{nat}}\|/L^{7/2} \to 8/\sqrt{5}$. Consequently

$$\frac{1}{j}\frac{\mathcal{T}_2}{\|\mathsf{I}t\|} \longrightarrow \frac{8\sqrt{5}\,\pi^2}{9\cdot 10^8}, \qquad \frac{\varepsilon_{\mathrm{prod}}}{j} \longrightarrow \frac{512\sqrt{5}\,\pi^2}{27\cdot 10^8} \tag{20.22}$$

at any fixed allowed $\xi > 0$. The coefficient in this *upper bound* grows on the prescribed path. This proves that (20.19) supplies no vanishing relative error there; it does not prove that the actual difference grows or that the original state has any particular continuum spectral measure. The local weighted result (20.14) is retained at its proved scope, but this triangle estimate alone does not control the centered product error (20.16). Section22 subsequently controls that same complete norm remainder by an exact exponential/cutoff identity. Section21 calculates the negative spectral component of that same product at all positive couplings, without deleting either sign sector or claiming that this alone controls the whole remainder.

20.7. Primary-source dependencies

Section20.1 uses the original full representation and class-averaging calculation in *Magnetic continuation, Gauss projection, and the interacting vacuum*, Section4, equations(4.1)–(4.5). The complete local-resolvent, score and physical-gap proofs used here are Sections2–4, 10 and15 of this repository, with their cited human sources.

The covariance input (17.10) is derived from Bruno Nachtergaele and Robert Sims, *On the dynamics of lattice systems with unbounded on-site terms in the Hamiltonian*, arXiv:1410.8174v1, Section3, and their *Lieb–Robinson Bounds and the Exponential Clustering Theorem*, arXiv:math-ph/0506030v3, Section3.2. The retained original TeX loci378–615 and440–723 were reread for this continuation. The exact dictionary uses link sites, $H_e = \kappa E_e$, $\Phi(\partial p) = -bW_p$, and the untouched scalar $2bM$. The crossing-set and Gaussian spectral-integral proofs are given in Sections13,16 and17; the filter is not a replacement Gaussian state. Their application here is to the full face blocks, with the already calculated support and row constants retained.

The prescribed sequence (20.21) is read directly from *The retained cusp, spatial refinement, and the spectral measure of the actual magnetic vacuum translation*, Section2 and subsection2.1, equations(3)–(12), file `spatial_continuum.md` in the companion lane. Only its displayed period dictionary and sequence are used; no later claimed continuum or Navier–Stokes conclusion is imported. The local result and the product-bound failure are proved here in full.

21. Negative magnetic-sector weights of the original interacting state

This section computes spectral weights of the original product, not only its possible eigenvalues. No negative eigenvalue is deleted, and no positivity condition is imposed on the magnetic class operator. The following estimates hold for every finite original box and every $\xi > 0$, with their full coupling polynomials retained. They do not use the small-coupling spectral gap from Sections12 and15.

21.1. Joint coordinates and exact spectral projections

Use the original Haar Hilbert space of Section1, the endpoint convention of Section15.1, and the class multipliers of Section20.1. A link label array is $\boldsymbol{\jmath} = (j_e)_{e\in\mathsf{E}_L}$, with $j_e \in \frac{1}{2}\mathbb{N}_0$. Write $\mathscr{H}_{\boldsymbol{\jmath}}$ for its full matrix-coefficient block, with the inherited Haar inner product, and $P_{\boldsymbol{\jmath}}$ for its orthogonal projection. Neither a representation nor a matrix-coefficient multiplicity is removed. The finite-graph construction in John C. Baez, *Spin Network States in Gauge Theory*, arXiv:gr-qc/9411007v1, Section2, gives this direct sum and its vertex-invariant subspaces. The inverse-link convention map and all boundary gauge actions are already written and proved in Section15.1.

For the actual angle array $s = (s_e)_e$, set

$$K_s = \prod_{e\in\mathsf{E}_L} C_{e,s_e}, \qquad k_s(\boldsymbol{\jmath}) = \prod_{e\in\mathsf{E}_L} c_{j_e}(s_e), \qquad \psi_{\boldsymbol{\jmath}} = P_{\boldsymbol{\jmath}}\psi.$$

Identity-angle factors are retained. The coefficients c_j are exactly the finite character sums in Section20.1, so

$$K_s|_{\mathscr{H}_{\boldsymbol{\jmath}}} = k_s(\boldsymbol{\jmath})I, \qquad \sum_{\boldsymbol{\jmath}} \|\psi_{\boldsymbol{\jmath}}\|^2 = 1, \qquad \bar{k}_s := \langle\psi, K_s\psi\rangle = \sum_{\boldsymbol{\jmath}} k_s(\boldsymbol{\jmath})\|\psi_{\boldsymbol{\jmath}}\|^2. \tag{21.1}$$

The series for $\bar{k}_s$ is absolutely convergent, since $|k_s(\boldsymbol{j})| \le 1$.

Define

$$\Pi_s^- F = \sum_{k_s(\boldsymbol{j})<0} P_{\boldsymbol{j}} F, \qquad \mathscr{H}_s^- = \widehat{\bigoplus_{k_s(\boldsymbol{j})<0}} \mathscr{H}_{\boldsymbol{j}}, \qquad \mathscr{H}_s^{\ge 0} = \widehat{\bigoplus_{k_s(\boldsymbol{j})\ge 0}} \mathscr{H}_{\boldsymbol{j}}. \tag{21.2}$$

The orthogonal series converge for every $F \in \mathcal{H}_L$. Consequently Π_s^- is a bounded self-adjoint idempotent, its image is exactly $\mathscr{H}_s^-$, and its kernel is exactly $\mathscr{H}_s^{\ge 0}$. The map $F \mapsto (\Pi_s^- F, (I - \Pi_s^-)F)$ is an isometric bijection $\mathcal{H}_L \to \mathscr{H}_s^- \oplus \mathscr{H}_s^{\ge 0}$; its inverse is $(F_-, F_{\ge 0}) \mapsto F_- + F_{\ge 0}$. These claims follow term by term from orthogonality, including the norm identity for the two components. They hold also on the gauge-invariant subspace: each link Casimir block is invariant under both endpoint transformations, so all displayed projections commute with every vertex action.

For any label array, $k_s(\boldsymbol{j}) < 0$ means that none of its factors is zero and an odd number of them is negative. In particular it implies at least one negative factor. Let $\Pi^-_{e,s_e} = \mathbf{1}_{(-\infty,0)}(C_{e,s_e})$. Comparison of the diagonal multipliers proves the bounded-operator inequality

$$0 \le \Pi_s^- \le \sum_e \Pi^-_{e,s_e}. \tag{21.3}$$

The right side counts all negative factors, not only their parity. It is an upper bound and is not substituted for the projector. Arrays with a zero factor stay in $\mathscr{H}_s^{\ge 0}$.

21.2. Full electric moments and the negative-sector bound

We first retain an all-positive-coupling version of the moment used in (13.2). For $S = \{e\}$, Section2 constructs $K_{\{e\}} \ge 0$ on the unchanged exterior space. In joint electric and exterior spectral variables, the reduced resolvent has multiplier $(\kappa\lambda + k)^{-1}$, $\lambda \ge 3/4,\ k \ge 0$. It follows from $\lambda^2/(\kappa\lambda + k) \le \lambda/\kappa$ and $Q_{\{e\}}\psi = bR_{\{e\}}W_e^\star\psi$ that

$$\|E_e^2\psi\| \le \xi \|E_e(W_e^\star\psi)\|.$$

The complete product rule is

$$E_e(W_e^\star\psi) = \tfrac{3}{4}W_e^\star\psi + W_e^\star E_e\psi - 2\sum_a (X_{e,a}W_e^\star)(X_{e,a}\psi).$$

The all-positive-coupling estimates (3.1) and (10.5) are $\|E_e\psi\| \le 2r_e\xi$ and $\|\nabla_e\psi\|^2 = \gamma_e \le 4r_e^2\xi^2$. Also $\|W_e^\star\|_\infty \le 2r_e$ and $\|\nabla_e W_e^\star\|_\infty \le r_e$, with the original radius-two group metric. Hence the three retained terms have norms at most $3r_e/2,\ 4r_e^2\xi,\ 4r_e^2\xi$, respectively. Thus, without a small-ξ assumption,

$$\|E_e^2\psi\| \le M_{2,e}(\xi) := \frac{3r_e}{2}\xi + 8r_e^2\xi^2 \le (6 + 128\xi)\xi. \tag{21.4}$$

All vectors used in this resolvent calculation are smooth; alternatively, finite joint spectral cutoffs followed by monotone summation prove the same domain statement and norm inequality.

Put $t_e = 1 - \cos s_e$. On a negative one-link class eigenspace, (20.1) gives

$$1 < 1 - c_{j_e}(s_e) \le \frac{4}{3}t_e\lambda_e, \qquad \lambda_e = j_e(j_e+1).$$

Therefore, on every link block, including nonnegative blocks,

$$\mathbf{1}_{\{c_{j_e}(s_e)<0\}} \le \left(\frac{4}{3}t_e\lambda_e\right)^4.$$

Apply this scalar inequality to the squared norms of the vacuum's orthogonal components and use (21.4). This proves

$$\begin{aligned} \|\Pi^-_{e,s_e}\psi\|^2 &\le \frac{256}{81}t_e^4 M_{2,e}(\xi)^2, \\ \|\Pi_s^-\psi\|^2 &\le \frac{256}{81}\sum_e t_e^4 M_{2,e}(\xi)^2 \le \frac{256}{81}(6+128\xi)^2\xi^2\sum_e t_e^4. \end{aligned} \tag{21.5}$$

The second line uses (21.3) before taking the vacuum expectation. At $t_e = 0$, the entire class operator is the identity and its negative projector is zero; no division by t_e was made. Large angles and negative coefficients remain allowed. The estimate may be large there, but its proof never replaces C_{e,s_e} by a positive operator.

The original centered product state is $\chi_s = Q_\psi K_s \psi = (K_s - \bar{k}_s)\psi$, since $\|\psi\| = 1$. Its exact negative-sector weight is

$$\begin{aligned} \|\Pi_s^- \chi_s\|^2 &= \sum_{k_s(\boldsymbol{j})<0} |k_s(\boldsymbol{j}) - \bar{k}_s|^2 \|\psi_{\boldsymbol{j}}\|^2, \\ \|\Pi_s^- K_s \psi\|^2 &= \sum_{k_s(\boldsymbol{j})<0} |k_s(\boldsymbol{j})|^2 \|\psi_{\boldsymbol{j}}\|^2. \end{aligned} \tag{21.6}$$

This keeps the actual centering term: Π_s^- need not commute with Q_ψ. Since $|\bar{k}_s| \le 1$, (21.5)–(21.6) give

$$\begin{aligned} \|\Pi_s^- K_s \psi\| &\le \frac{16}{9}(6 + 128\xi)\xi \left(\sum_e t_e^4 \right)^{1/2}, \\ \|\Pi_s^- \chi_s\| &\le \frac{32}{9}(6 + 128\xi)\xi \left(\sum_e t_e^4 \right)^{1/2}. \end{aligned} \tag{21.7}$$

These are actual-vacuum estimates for the complete product at every positive coupling. They do not require any denominator estimate for χ_s.

21.3. An exact bound on the retained vacuum coefficient

For scalars $c_i \in [-1, 1]$, the complete telescope gives

$$1 - \prod_i c_i = \sum_i (1 - c_i) \prod_{k<i} c_k, \qquad 0 \le 1 - \prod_i c_i \le \sum_i (1 - c_i).$$

The lower bound uses $\prod_i c_i \le 1$; the upper bound uses absolute values of every prefix, which are at most one. Consequently, by the same joint spectral sums,

$$0 \le I - K_s \le \sum_e J_e, \qquad 0 \le 1 - \bar{k}_s \le \frac{4}{3} \sum_e t_e \gamma_e \le \frac{16}{3} \xi^2 \sum_e r_e^2 t_e \le \frac{256}{3} \xi^2 \sum_e t_e. \tag{21.8}$$

No sign of a prefix has been assumed. The middle inequality is the expectation of (20.1); all first and second electric moments are those of ψ, not the free vacuum. In particular (21.8) quantifies the vacuum component removed by Q_ψ in (21.6).

21.4. Exact native counts and the unchanged cusp sequence

Specialize only now to the original direction-one angles $s_{(n,1)} = \theta n_2$, with identity factors on the other links. The exact additional count is

$$S_{4,L}(\theta) := \sum_e t_e^4 = 2L(2L+1) \sum_{r=-L}^{L} \big(1 - \cos(\theta r)\big)^4. \tag{21.9}$$

Together with (20.20), this is a finite formula for all bounds above. For the prescribed L_j, a_j, D_j, θ_j of (20.21), set $z_j = \mathsf{I} t^{(j)}$. For all sufficiently large j, $0 < |\theta_j| L_j < \pi$. Thus $t^{(j)} \neq 0$ and $\|z_j\| > 0$, because all coefficients are nonnegative and every link is in a face.

Here is the full power count in the original coordinates. The exact polynomial

$$Q_8(L) = \frac{2}{9}L^9 + L^8 + \frac{4}{3}L^7 - \frac{14}{15}L^5 + \frac{4}{9}L^3 - \frac{1}{15}L$$

satisfies $Q_8(0) = 0$ and $Q_8(L) - Q_8(L-1) = 2L^8$: expanding each $(L-1)^m = \sum_{r=0}^m \binom{m}{r} L^r (-1)^{m-r}$ gives the stated difference with all other coefficients zero. Induction therefore proves $Q_8(L) = \sum_{r=-L}^{L} r^8$.

For $r \neq 0$ the Taylor bound in Section20.6 writes

$$t_r = \frac{\theta_j^2 r^2}{2}(1+\epsilon_{j,r}), \qquad |\epsilon_{j,r}| \le \delta_j := \frac{\theta_j^2 L_j^2}{12} \longrightarrow 0.$$

The $r = 0$ term is exactly zero. Once $\delta_j < 1$, termwise comparison gives the explicit two-sided bounds

$$\begin{aligned} (1-\delta_j)^4 A_j \le S_{4,L_j}(\theta_j) &\le (1+\delta_j)^4 A_j, \\ A_j &= \frac{\theta_j^8}{16}\, 2L_j(2L_j+1)Q_8(L_j), \\ (1-\delta_j)\frac{\theta_j^2}{2}\|\mathsf{I} w^{\mathrm{nat}}\| \le \|z_j\| &\le (1+\delta_j)\frac{\theta_j^2}{2}\|\mathsf{I} w^{\mathrm{nat}}\|. \end{aligned} \tag{21.10}$$

The last pair follows first coordinatewise through the nonnegative incidence matrix and then by the squared Euclidean norm. In particular, using $Q_8(L)/L^9 \to 2/9$ and the exact Section8 native norm,

$$\begin{aligned} \frac{S_{4,L_j}(\theta_j)}{\theta_j^8 L_j^{11}} &\longrightarrow \frac{1}{18}, \\ \frac{\|z_j\|}{\theta_j^2 L_j^{7/2}} &\longrightarrow \frac{4}{\sqrt{5}}, \\ j^8 \frac{\sqrt{S_{4,L_j}(\theta_j)}}{\|z_j\|} &\longrightarrow \frac{\sqrt{10}\,\pi^2}{6\cdot 10^8}. \end{aligned} \tag{21.11}$$

For the last limit, the quotient of the first two square-root scale constants is $\sqrt{10}/24$, and the unchanged period relation $D_j = j^4 + O(j^2)$ gives $j^8\theta_j^2 L_j^2 \to 4\pi^2/10^8$. No coefficient of D_j, no lattice link, and no cover parameter has been substituted out of the finite formulas.

Define the positive *coefficient size* $h_j^{1/2} = \xi\|z_j\|/3$, as in Section20. It is not a replacement for $\|\chi_{\theta_j}\|$. Combining (21.7) and (21.11), for every fixed $\xi > 0$, proves

$$\limsup_{j\to\infty} j^8 \frac{\|\Pi^-_{\theta_j}\chi_{\theta_j}\|}{\xi\|z_j\|/3} \le \frac{16(6+128\xi)\sqrt{10}\,\pi^2}{9\cdot 10^8}. \tag{21.12}$$

Thus this component is $O(\xi j^{-13})$ at each fixed positive coupling, whereas the displayed coefficient size is of order ξj^{-5}. Equation (21.12) is not a statement about its fraction of the actual state norm; that norm has not been divided out. At $0 < \xi \le 1/49152$ the coefficient $6 + 128\xi$ is at most seven, but that restriction is unnecessary for this result.

The original first sum also satisfies $j^2 \sum_e t_e^{(j)} \to 16\pi^2/(3\cdot 10^8)$, by (20.20) and the same termwise bounds. Hence (21.8) gives, still at every fixed positive coupling,

$$0 \le 1 - \bar{k}_{\theta_j}, \qquad \limsup_{j\to\infty} j^2(1-\bar{k}_{\theta_j}) \le \frac{4096\pi^2}{9\cdot 10^8}\xi^2. \tag{21.13}$$

Both the vacuum coefficient and the negative-sector weight were computed for the original product, not its additive substitute.

21.5. The full Hamiltonian map and the remaining product calculation

The spectral projection in (21.2) commutes with $\kappa\sum_e E_e$, since both are diagonal on every full link-label block. In particular it preserves the domain of that operator and, because the original potential is bounded at each finite box, the domain of H. For every $F \in \operatorname{Dom} H$, the exact map is

$$(H-\mathcal{E})\Pi_s^- F = \Pi_s^-(H-\mathcal{E})F - b\sum_{p\in \mathsf{P}_L}[W_p, \Pi_s^-]F. \tag{21.14}$$

This follows by expanding $H = \kappa \sum E_e + 2bM - b \sum W_p$; the scalar $2bM$ and the full kinetic term commute with the projection, and every face commutator remains. Each commutator is bounded, with norm at most four, so the finite sum is defined on the whole Hilbert space. Projecting (21.14) onto the complementary spectral subspace displays the actual off-diagonal Hamiltonian block. No assertion that this block vanishes is made. The same calculation holds on the gauge-invariant subspace.

Equations (21.5)–(21.14) retain the negative magnetic eigenspaces and compute their actual weights and Hamiltonian coupling. They neither identify magnetic eigenvalues with Hamiltonian energies nor remove the negative eigenvectors of Section19. In particular they do not prove that χ_{θ_j} has a nonzero limiting normalization or a particular excitation spectral measure. This negative-sector estimate alone does not control the complete centered remainder (20.16); its upper bound in (20.22) is not improved merely by deleting the negative sector. Section22 subsequently supplies the complete norm remainder and actual denominator on its stated coupling interval, with the original cusp and both sign sectors retained.

22. The complete product remainder on the unchanged cusp

We now bound the complete centered remainder (20.16), not just one magnetic sign component. The Hamiltonian, its actual vacuum and all angles remain unchanged. An auxiliary exponential and an electric spectral cutoff enter only inside exact difference identities; all differences are bounded before the original product is recovered. The general comparison holds for every positive coupling. The nonzero denominator and spectral-probability comparison use the already proved interval $0 < \xi \le 10^{-16}$.

22.1. Exact two-band multiplication and a third electric moment

Here is a coordinate proof of the fundamental tensor-product rule needed for the new domain estimate. Let $\mathcal{P}_n = \operatorname{Sym}^n \mathbb{C}^2$, realized as formal homogeneous polynomials in z_1, z_2, with the induced $SU(2)$ action. For $n \ge 1$, write an element of $\mathcal{P}_n \otimes \mathbb{C}^2$ as $v = p_1 \otimes e_1 + p_2 \otimes e_2$, and define

$$\begin{aligned} m_n(v) &= z_1 p_1 + z_2 p_2 \in \mathcal{P}_{n+1}, \\ s_n(r) &= \frac{1}{n+1}\big((\partial_1 r) \otimes e_1 + (\partial_2 r) \otimes e_2\big), \\ \iota_n(q) &= -z_2 q \otimes e_1 + z_1 q \otimes e_2, \qquad q \in \mathcal{P}_{n-1}. \end{aligned} \tag{22.1}$$

Multiplication is equivariant, and s_n is the polarization map: the chain rule proves equivariance under each linear change of the two basis vectors. The tensor $z_1 \otimes e_2 - z_2 \otimes e_1$ is multiplied by the determinant, which is one for $SU(2)$; therefore ι_n is equivariant too. Euler's identity for homogeneous polynomials gives $m_n s_n = I$, and direct substitution gives $m_n \iota_n = 0$.

If $z_1 p_1 + z_2 p_2 = 0$, evaluation at $z_1 = 0$ shows that z_1 divides p_2; evaluation at $z_2 = 0$ shows that z_2 divides p_1. Thus there is a unique polynomial $q \in \mathcal{P}_{n-1}$ with $p_1 = -z_2 q,\ p_2 = z_1 q$. This proves $\ker m_n = \operatorname{im} \iota_n$, and ι_n is injective. In full coordinates the bijection and inverse are

$$\begin{aligned} v &\longmapsto (r, q), \qquad r = m_n(v), \\ q &= \frac{p_2 - (n+1)^{-1}\partial_2 r}{z_1} = -\frac{p_1 - (n+1)^{-1}\partial_1 r}{z_2}, \\ (r, q) &\longmapsto s_n(r) + \iota_n(q). \end{aligned} \tag{22.2}$$

The quotients in (22.2) are polynomial divisions proved exact above, not pointwise divisions requiring $z_1 z_2 \neq 0$. For $n = 0$, $m_0 : \mathcal{P}_0 \otimes \mathbb{C}^2 \to \mathcal{P}_1$ is already a bijection with inverse s_0, and there is no $\mathcal{P}_{-1}$ summand.

For completeness, $\mathcal{P}_n$ is irreducible: the diagonal torus has the distinct weights of the monomials $z_1^{n-r} z_2^r$. A nonzero invariant subspace contains a weight vector by finite torus projection. The complexified raising and lowering operators $z_1\partial_2, z_2\partial_1$ connect that monomial to every other one with nonzero displayed integer coefficients. Consequently the subspace is all of $\mathcal{P}_n$. These are the original spin-$n/2$ spaces with electric eigenvalue $\lambda_n = n(n+2)/4$, in the unchanged $T_a = -i\sigma_a/2$ convention. Thus (22.1)–(22.2) prove the two

tensor-product bands $n+1, n-1$. No isometry of the polynomial splitting is assumed; all Hilbert estimates below use the original Haar orthogonal Peter–Weyl projections.

Let $P_n^{(e)}$ denote the one-link electric projection on label $n = 2j_e$, tensored with the unchanged exterior identity. Each W_p at e is linear in the entries of U_e or U_e^{-1}. For a determinant-one matrix $U_e = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, its inverse is $\left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$, so these are all fundamental matrix coefficients. The product of two matrix coefficients is a coefficient of the tensor product. Equation (22.2) therefore proves

$$P_m^{(e)} W_e^\star P_n^{(e)} = 0 \quad \text{unless } m = n+1 \text{ or } m = n-1. \tag{22.3}$$

This retains the sum over every face at e.

For each fixed sign define $T_\pm = \sum_{n \geq 0} P_{n\pm1}^{(e)} W_e^\star P_n^{(e)}$, omitting the negative label. For a finite sum of input blocks, the corresponding output blocks of a fixed $T_\pm$ are orthogonal and distinct. Hence $\|T_\pm F\|^2 \leq \|W_e^\star\|_\infty^2 \sum_n \|P_n^{(e)} F\|^2$. This proves bounded extension with norm at most $2r_e$, and (22.3) gives $W_e^\star = T_+ + T_-$. The retained eigenvalues obey

$$\lambda_{n+1} + 1 \leq 2(\lambda_n + 1), \qquad \lambda_{n-1} + 1 \leq 2(\lambda_n + 1) \quad (n \geq 1).$$

For the plus sign the difference on the right is $(n^2+1)/4 > 0$; the minus sign follows from $(n^2+3)/4 \leq (n^2+2n+4)/2$. Applying these scalar bounds to each of the orthogonal bands gives

$$\|E_e^2(W_e^\star F)\| \leq 8\|W_e^\star\|_\infty \|(E_e + I)^2 F\| \leq 16 r_e \|(E_e + I)^2 F\|, \quad F \in \operatorname{Dom} E_e^2. \tag{22.4}$$

Indeed the norm of each band is at most $4\|W_e^\star\|_\infty \|(E_e+I)^2 F\|$. Finite electric truncations and closedness of E_e^2 justify the full domain statement in (22.4).

The same exact one-link comparison as in (21.4) has multiplier $\lambda^3/(\kappa\lambda + k) \leq \lambda^2/\kappa$. Using ψ's smoothness and the full product estimate (22.4), with no exterior gap assumption, proves for every $\xi > 0$

$$\begin{aligned} \|E_e^3 \psi\| &\leq \xi \|E_e^2 (W_e^\star \psi)\| \\ &\leq 16 r_e \xi \big(1 + 4 r_e \xi + M_{2,e}(\xi)\big) \\ &\leq M_3(\xi) := 64\xi(1 + 22\xi + 128\xi^2). \end{aligned} \tag{22.5}$$

Here $\|(E_e+I)^2\psi\| \leq \|E_e^2\psi\| + 2\|E_e\psi\| + \|\psi\|$; every term is kept. This is a volume-independent moment of the interacting vacuum.

22.2. Complete product-to-exponential identity, with negative factors

Keep $J_e = I - C_{e,s_e}$, $t_e = 1 - \cos s_e$, and define

$$S = \sum_e J_e, \qquad T_1 = \sum_e t_e, \qquad T_2 = \sum_e t_e^2, \qquad K_s = \prod_e C_{e,s_e}.$$

The symbol T_2 here is the diagonal sum, not the pair sum $\mathcal{T}_2$ in Section20. All these sums are finite at the original regulator. On the exact link-label coordinates, J_e has a scalar $x_e \in [0,2]$. For $x \geq 0$, two integrations of the derivative of e^{-x} give

$$1 - x - e^{-x} = -\int_0^x (x - v) e^{-v}\, dv, \qquad |1 - x - e^{-x}| \leq x^2/2.$$

For any enumeration of the links the complete product identity is

$$K_s - e^{-S} = \sum_i (C_i - e^{-J_i}) \prod_{k<i} C_k \prod_{k>i} e^{-J_k}. \tag{22.6}$$

It follows by replacing one factor at a time; the commutation between distinct links also proves $e^{-S} = \prod_i e^{-J_i}$. Every factor other than the displayed difference is a contraction, including negative C_k eigenvalues. Those

factors commute with the difference and can be moved after it when taking its norm. Since $\|J_e^2\psi\| \le (16/9)t_e^2\|E_e^2\psi\|$, (21.4) gives

$$\|(K_s - e^{-S})\psi\| \le A_s := \frac{8}{9}(6 + 128\xi)\xi T_2. \tag{22.7}$$

No product factor is omitted. The auxiliary exponential has no interpretation here as a replacement Hamiltonian or vacuum.

22.3. Cutoff remainder and an exact centered state identity

For a positive real number Λ, let

$$B = \sum_e J_e \mathbf{1}_{[0,\Lambda]}(E_e), \qquad D = S - B, \qquad R_\Lambda = \frac{4}{3}\Lambda T_1.$$

These are functions of the same commuting link spectra, and $0 \le B \le R_\Lambda I,\ D \ge 0$. The vacuum is not truncated. On the tail $\lambda > \Lambda$, $\lambda \le \lambda^3/\Lambda^2$; thus (20.1),(22.5) give

$$\|D\psi\| \le d_\Lambda := \frac{4}{3}\, M_3(\xi)\frac{T_1}{\Lambda^2}, \qquad \|(e^{-S} - e^{-B})\psi\| \le d_\Lambda. \tag{22.8}$$

For the second inequality, on the common spectrum $0 \le e^{-B} - e^{-S} \le S - B$, since both B, S are nonnegative. The norm inequality follows by squaring this scalar comparison and summing the original vacuum spectral weights.

Put $\beta = \langle\psi, B\psi\rangle$. Then

$$0 \le \beta \le \min\{R_\Lambda, m_s\}, \qquad m_s := \frac{256}{3}\xi^2 T_1, \tag{22.9}$$

by $\langle\psi, J_e\psi\rangle \le (4/3)t_e\gamma_e$ and the all-coupling score bound. For $0 \le x \le R_\Lambda$, Taylor's integral formula about β gives

$$F_\beta(x) = e^{-x} - e^{-\beta} + e^{-\beta}(x - \beta), \qquad |F_\beta(x)| \le \frac{1}{2}|x - \beta|^2 \le \frac{R_\Lambda}{2}|x - \beta|.$$

The same formula holds for $x < \beta$: the second derivative e^{-v} is at most one throughout the interval between two nonnegative endpoints. Applying it to B proves

$$\|Q_\psi F_\beta(B)\psi\| \le \frac{R_\Lambda}{2}\|Q_\psi B\psi\|. \tag{22.10}$$

Write $\mathcal{U} = Q_\psi S\psi = \mathcal{U}_s(\mathbf{1})$ and $\chi = Q_\psi K_s\psi$, exactly as in Section20. All terms of the following identity are in the original Hilbert space:

$$\begin{aligned}\chi + \mathcal{U} = {} & Q_\psi(K_s - e^{-S})\psi + Q_\psi(e^{-S} - e^{-B})\psi \\ & + Q_\psi F_\beta(B)\psi + (1 - e^{-\beta})\mathcal{U} + e^{-\beta}Q_\psi D\psi.\end{aligned} \tag{22.11}$$

To verify it, substitute $Q_\psi e^{-B}\psi = -e^{-\beta}Q_\psi B\psi + Q_\psi F_\beta(B)\psi$ and $Q_\psi B\psi = \mathcal{U} - Q_\psi D\psi$; the scalar term is killed only by the actual vacuum projection. Using $1 - e^{-\beta} \le \beta$, (22.7)–(22.10), and $\|Q_\psi B\psi\| \le \|\mathcal{U}\| + d_\Lambda$, gives

$$\boxed{\|\chi + \mathcal{U}\| \le A_s + (2 + R_\Lambda/2)d_\Lambda + (R_\Lambda/2 + m_s)\|\mathcal{U}\|.} \tag{22.12}$$

This holds at every positive coupling, every real angle array and every positive cutoff. The right side controls the *complete* centered product remainder, equal to Q_ψ applied to the right side of (20.16). Both sign sectors and all pair/prefix terms remain in the exact identity; no independence of vacuum variables has been assumed.

22.4. A vanishing bound on the prescribed, unaltered sequence

Now fix $0 < \xi \le 10^{-16}$. Let $z = \mathsf{I}t$, $c_s = \xi\|z\|/3$, for an angle array with $t \neq 0$. Since $t \ge 0$, $\|z\| > 0$. The already proved (20.12) gives

$$c_s\sqrt{1 - b_C(\xi)} \le \|\mathcal{U}\| \le c_s\sqrt{1 + b_C(\xi)}.$$

Define, with every finite-regulator sum still present,

$$\begin{aligned} p(\xi) &= 1 + 22\xi + 128\xi^2, \qquad q(\xi) = 6 + 128\xi, \\ a_s &= \frac{8}{3}q(\xi)\frac{T_2}{\|z\|}, \qquad e_\Lambda = 256p(\xi)\frac{T_1}{\Lambda^2\|z\|}, \\ \varepsilon_s(\Lambda) &= a_s + (2 + R_\Lambda/2)e_\Lambda + (R_\Lambda/2 + m_s)\sqrt{1 + b_C(\xi)}. \end{aligned} \tag{22.13}$$

These quantities are respectively $A_s/c_s, d_\Lambda/c_s$ and the full bound obtained from (22.12). Hence

$$\|\chi + \mathcal{U}\| \le c_s\varepsilon_s(\Lambda). \tag{22.14}$$

For identity angles everywhere, the un-divided identity gives $\chi = \mathcal{U} = 0$; no definition of c_s^{-1} is needed.

Use the exact original sequence (20.21), including $D_j = L_{j^2}q_{j^2} + 6m_{j^2}^2$, and choose only the auxiliary spectral cutoff $\Lambda_j = j^{5/3}$. This does not change the original L_j, a_j, θ_j, D_j, state, or Hamiltonian. Write $c_j = \xi\|z_j\|/3$ and $\varepsilon_j = \varepsilon_{s^{(j)}}(\Lambda_j)$. The exact finite sums of Section20 give

$$\begin{aligned} j^2 T_{1,j} &\longrightarrow \frac{16\pi^2}{3\cdot 10^8}, \\ j^{-3}\frac{T_{1,j}}{\|z_j\|} &\longrightarrow \frac{\sqrt{5}}{3}, \\ j^5\frac{T_{2,j}}{\|z_j\|} &\longrightarrow \frac{2\pi^2}{\sqrt{5}\cdot 10^8}, \\ j^5 c_j &\longrightarrow \frac{16\pi^2\xi}{3\sqrt{5}\cdot 10^8}. \end{aligned} \tag{22.15}$$

Here the first two follow from $T_1 = (\theta_j^2/2)S_{1,L_j}(1 + o(1))$ and $\|z_j\| = (\theta_j^2/2)\|\mathsf{I}w^{\mathrm{nat}}\|(1 + o(1))$. For the third retain $T_2 = (\theta_j^4/4)S_{2,L_j}(1 + o(1))$, $S_{2,L}/L^7 \to 8/5$, and $\|\mathsf{I}w^{\mathrm{nat}}\|/L^{7/2} \to 8/\sqrt{5}$. All relative errors here are bounded by the termwise Taylor errors in (21.10); the original finite formulas are not replaced by limits.

Substitution of these four limits into every term of (22.13) now proves

$$j^{1/3}\varepsilon_j \longrightarrow \frac{32\pi^2}{9\cdot 10^8}\sqrt{1 + b_C(\xi)} + \frac{512\sqrt{5}}{3}p(\xi). \tag{22.16}$$

The term a_s is $O(j^{-5})$, m_s is $O(j^{-2})$, and the product $R_\Lambda e_\Lambda$ is $O(j^{-2/3})$; none contributes to this limit. Thus $\varepsilon_j \to 0$ on the original sequence at each fixed allowed positive coupling. This improves the bound failure (20.22) by a proved estimate on the same complete remainder, not by a different cusp.

Here is also an explicit eventual nonzero-state threshold, so the existence of a denominator is not an additional hypothesis. Let C_{per} be the finite bound $\sup_{T \ge T_0}\|C_T\|$ for the retained period correction $B_T = TJ + C_T$ in the companion's original Section2. Take a dyadic j with

$$j \ge 2, \qquad j^2 \ge T_0, \qquad j^2 \ge 2C_{\mathrm{per}}.$$

Then $D_j \ge j^4/4$, and $\theta_j \le (8\pi/10^4)j^{-6}$. In particular $0 < \theta_j L_j < 1$, so $t^{(j)} \neq 0$. The original direction-one edge count is $N_1 = 2L(2L + 1)^2 \le 18L^3$. Cauchy–Schwarz and the nonnegative incidence inequality prove

$$\frac{T_1}{\|z\|} \le \sqrt{N_1/2} \le 3j^3.$$

Also $S_{1,L} \le 12L^5$, since $2L(2L+1) \le 6L^2$ and $\sum_{r=-L}^{L} r^2 \le 2L^3$. Using $t_r \le \theta_j^2 r^2/2$ and the retained $\pi < 4$ bound,

$$T_{1,j} \le \frac{6144}{10^8} j^{-2}, \qquad R_{\Lambda_j}/2 \le \frac{4096}{10^8} j^{-1/3} < 1,$$
$$m_{s^{(j)}} \le \frac{524288}{10^8} j^{-2}, \qquad e_{\Lambda_j} \le 1536 j^{-1/3}.$$

We used only $p(\xi) < 2,\ q(\xi) < 7,\ \xi^2 \le 1$ here. For the other term, $T_2/\|z\| \le \sqrt{T_2/2}$, $\sqrt{T_2} \le (\theta_j^2 L_j^2/2)\sqrt{N_1}$, so $a_s \le 28\theta_j^2 L_j^{7/2} \le (28672/10^8) j^{-5}$. Finally $\sqrt{1 + b_C(\xi)} < 2$. Adding the three retained terms of (22.13) therefore proves the convenient explicit bound

$$\varepsilon_j \le 5000 j^{-1/3}. \tag{22.17}$$

For example it holds for every dyadic j satisfying the preceding period thresholds, and makes $\varepsilon_j \le 1/4$ whenever $j \ge 20000^3$.

Since $b_C < 1/500$, the reverse triangle inequality in (22.14) then proves, with the actual unmodified states,

$$\frac{1}{2} c_j \le \|\chi_{\theta_j}\| \le \frac{3}{2} c_j, \qquad \|\chi_{\theta_j} + \mathcal{U}_j\| \le c_j \varepsilon_j \quad \left(j \ge \max\{2, \sqrt{\max\{T_0, 0\}}, \sqrt{2C_{\mathrm{per}}}, 20000^3\} \right). \tag{22.18}$$

All j in this statement are dyadic. This is a proved threshold, not an assumed nonvanishing denominator. The raw norm is of order ξj^{-5}; it has not been set to one.

22.5. Full bounded spectral measurements, with both masses retained

At each original regulator let $A_j = H_j - \mathcal{E}_j$ be the actual self-adjoint excitation operator and $\mathsf{P}_j(B)$ its spectral projection for a Borel set $B \subset \mathbb{R}$. For a state X define its raw finite measure and its total mass

$$\eta_{X,j}(B) = \langle X, \mathsf{P}_j(B) X \rangle, \qquad C_{X,j} = \eta_{X,j}(\mathbb{R}) = \|X\|^2.$$

No eigenvalue, energy unit or state is rescaled. For every bounded operator O of norm at most one, write $r_j = \chi_{\theta_j} + \mathcal{U}_j$. Expanding $\chi_{\theta_j} = -\mathcal{U}_j + r_j$ and keeping all three difference terms gives

$$\begin{aligned} \langle \chi_{\theta_j}, O\chi_{\theta_j} \rangle - \langle \mathcal{U}_j, O\mathcal{U}_j \rangle &= -\langle \mathcal{U}_j, O r_j \rangle - \langle r_j, O\mathcal{U}_j \rangle + \langle r_j, O r_j \rangle, \\ \left| \langle \chi_{\theta_j}, O\chi_{\theta_j} \rangle - \langle \mathcal{U}_j, O\mathcal{U}_j \rangle \right| &\le c_j^2 \Theta_j, \qquad \Theta_j = \varepsilon_j (2\sqrt{1 + b_C(\xi)} + \varepsilon_j). \end{aligned} \tag{22.19}$$

This proves the same estimate for $O = f(A_j)$, uniformly over all complex Borel functions with $|f| \le 1$, and for every original spectral projection. In particular it holds at $O = I$, so both total masses are controlled with exactly the same error.

The original spectral-probability quotient retains those masses explicitly as $\mu_{X,j}(B) = \eta_{X,j}(B)/C_{X,j}$. No raw state or physical parameter is changed by recording that quotient. For all the dyadic j in (22.18), both denominators are now proved positive. Subtract the two fractions, using (22.19) once for $f(A_j)$ and once for I, to obtain

$$\begin{aligned} \left| \frac{\int f \, d\eta_{\chi,j}}{C_{\chi,j}} - \frac{\int f \, d\eta_{\mathcal{U},j}}{C_{\mathcal{U},j}} \right| &\le \frac{2c_j^2 \Theta_j}{C_{\chi,j}} \le 8\Theta_j, \qquad |f| \le 1, \\ \|\mu_{\chi,j} - \mu_{\mathcal{U},j}\|_{\mathrm{TV}} &\le 8\Theta_j \longrightarrow 0. \end{aligned} \tag{22.20}$$

Here the total-variation convention is the supremum over all measurable $|f| \le 1$. To verify the first line explicitly, the first fraction difference contributes at most $c_j^2 \Theta_j / C_{\chi,j}$. The remaining denominator difference is at most $C_{\mathcal{U},j} |C_{\mathcal{U},j} - C_{\chi,j}| / (C_{\chi,j} C_{\mathcal{U},j})$, which has that same bound. Finally $C_{\chi,j} \ge c_j^2/4$ by (22.18). This is comparison of the entire spectral probabilities of two specified states of the *same original Hamiltonian*, not convergence to a newly selected spectrum or a construction of a continuum theory.

Section21's retained negative magnetic projection can also now be compared to the actual norm, on this proved coupling interval:

$$\frac{\|\Pi_{\theta_j}^{-} \chi_{\theta_j}\|^2}{C_{\chi,j}} \le 4 \left[\frac{32}{3} (6 + 128\xi) \frac{\sqrt{S_{4,L_j}(\theta_j)}}{\|z_j\|} \right]^2 = O_\xi(j^{-16}). \tag{22.21}$$

This follows from (21.7) and the lower denominator in (22.18). It neither deletes that subspace nor reads its magnetic sign as an energy sign.

22.6. The original weighted electric state and its entire spectral measure

The finite-angle sum itself has an exact controlled relation to the original native electric state. In the spin-$n/2$ block, retain $k_r = 2r - n$ and $\lambda_n = n(n+2)/4$. Direct expansion gives

$$\frac{1}{n+1}\sum_{r=0}^{n} k_r^4 = \frac{n(n+2)(3n^2+6n-4)}{15} = \frac{16}{15}\lambda_n(3\lambda_n - 1).$$

For instance expand the fourth power and use $\sum r = n(n+1)/2$, $\sum r^2 = n(n+1)(2n+1)/6$, $\sum r^3 = n^2(n+1)^2/4$, and $\sum r^4 = n(n+1)(2n+1)(3n^2+3n-1)/30$. Each of these finite-sum identities follows by subtracting its polynomial at $n-1$ from its value at n, obtaining respectively n, n^2, n^3, n^4, and checking the value zero at $n = 0$. The same expansion of k_r^2 gave $4\lambda_n/3$ in Section20. The full cosine remainder therefore proves

$$\begin{aligned} \left| 1 - c_{n/2}(s) - \frac{2}{3}s^2\lambda_n \right| &\le \frac{s^4}{24}\frac{16}{15}\lambda_n(3\lambda_n - 1) \le \frac{2}{15}s^4\lambda_n^2, \\ \left\| \left(J_e - \frac{2}{3}s_e^2 E_e \right)\psi \right\| &\le \frac{2}{15}s_e^4 M_{2,e}(\xi). \end{aligned} \tag{22.22}$$

The last line is obtained by squaring the scalar inequality and summing every original link block. It keeps all spins and every exceptional angle; no equality is inferred from the Taylor estimate.

For the original native weights define, without changing the vector,

$$v_j = Q_\psi \sum_{e=(n,1)} n_2^2 E_e \psi, \qquad \mathcal{Z}_j = \frac{2}{3}\theta_j^2 v_j.$$

Retain $S_{2,L} = \sum_{e=(n,1)} n_2^4$ from Section8. Summing (22.22) before centering proves

$$\begin{aligned} \|\mathcal{U}_j - \mathcal{Z}_j\| &\le \frac{2}{15}q(\xi)\xi\theta_j^4 S_{2,L_j} = c_j\delta_j^E, \\ \delta_j^E &= \frac{2}{5}q(\xi)\frac{\theta_j^4 S_{2,L_j}}{\|z_j\|}, \\ j^5\delta_j^E &\longrightarrow \frac{16q(\xi)\pi^2}{5\sqrt{5}\cdot 10^8}. \end{aligned} \tag{22.23}$$

The last limit follows from the same exact sums as (22.15): $\theta_j^4 S_{2,L_j}/\|z_j\|$, multiplied by j^5, tends to $8\pi^2/(\sqrt{5}\cdot 10^8)$. Thus the retained finite-angle correction is of higher order than the full product comparison.

Here nonvanishing of v_j can also be checked quantitatively. Under the period thresholds used in (22.17), $\theta_j L_j < 1$. Termwise Taylor bounds imply $\|z_j\| \ge (\theta_j^2/3)\|\mathsf{I}w^{\mathrm{nat}}\|$. Also $S_{2,L}/\|\mathsf{I}w^{\mathrm{nat}}\| \le \sqrt{S_{2,L}/2} \le 3L^{7/2}$. Consequently

$$\delta_j^E \le \frac{18}{5}q(\xi)\theta_j^2 L_j^{7/2} \le \frac{129024}{5\cdot 10^8}j^{-5} < \frac{1}{4}.$$

Combining this with $\|\mathcal{U}_j\| \ge c_j\sqrt{1-b_C}$ proves $\|\mathcal{Z}_j\| \ge c_j/2 > 0$. Thus $v_j \ne 0$ at the same explicit eventual threshold, with its original amplitude retained.

Set

$$\Theta_j^E = (\varepsilon_j + \delta_j^E)\big(2\sqrt{1+b_C(\xi)} + \varepsilon_j + 3\delta_j^E\big).$$

The exact difference $\chi_{\theta_j} + \mathcal{Z}_j = (\chi_{\theta_j} + \mathcal{U}_j) - (\mathcal{U}_j - \mathcal{Z}_j)$ has norm at most $c_j(\varepsilon_j + \delta_j^E)$. Apply the three-term expansion from (22.19), now with $\|\mathcal{Z}_j\| \le c_j(\sqrt{1+b_C} + \delta_j^E)$, to obtain

$$\left| \int f\, d\eta_{\chi,j} - \int f\, d\eta_{\mathcal{Z},j} \right| \le c_j^2\Theta_j^E, \qquad |f| \le 1. \tag{22.24}$$

This includes the difference of the two raw total masses. In the actual spectral-probability fractions the coefficient is retained in both numerator and denominator:

$$\eta_{\mathcal{Z},j} = \frac{4}{9}\theta_j^4 \eta_{v,j}, \qquad C_{\mathcal{Z},j} = \frac{4}{9}\theta_j^4 C_{v,j}, \qquad \mu_{\mathcal{Z},j} = \mu_{v,j}.$$

The equality of the fractions follows from these exact two identities and their already proved positive denominators; it does not set the original vector amplitude to one. The proof of (22.20) therefore gives the complete original-state comparison

$$\boxed{\|\mu_{\chi,j} - \mu_{v,j}\|_{\mathrm{TV}} \le 8\Theta_j^E \longrightarrow 0.} \tag{22.25}$$

The Hamiltonian, its spectral projections, original physical units, fixed positive coupling, raw native n_2^2 weights and original cusp sequence are the same on both sides. This is not an assertion that the electric-state spectral probabilities themselves have a continuum limit.

22.7. Sources, scope, and the next energy calculation

The original finite-graph Haar and invariant Hilbert decomposition is the Baez source cited in Section21.1; its retained TeX174–321 was reread for this calculation. The tensor-product splitting has been proved here in polynomial coordinates, including its inverse and kernel, rather than inferred from a name or a routed literature hit. The canonical index was also queried for Clebsch–Gordan material; the returned adjacent books were not used as unread theorem evidence. The higher electric moments come from the full local resolvent and score proofs in Sections2,3,10,13 and21. The covariance estimate used for the nonzero denominator is (20.12), with its original Nachtergaele–Sims primary-source dependencies and complete Sections13–17 proof. The original period input is the complete used Section2/2.1 of the companion source cited in Section20.7.

The previously unresolved whole-product *norm* remainder is now controlled on the unchanged cusp, with a proved denominator and complete bounded spectral-measure comparison at fixed $0 < \xi \le 10^{-16}$. The generic comparison (22.12) remains valid at every positive coupling. No claim that its denominator estimate extends to all such couplings is made. Bounds for all bounded spectral tests do not themselves bound an unbounded first energy moment. Section23 supplies a separate full $A_j^{1/2}$-form calculation and an energy-weighted comparison. Section23.8 also proves that the individual first-energy tails are not uniformly integrable on this fixed-coupling sequence. Coupling continuation and the spatial continuum construction remain unfinished; neither the Millennium mass gap nor its negation follows from (22.20) alone.

23. Full-product energy form and first-energy-weighted spectral comparison

Retain the complete finite box, Hilbert space, physical gauge restriction and Hamiltonian (1.1)–(1.3). Put $H_0 = \sum_e E_e$, $A = H - \mathcal{E}$, where $H\psi = \mathcal{E}\psi$, and $q_A[x] = \|A^{1/2}x\|^2$. The vacuum is the same smooth positive unit vector as in Section1. Assign an arbitrary real angle s_e to every link, permitting zero angles on all unused links, and retain

$$C_e = C_{e,s_e}, \quad J_e = I - C_e, \quad S = \sum_e J_e, \quad K = \prod_e C_e, \quad t_e = 1 - \cos s_e, \quad T_1 = \sum_e t_e, \quad z = \mathsf{l}t. \tag{23.1}$$

Thus $z_p = \sum_{e\in p} t_e \ge 0$. The incidence matrix is the original unsigned face incidence, not a replacement graph. Set $\mathcal{U} = Q_\psi S\psi$ and $\chi = Q_\psi K\psi$. All products below include the negative C_e multipliers.

23.1. Joint moments needed by the complete product

The first, second and mixed electric estimates already proved give, for every $\xi > 0$,

$$\begin{aligned} \|E_e\psi\| &\le 8\xi, & \|E_e^2\psi\| &\le 6\xi + 128\xi^2, \\ \|E_eE_f\psi\| &\le 3\xi + 96\xi^2 \quad (e \ne f). \end{aligned} \tag{23.2}$$

For the last inequality use the middle bound in (20.15), not its last small-coupling bound 4ξ. Its proof used the full local comparison, score estimate and product rule, all valid for every positive ξ. Define

$$B_1(\xi) = \xi(14 + 128\xi), \qquad p(\xi) = 1 + 22\xi + 128\xi^2.$$

For all links e, f, including $e = f$, the triangle inequality applied to the full expansions proves

$$\|(E_e + I)E_f\psi\| \le B_1(\xi), \qquad \|(E_e + I)(E_f + I)\psi\| \le p(\xi). \tag{23.3}$$

For distinct links the actual bounds before enlargement are $11\xi + 96\xi^2$ and $1 + 19\xi + 96\xi^2$. For equal links they are $14\xi + 128\xi^2$ and $1 + 22\xi + 128\xi^2$. This explicitly accounts for the diagonal. All these operators commute in the original joint electric resolution. The all-angle inequality $0 \le J_f \le (4/3)t_f E_f$, proved in (20.1), therefore also gives

$$\|(E_e + I)S\psi\| \le \frac{4}{3}B_1(\xi)T_1. \tag{23.4}$$

Indeed square the scalar multiplier inequality on each joint block for each summand, sum its squared action on ψ, and then use the triangle inequality over f. No independence of vacuum marginals and no spectral gap are used in (23.2)–(23.4).

23.2. The sixteen full plaquette bands and their exact commutators

For a face p, order its four distinct physical links and write $\boldsymbol{n} = (n_e)_{e\in p} \in \mathbb{N}_0^4$, where $n_e = 2j_e$ and $\lambda_n = n(n+2)/4$. Let $P^p_{\boldsymbol{n}} = \prod_{e\in p} P^{(e)}_{n_e}$, with the exterior identity retained. For $\sigma \in \{-1, 1\}^4$ define

$$W_p^\sigma = \sum_{\substack{\boldsymbol{n}\in\mathbb{N}_0^4 \\ \boldsymbol{n}+\sigma\in\mathbb{N}_0^4}} P^p_{\boldsymbol{n}+\sigma}\, W_p\, P^p_{\boldsymbol{n}}. \tag{23.5}$$

The source polynomial splitting (22.1)–(22.3), applied separately to each of the four matrix-coefficient factors in W_p, proves $W_p = \sum_{\sigma\in\{-1,1\}^4} W_p^\sigma$. Inverse occurrences of links remain fundamental coefficients by the explicit adjugate formula in Section22.1. For fixed σ, distinct input labels have distinct orthogonal output blocks. For a finite block vector v,

$$\|W_p^\sigma v\|^2 \le \|W_p\|_\infty^2 \sum_{\boldsymbol{n}} \|P^p_{\boldsymbol{n}} v\|^2 \le 4\|v\|^2.$$

This proves the strong extension and $\|W_p^\sigma\| \le 2$. Invalid input or output labels give the zero operator. The bands are not claimed invertible. Each electric projection and W_p commutes with the vertex gauge action, so each band preserves the physical subspace, including every boundary constraint.

Let F be any bounded real function of the joint electric labels, with scalar value $f(\boldsymbol{n}, \boldsymbol{n}_{\mathrm{out}})$. Define $D^{\mathrm{in}}_{p,\sigma}$ to multiply a valid input block by $f(\boldsymbol{n} + \sigma, \boldsymbol{n}_{\mathrm{out}}) - f(\boldsymbol{n}, \boldsymbol{n}_{\mathrm{out}})$, and zero on invalid inputs. Define $D^{\mathrm{out}}_{p,\sigma}$ on a valid output block $\boldsymbol{m}$ by $f(\boldsymbol{m}, \boldsymbol{n}_{\mathrm{out}}) - f(\boldsymbol{m} - \sigma, \boldsymbol{n}_{\mathrm{out}})$, and zero otherwise. These are real diagonal operators. Direct multiplication of each input-output block proves the exact identities

$$\begin{aligned} [F, W_p^\sigma] &= W_p^\sigma D^{\mathrm{in}}_{p,\sigma} = D^{\mathrm{out}}_{p,\sigma} W_p^\sigma, \\ [F, [F, W_p^\sigma]] &= W_p^\sigma (D^{\mathrm{in}}_{p,\sigma})^2 = D^{\mathrm{out}}_{p,\sigma} W_p^\sigma D^{\mathrm{in}}_{p,\sigma}. \end{aligned} \tag{23.6}$$

All exterior labels and matrix-coefficient multiplicities are unchanged. The same difference estimate on a reverse shift $-\sigma$ bounds the output multiplier on ψ; one does not replace its vacuum distribution by the input distribution.

23.3. Complete signed-product difference on one plaquette

Use specifically $F = K - I + S$, so that $Q_\psi F\psi = \chi + \mathcal{U}$. On a joint block write $c_e = c_{n_e/2}(s_e)$ and, for the shifted face labels, $c'_e = c_{(n_e+\sigma_e)/2}(s_e)$. Put $K_{\mathrm{out}} = \prod_{e\notin p} c_e$ and $S_{\mathrm{out}} = \sum_{e\notin p}(1 - c_e)$. In the chosen face

order the full product difference is exactly

$$\Delta f = \sum_{e\in p}(c'_e - c_e)\left[K_{\text{out}}\prod_{\substack{f\in p\\ f<e}} c'_f \prod_{\substack{f\in p\\ f>e}} c_f - 1\right]. \tag{23.7}$$

To verify it, telescope $K_{\text{out}}(\prod c'_e - \prod c_e)$; the change of S is exactly $-\sum_{e\in p}(c'_e - c_e)$. This proves (23.7) including every mixed term.

All c_e, c'_e lie in $[-1,1]$. For real contractions a_i, telescoping gives $|1-\prod_i a_i| \le \sum_i (1-a_i)$: the absolute value of each prefix product is at most one and $1-a_i \ge 0$. This proof requires no positive-product restriction. The one-link Casimir inequalities from Section22.1 imply

$$\begin{aligned} |c'_e - c_e| &\le \frac{4}{3} t_e(\lambda_{n_e+\sigma_e} + \lambda_{n_e}) \le 4t_e(\lambda_{n_e}+1), \\ \left|K_{\text{out}}\prod_{f<e} c'_f \prod_{f>e} c_f - 1\right| &\le S_{\text{out}} + \frac{8}{3}\sum_{\substack{f\in p\\ f\neq e}} t_f(\lambda_{n_f}+1). \end{aligned} \tag{23.8}$$

For the first line, $\lambda_{n+\sigma} \le 2(\lambda_n+1)$ and $\lambda_n + \lambda_{n+\sigma} \le 3(\lambda_n+1)$. For the second, an unshifted $1-c_f$ is at most $(4/3)t_f\lambda_{n_f}$, and a shifted one is at most $(8/3)t_f(\lambda_{n_f}+1)$. The negative label is never introduced.

Retain the ordered off-diagonal face sum

$$\omega_p = \sum_{\substack{e,f\in p\\ e\neq f}} t_e t_f = z_p^2 - \sum_{e\in p} t_e^2.$$

Multiplying the pointwise nonnegative majorants in (23.8), using (23.3)–(23.4) on the original ψ, and taking the triangle inequality proves both bounds

$$\begin{aligned} \|D^{\text{in}}_{p,\sigma}\psi\|,\ \|D^{\text{out}}_{p,\sigma}\psi\| &\le B_p, \\ B_p &= \frac{16}{3}B_1(\xi)T_1 z_p + \frac{32}{3}p(\xi)\omega_p. \end{aligned} \tag{23.9}$$

For example the exterior term for link e contributes at most $4t_e(4/3)B_1T_1$; an ordered distinct face pair contributes $4t_e(8/3)t_f p$. This verifies every coefficient in (23.9). For the output multiplier apply the same proof with $-\sigma$ and regard its output label as the new unshifted label.

23.4. Exact full-Hamiltonian energy identity and volume estimate

Every bounded real electric multiplier preserves $\operatorname{Dom} H_0$, commutes with H_0, and preserves $\operatorname{Dom} H = \operatorname{Dom} H_0$. Consequently the following double commutator identity is valid on the smooth vacuum; its final commutators are bounded operators:

$$\begin{aligned} q_A[Q_\psi F\psi] &= \langle F\psi, (H-\mathcal{E})F\psi\rangle \\ &= \frac{1}{2}\langle \psi, [F,[H,F]]\psi\rangle \\ &= \frac{b}{2}\sum_{p\in \mathsf{P}_L}\langle \psi, [F,[F,W_p]]\psi\rangle. \end{aligned} \tag{23.10}$$

Indeed $[F,[H,F]] = 2FHF - F^2H - HF^2$; insertion of $H\psi = \mathcal{E}\psi$ proves the second line. The full Hamiltonian is $\kappa H_0 + 2bM - b\sum_p W_p$. Its electric and scalar terms commute with F, while $[H,F] = b\sum_p [F,W_p]$. This gives the plus sign $b/2$ in the last line without deleting the original $-bW_p$ term or changing the energy origin.

Equations (23.6) and (23.9) give

$$|\langle \psi, [F,[F,W_p^\sigma]]\psi\rangle| \le 2\|D^{\text{out}}_{p,\sigma}\psi\|\|D^{\text{in}}_{p,\sigma}\psi\| \le 2B_p^2.$$

There are sixteen signs. Thus, with $z_{\max} = \max_{p\in \mathsf{P}_L} z_p$,

$$\begin{aligned} q_A[\chi+\mathcal{U}] &\le 16b\sum_p B_p^2 \\ &\le 16b\|z\|^2\left[\frac{16}{3}B_1(\xi)T_1+\frac{32}{3}p(\xi)z_{\max}\right]^2. \end{aligned} \tag{23.11}$$

The second inequality uses only $0 \le \omega_p \le z_p^2 \le z_{\max} z_p$. In particular it does not cost a factor equal to the number of faces.

For $t \neq 0$ every link belongs to at least one face, whence $\|z\| > 0$. Retain the original coefficient $c_s = \xi\|z\|/3$ and define

$$\begin{aligned} \eta_F &= 64\sqrt{\xi}(14+128\xi)T_1 + \frac{128p(\xi)}{\sqrt{\xi}}z_{\max}, \\ \|A^{1/2}(\chi+\mathcal{U})\| &\le \sqrt{\kappa}\, c_s \eta_F. \end{aligned} \tag{23.12}$$

Taking the square root of (23.11), inserting $b = \kappa\xi$, and dividing the bound by $\sqrt{\kappa}c_s$ proves this formula. No vector is divided by a norm. When $t = 0$, (20.1) gives $J_e = 0$ for every link, so the un-divided statement is the identity $\chi = \mathcal{U} = 0$. All statements (23.1)–(23.12) hold at every positive coupling.

23.5. Form-domain map back to the unchanged electric weights

Define the real diagonal remainder from (22.22),

$$h_e(n) = 1 - c_{n/2}(s_e) - \frac{2}{3}s_e^2\lambda_n, \qquad G = \sum_e h_e(E_e),$$

where $h_e(E_e)$ denotes its value $h_e(n)$ on label n, not substitution of λ_n for the label. To avoid any bounded-commutator assumption for this unbounded operator, first take the actual electric cutoff

$$h_{e,\Lambda}(n) = h_e(n)\mathbf{1}_{\{\lambda_n\le\Lambda\}}, \qquad G_\Lambda = \sum_e h_{e,\Lambda}(E_e).$$

The cutoff is auxiliary to this proof only. The absolute remainder bound $|h_e(n)| \le (2/15)s_e^4\lambda_n^2$ proves, including a jump across the cutoff,

$$|h_{e,\Lambda}(n+\sigma) - h_{e,\Lambda}(n)| \le \frac{2}{15}s_e^4(\lambda_{n+\sigma}^2+\lambda_n^2) \le \frac{2}{3}s_e^4(\lambda_n+1)^2. \tag{23.13}$$

There is no differentiation of a discontinuous cutoff. The last coefficient follows from $\lambda_{n+\sigma}^2+\lambda_n^2 \le 5(\lambda_n+1)^2$. Consequently the input and output differences for G_Λ on every band have vacuum norm at most $(2/3)p(\xi)\sum_{e\in p}s_e^4$, by (23.3) with $e = f$. The bounded identity (23.10) and all sixteen bands now give

$$q_A[Q_\psi G_\Lambda\psi] \le \frac{64}{9}b\,p(\xi)^2\|\mathsf{I}(s^4)\|^2. \tag{23.14}$$

Here $(s^4)_e = s_e^4$; the complete face norm, rather than a maximum over a selected subset of faces, is retained.

The limit $\Lambda\to\infty$ is a form-domain limit at each original finite box. To prove it, put $h_0 = \sum_e \lambda_{n_e}$ on a joint block. The bound

$$\left|\sum_e h_{e,\Lambda}(n_e)\right| \le \frac{2}{15}\left(\max_e s_e^4\right)h_0^2$$

is uniform in Λ. The smooth vacuum belongs to $\operatorname{Dom} H_0^k$ for every integer k. Dominated summation with weight $1+h_0$ therefore proves $G_\Lambda\psi\to G\psi$ in the $H_0^{1/2}$ graph norm. Since $H-\kappa H_0$ is bounded at this fixed box, that graph norm and the $A^{1/2}$ graph norm are equivalent. Centering subtracts a multiple of ψ and is continuous in both. Passing to the limit in (23.14) proves

$$\|A^{1/2}Q_\psi G\psi\| \le \frac{8}{3}\sqrt{b}\,p(\xi)\|\mathsf{I}(s^4)\|. \tag{23.15}$$

No uniform cutoff convergence was presumed; the uniform final bound follows from (23.14), whose right side has no cutoff.

On the native angles $s_{(n,1)} = \theta n_2$, zero on the other directions, retain $v = Q_\psi \sum_{e=(n,1)} n_2^2 E_e \psi$ and $\mathcal{Z} = (2/3)\theta^2 v$. The exact identity and bound are

$$\begin{aligned} \mathcal{U} - \mathcal{Z} &= Q_\psi G \psi, \\ \eta_G &= \frac{8p(\xi)}{\sqrt{\xi}} \frac{\|\mathsf{I}(s^4)\|}{\|\mathsf{I}t\|}, \\ \|A^{1/2}(\mathcal{U} - \mathcal{Z})\| &\le \sqrt{\kappa}\, c_s \eta_G. \end{aligned} \tag{23.16}$$

Thus the electric state and the full product are connected by the proved form-domain difference, not only by a formal series.

23.6. The same cusp: exact bound and its physical scale

Use exactly (20.21): $L_j = j^2$, $a_j = 1/(100j)$, $\theta_j = 2\pi/(10^4 j^2 D_j)$, with $D_j = L_{j^2} q_{j^2} + 6m_{j^2}^2$ retained. Coupling g_{YM} is fixed and positive, and $\kappa_j = 200 g_{\mathrm{YM}}^2 j$. For the period thresholds in Section22.4, $\theta_j L_j < 1$. Every face has at most two direction-one links, so

$$z_{\max,j} \le 2 \max_e t_e \le \theta_j^2 L_j^2.$$

For $w_e^{\mathrm{nat}} = n_2^2$ on the direction-one links, $(w_e^{\mathrm{nat}})^2 \le L_j^2 w_e^{\mathrm{nat}}$. Every entry of I is nonnegative. The termwise bound $1 - \cos(\theta_j n_2) \ge \theta_j^2 n_2^2/3$ therefore gives

$$\frac{\|\mathsf{I}(s^4)\|}{\|\mathsf{I}t\|} \le 3\theta_j^2 L_j^2, \qquad \eta_{G,j} \le \frac{24p(\xi)}{\sqrt{\xi}} \theta_j^2 L_j^2. \tag{23.17}$$

The numerator before this inequality is $\theta_j^4 \|\mathsf{I}((w^{\mathrm{nat}})^2)\|$; the denominator is the original finite-angle incidence norm.

Set

$$\eta_j = \eta_{F,j} + \eta_{G,j}, \qquad r_j = \chi_{\theta_j} + \mathcal{Z}_j.$$

Combining the exact identity $r_j = (\chi_{\theta_j} + \mathcal{U}_j) - (\mathcal{U}_j - \mathcal{Z}_j)$ with (23.12) and (23.16) proves

$$\boxed{\|A_j^{1/2} r_j\| \le \sqrt{\kappa_j}\, c_j \eta_j.} \tag{23.18}$$

This estimate is valid at every fixed $\xi > 0$. The original finite sums (22.15) and the period expansion $\theta_j^2 L_j^2 = O(j^{-8})$ give

$$\begin{aligned} \eta_{G,j} &= O_\xi(j^{-8}), \\ j^2 \eta_j &\longrightarrow \frac{1024\pi^2 \sqrt{\xi}(14 + 128\xi)}{3 \cdot 10^8}. \end{aligned} \tag{23.19}$$

Both terms of η_F and the entire η_G were bounded before taking this limit. In particular no faster diagonal sequence or rescaled Hamiltonian was substituted.

Now restrict to $0 < \xi \le 10^{-16}$, the already proved denominator and covariance interval. The independent physical gap (15.13) gives $\gamma_\xi = 3(1 - 512\xi/3) > 0$ and $A_j \ge \kappa_j \gamma_\xi$ on the physical vacuum complement. The vector r_j lies there, since all of its factors preserve gauge invariance and it is centered. Hence

$$\|r_j\| \le c_j \epsilon_j^{\mathrm{en}}, \qquad \epsilon_j^{\mathrm{en}} = \eta_j/\sqrt{\gamma_\xi} = O_\xi(j^{-2}). \tag{23.20}$$

This improves the earlier norm rate by a separate energy proof; it is not used circularly to prove the physical gap. At the explicit dyadic threshold of (22.18) retain $\|\chi_{\theta_j}\| \ge c_j/2$ and $\|\mathcal{Z}_j\| \ge c_j/2$, already proved independently.

23.7. Raw measures, mass fractions and the unbounded energy test

For $X = \chi_{\theta_j}$ or $\mathcal{Z}_j$, retain

$$\eta_{X,j}(B) = \langle X, P_{A_j}(B)X\rangle, \quad C_{X,j} = \|X\|^2, \quad \mu_{X,j} = \eta_{X,j}/C_{X,j}.$$

The scalar notation η_j in (23.18) is an error coefficient; the notation $\eta_{X,j}$ here is a measure and carries its vector index. No raw vector or its amplitude is changed. Define, using the already proved b_C, b_N, δ_j^E,

$$\begin{aligned} B_{Z,j} &= \sqrt{1 + b_C(\xi)} + \delta_j^E, \\ A_{Z,j} &= \sqrt{3 + b_N(\xi)} + \eta_{G,j}, \\ d_{C,j} &= \epsilon_j^{\mathrm{en}}(2B_{Z,j} + \epsilon_j^{\mathrm{en}}), \\ d_{N,j} &= \eta_j(2A_{Z,j} + \eta_j). \end{aligned} \tag{23.21}$$

By (22.23), $\|\mathcal{Z}_j\| \le c_j B_{Z,j}$. By (20.12) and (23.16), $\|A_j^{1/2}\mathcal{Z}_j\| \le \sqrt{\kappa_j} c_j A_{Z,j}$. The two upper coefficients stay bounded at fixed ξ.

For any complex Borel function with $|f| \le 1$, use $\chi_{\theta_j} = -\mathcal{Z}_j + r_j$. Expand the quadratic expressions into the two cross terms and the r_j term and apply the operator bound $\|f(A_j)\| \le 1$. The same expansion on $A_j^{1/2}\chi_{\theta_j} = -A_j^{1/2}\mathcal{Z}_j + A_j^{1/2}r_j$ gives

$$\begin{aligned} \left| \int f \, d\eta_{\chi,j} - \int f \, d\eta_{\mathcal{Z},j} \right| &\le c_j^2 d_{C,j}, \\ \left| \int \lambda f(\lambda) \, d\eta_{\chi,j} - \int \lambda f(\lambda) \, d\eta_{\mathcal{Z},j} \right| &\le \kappa_j c_j^2 d_{N,j}. \end{aligned} \tag{23.22}$$

The second integrals are finite since the vectors lie in $\mathrm{Dom}\, A_j^{1/2}$. The spectral theorem identifies them with the corresponding bounded-f quadratic forms of the $A_j^{1/2}$ vectors. This is a new estimate for the unbounded energy weight, not an inference from (22.25). The first line with $f = 1$ also bounds the difference of the two actual masses by $c_j^2 d_{C,j}$.

To check the division explicitly, for any integrand g write

$$\frac{\int g \, d\eta_{\chi,j}}{C_{\chi,j}} - \frac{\int g \, d\eta_{\mathcal{Z},j}}{C_{\mathcal{Z},j}} = \frac{\int g \, d(\eta_{\chi,j} - \eta_{\mathcal{Z},j})}{C_{\chi,j}} + \frac{C_{\mathcal{Z},j} - C_{\chi,j}}{C_{\chi,j}C_{\mathcal{Z},j}} \int g \, d\eta_{\mathcal{Z},j}. \tag{23.23}$$

For $g = f$, the first term has absolute value at most $4d_{C,j}$; the second is also at most $4d_{C,j}$. For $g(\lambda) = \lambda f(\lambda)$, the first is at most $4\kappa_j d_{N,j}$. The second is at most $16\kappa_j A_{Z,j}^2 d_{C,j}$, because $\int \lambda \, d\eta_{\mathcal{Z},j} \le \kappa_j c_j^2 A_{Z,j}^2$ and both masses are at least $c_j^2/4$. The exact earlier identities $\eta_{\mathcal{Z},j} = (4/9)\theta_j^4 \eta_{v,j}$ and $C_{\mathcal{Z},j} = (4/9)\theta_j^4 C_{v,j}$ give $\mu_{\mathcal{Z},j} = \mu_{v,j}$. With total variation defined as the supremum over complex Borel $|f| \le 1$, this proves

$$\begin{aligned} \|\mu_{\chi,j} - \mu_{v,j}\|_{\mathrm{TV}} &\le 8d_{C,j} = O_\xi(j^{-2}), \\ \|\lambda(\mu_{\chi,j} - \mu_{v,j})\|_{\mathrm{TV}} &\le \kappa_j(4d_{N,j} + 16A_{Z,j}^2 d_{C,j}) = O_{\xi, g_{\mathrm{YM}}}(j^{-1}) \longrightarrow 0. \end{aligned} \tag{23.24}$$

The energy variable is that of the original A_j, not A_j/κ_j. The final rate follows from $d_C, d_N = O_\xi(j^{-2})$ and $\kappa_j = 200 g_{\mathrm{YM}}^2 j$. Thus it remains a vanishing bound even after the original physical energy factor is restored.

For completeness let $s \ge 0$ be a common spectral cutoff. Use $f = \mathbf{1}_{[s,\infty)}$ in (23.22)–(23.24). The difference of the two first-energy tails is bounded uniformly in s by the second line of (23.24). This does not bound either individual tail uniformly in j. The distinction follows directly from (23.23): its left side is a difference. Section23.8 proves that both individual means actually diverge on the present fixed-coupling sequence.

There is also a direct observable consequence. Write

$$\begin{aligned} \varphi_{X,j}(u) &= \int e^{-iu\lambda} \, d\mu_{X,j}(\lambda), \quad u \in \mathbb{R}, \\ \ell_{X,j}(u) &= \int e^{-u\lambda} \, d\mu_{X,j}(\lambda), \quad u \ge 0. \end{aligned}$$

Every first energy moment is finite at each fixed j. Dominated differentiation gives $\varphi' = -i\int \lambda e^{-iu\lambda} d\mu$ and $\ell' = -\int \lambda e^{-u\lambda} d\mu$, with the right derivative at zero for ℓ. The support is $[0,\infty)$, so all four bounded test functions have absolute value at most one before the explicitly retained energy factor. Consequently

$$\begin{aligned} &\sup_{u\in\mathbb{R}} |\varphi_{\chi,j} - \varphi_{v,j}|,\ \sup_{u\geq 0} |\ell_{\chi,j} - \ell_{v,j}| \leq 8 d_{C,j}, \\ &\sup_{u\in\mathbb{R}} |\varphi'_{\chi,j} - \varphi'_{v,j}|,\ \sup_{u\geq 0} |\ell'_{\chi,j} - \ell'_{v,j}| \leq \kappa_j (4 d_{N,j} + 16 A_{Z,j}^2 d_{C,j}). \end{aligned} \tag{23.25}$$

In each supremum both functions are evaluated at the same u. Thus the exact two state families have uniformly close real-time and Euclidean spectral functions and their first derivatives. This difference bound alone does not infer an individual limiting function; the next subsection calculates their fixed-coupling escape.

23.8. Exact fixed-coupling spectral escape and failure of uniform tails

The independent physical gap supplies more than a norm bound. On the original physical vacuum complement, $A_j \geq \kappa_j \gamma_\xi$. The spectral theorem on that reducing subspace therefore gives, for both $X = \chi_{\theta_j}$ and $X = v_j$,

$$\begin{aligned} \operatorname{supp} \eta_{X,j} &\subset [\kappa_j \gamma_\xi, \infty), \\ \mu_{X,j}([0,E]) &= 0 \quad \text{for} \quad j > \frac{E}{200 g_{\mathrm{YM}}^2 \gamma_\xi}, \quad E \geq 0. \end{aligned} \tag{23.26}$$

Here j also exceeds the explicit nonzero-state threshold of (22.18). To justify the spectral implication directly, a nonzero vector in a spectral interval lying strictly below $\kappa_j \gamma_\xi$ would have Rayleigh value below that lower bound, a contradiction. The union of such intervals exhausts $[0, \kappa_j \gamma_\xi)$. The full product and electric vectors are physical and centered; no component at the vacuum eigenvalue remains.

Every compactly supported continuous test function on $[0,\infty)$ has integral zero for all sufficiently large dyadic j. Thus these probabilities converge vaguely to the zero measure. They have no weakly convergent probability subsequence on this unscaled energy half-line. Indeed, any such weak limit would integrate every compactly supported continuous test to zero, so would be the zero Radon measure; weak convergence also tests the constant function one and would give total mass one. These conclusions are incompatible. This is a proof about the displayed states at the displayed fixed coupling, not a change of their measure definition.

The exact lower bound on the individual mean and tail is

$$\begin{aligned} \int \lambda \, d\mu_{X,j}(\lambda) &\geq 200 g_{\mathrm{YM}}^2 \gamma_\xi j, \\ \sup_j \int_{[E,\infty)} \lambda \, d\mu_{X,j}(\lambda) &= \infty \qquad \text{for every finite } E \geq 0. \end{aligned} \tag{23.27}$$

For the second line take arbitrarily large dyadic j for which the whole support lies above E, and apply the first line. In particular neither family of first-energy tails is uniformly integrable. Their difference nevertheless tends to zero in the weighted variation norm (23.24). The latter therefore compares two escaping measures; it is not a hidden tightness statement.

For the Euclidean functions at each fixed $u > 0$, (23.26) also gives the actual individual limit

$$0 \leq \ell_{X,j}(u) \leq e^{-200 g_{\mathrm{YM}}^2 \gamma_\xi j u} \longrightarrow 0, \qquad \ell_{X,j}(0) = 1. \tag{23.28}$$

This pointwise limit is discontinuous at zero and hence is not the Laplace transform of any probability measure on the unscaled nonnegative energy half-line. A probability Laplace transform is continuous at zero by dominated convergence applied to $e^{-u\lambda}$. No analogous pointwise real-time limit is asserted. The actual four-dimensional continuum programme must still calculate the original interacting theory's coupling/scale continuation; (23.26)–(23.28) establish precisely what the present fixed-coupling state sequence does.

23.9. Human sources and the remaining continuum mathematics

The Hilbert decomposition used in (23.5) is John C. Baez, *Spin Network States in Gauge Theory*, Advances in Mathematics 117 (1996), 253–272, arXiv:gr-qc/9411007v1, Section2, Lemmas1–3; the retained author TeX174–321 was read in full for this continuation. Section15 gives the exact inverse-link map from his vertex-action convention to the present one. The polynomial maps, kernels and inverses in Section22.1 give the four-link selection rule in (23.5). The sixteen-band norm, double commutator, full product difference, form-domain limit and all new constants are proved above, not imported from a similarly named result. The canonical literature index was queried for double-commutator sources; its adjacent Kubo and Yang–Mills routing hits were not treated as read theorem evidence. The inherited covariance and physical-gap estimates retain their original primary-source crosswalks in Sections13–17.

The original-product energy-form comparison is now proved, including the first-energy-weighted variation difference in unscaled physical units. Its nonzero-denominator and probability conclusions are at fixed $0 < \xi \le 10^{-16}$; the un-divided form estimates hold at every $\xi > 0$. The same fixed-coupling physical gap proves failure of individual first-energy uniform integrability and escape from every finite energy interval. This does not extend the coupling range of that gap, construct a spatial continuum theory under coupling/scale continuation, or decide the four-dimensional Millennium conclusion. Those remaining tasks require further mathematics beyond the complete comparisons and fixed-coupling escape proved here.

Section24 continues the un-divided energy calculation on the original logarithmic and fixed-electric-coefficient coupling paths. Its positive-energy transport theorem does not use the fixed-coupling gap.

24. The original logarithmic coupling path: energy-weighted spectral transport

Section23.8 concerned fixed coupling in its stated interval. Here the coupling varies along the already specified logarithmic and fixed-electric-coefficient paths. The full positive-coupling vacuum and all physical coefficients are retained. No small-ξ gap, weak-coupling covariance denominator, or faster cusp is used.

24.1. A full electric-state energy bound at every positive coupling

For the original real angles define the nonnegative electric multiplier

$$\mathsf{L}_s = \frac{2}{3}\sum_e s_e^2 E_e, \qquad \mathcal{Z}_s = Q_\psi \mathsf{L}_s \psi. \tag{24.1}$$

On native angles this is exactly $\mathcal{Z}_s = (2/3)\theta^2 v_{w^{\mathrm{nat}}}$, including its raw coefficient. On a single valid spin shift, the original Casimir obeys

$$\begin{aligned} \lambda_{n+1} - \lambda_n &= (2n+3)/4, \\ \lambda_{n-1} - \lambda_n &= -(2n+1)/4 \quad (n \ge 1), \\ |\lambda_{n+\sigma} - \lambda_n| &\le \lambda_n + 1. \end{aligned} \tag{24.2}$$

For the plus sign the difference between the upper bound and the left side is $(n^2+1)/4$; for the minus sign it is $(n^2+3)/4$. Both are nonnegative, including $n = 0$ in the valid plus case.

Use the auxiliary bounded multiplier $\mathsf{L}_{s,\Lambda} = (2/3)\sum_e s_e^2 \min(E_e, \Lambda)$, where the minimum is spectral calculus. The function $x \mapsto \min(x, \Lambda)$ is increasing and 1-Lipschitz on $[0, \infty)$, so its neighboring difference is bounded by the absolute Casimir difference in (24.2), even across the cutoff. In each of the sixteen exact plaquette bands of (23.5), both the input and output difference applied to ψ therefore have norm at most

$$\frac{2}{3}\sum_{e\in p} s_e^2 \|(E_e + I)\psi\| \le \frac{2}{3}(1 + 8\xi)\sum_{e\in p} s_e^2.$$

The first electric moment (23.2) proves the last step for every $\xi > 0$. The complete band identity and double commutator (23.6), (23.10) give

$$q_A[Q_\psi \mathsf{L}_{s,\Lambda}\psi] \le \frac{64}{9} b(1+8\xi)^2 \|\mathsf{I}(s^2)\|^2. \tag{24.3}$$

The coefficient is exactly $16(2/3)^2$, before substituting any physical parameter. Every face is present in the incidence norm.

For each fixed box, $\mathsf{L}_{s,\Lambda}\psi \to \mathsf{L}_s\psi$ in the $H_0^{1/2}$ graph norm: on a joint label the multipliers are bounded by $(2/3)(\max_e s_e^2)h_0$, where $h_0 = \sum_e \lambda_{n_e}$, and converge pointwise. Smoothness of ψ permits dominated summation with weight $1 + h_0$. The same bounded-potential and centering argument as in Section23.5 transfers this to the $A^{1/2}$ graph norm. Passing to the limit proves the actual, uncut bound

$$\|A^{1/2}\mathcal{Z}_s\| \le \frac{8}{3}\sqrt{b}(1+8\xi)\|\mathsf{I}(s^2)\|. \tag{24.4}$$

There is no assumption that L_s is bounded and no cutoff state is substituted for $\mathcal{Z}_s$.

For $t_e = 1 - \cos s_e$ not identically zero, retain $c_s = \xi\|\mathsf{I}t\|/3 > 0$, and put

$$\begin{aligned} R_{2,s} &= \frac{\|\mathsf{I}(s^2)\|}{\|\mathsf{I}t\|}, \\ a_s^{\mathrm{el}} &= \frac{8(1+8\xi)}{\sqrt{\xi}} R_{2,s}. \end{aligned} \tag{24.5}$$

Thus (24.4) is $\|A^{1/2}\mathcal{Z}_s\| \le \sqrt{\kappa} c_s a_s^{\mathrm{el}}$. This is a coefficient-relative estimate, not a probability denominator or a change of the electric vector's amplitude. If all $t_e = 0$, use (24.4) without division; angles that are nonzero multiples of 2π can have $\mathcal{Z}_s \neq 0$. No assertion of $\mathcal{Z}_s = 0$ is made at those exceptional angles.

24.2. Energy-weighted raw measures without a spectral-gap assumption

The full product is still $K_s = \prod_e C_{e,s_e}$ and $\chi_s = Q_\psi K_s \psi$. Define the all-angle form coefficients from Section23 without a fixed-coupling asymptotic replacement:

$$\begin{aligned} \eta_{F,s} &= 64\sqrt{\xi}(14+128\xi)T_1 + \frac{128p(\xi)}{\sqrt{\xi}} z_{\max}, \\ \eta_{G,s} &= \frac{8p(\xi)}{\sqrt{\xi}} \frac{\|\mathsf{I}(s^4)\|}{\|\mathsf{I}t\|}, \\ \eta_s &= \eta_{F,s} + \eta_{G,s}, \qquad \mathscr{D}_s = \kappa\eta_s(2a_s^{\mathrm{el}} + \eta_s). \end{aligned} \tag{24.6}$$

The un-divided estimates (23.12), (23.15) apply to all real angles, and the exact identity

$$\chi_s + \mathcal{Z}_s = (\chi_s + \mathcal{U}_s) - (\mathcal{U}_s - \mathcal{Z}_s)$$

has the same diagonal remainder G from Section23.5. Consequently, for every $\xi > 0$,

$$\|A^{1/2}(\chi_s + \mathcal{Z}_s)\| \le \sqrt{\kappa} c_s \eta_s. \tag{24.7}$$

The source of the possible negative product factors has not been removed; (23.7) kept them in every prefix product.

For any vector X here define its raw spectral measure $\eta_X(B) = \langle X, P_A(B)X\rangle$. It is finite and has finite first moment. Put $\Delta_s = \eta_{\chi_s} - \eta_{\mathcal{Z}_s}$. Expand the two quadratic forms of $f(A)$ on $A^{1/2}\chi_s = -A^{1/2}\mathcal{Z}_s + A^{1/2}(\chi_s + \mathcal{Z}_s)$. For every complex Borel $|f| \le 1$, the two cross terms are bounded by the product of the two norms, and the residual term by its squared norm. Equations (24.4)–(24.7) give

$$\boxed{\|\lambda\Delta_s\|_{\mathrm{TV}} := \sup_{|f|\le 1} \left| \int \lambda f(\lambda)\, d\Delta_s \right| \le c_s^2 \mathscr{D}_s.} \tag{24.8}$$

Here λ is the physical spectral variable of A, not of A/κ. In native variables the signed measure is exactly

$$\Delta_s = \eta_{\chi_s} - \frac{4}{9}\theta^4 \eta_{v_{w^{\mathrm{nat}}}}. \tag{24.9}$$

Both measures retain their total masses. Neither is divided by its total mass in (24.8)–(24.9).

For an arbitrary physical threshold $\epsilon > 0$, the original state difference itself obeys

$$\|P_A([\epsilon,\infty))(\chi_s + \mathcal{Z}_s)\| \le \frac{\sqrt{\kappa} c_s \eta_s}{\sqrt{\epsilon}}. \tag{24.10}$$

This follows by integrating $\mathbf{1}_{[\epsilon,\infty)}(\lambda) \le \lambda/\epsilon$ against the positive raw measure of that difference. No whole-spectrum gap is used. The phase relation is the displayed plus sign between the original vectors.

24.3. Exact information retained by energy weighting

To specify the measure map precisely, let $\mathfrak{M}_1$ be the complex finite Borel measures ρ on $[0,\infty)$ with $\int \lambda \, d|\rho| < \infty$. The linear map

$$\mathsf{T} : \mathfrak{M}_1 \longrightarrow \mathfrak{M}, \qquad (\mathsf{T}\rho)(B) = \int_B \lambda \, d\rho(\lambda) \tag{24.11}$$

takes values in finite complex Borel measures. It has kernel $\mathbb{C}\delta_0$, and its image consists exactly of measures ν with

$$\nu(\{0\}) = 0, \qquad \int_{(0,\infty)} \lambda^{-1} \, d|\nu|(\lambda) < \infty. \tag{24.12}$$

To prove the kernel assertion, on every interval $[\epsilon, R]$, multiplication by $1/\lambda$ inverts T, so $\mathsf{T}\rho = 0$ implies that ρ is supported at zero. The converse is immediate. For the image, polar decomposition gives $|\lambda\rho| = \lambda|\rho|$ on $(0,\infty)$. This proves (24.12) for every image measure. Conversely, for a ν satisfying (24.12), set

$$\rho(B) = \int_{B\cap(0,\infty)} \lambda^{-1} \, d\nu(\lambda) + a\, \mathbf{1}_B(0), \qquad a \in \mathbb{C}. \tag{24.13}$$

Its total variation is finite by (24.12), its first absolute moment equals $|\nu|((0,\infty))$, and $\mathsf{T}\rho = \nu$. This proves the image, the complete fibre of the map and its inverse on measures having no atom at zero.

On $[\epsilon,\infty)$ the inverse has norm at most $1/\epsilon$. Thus (24.8) proves

$$\|\Delta_s|_{[\epsilon,\infty)}\|_{\mathrm{TV}} \le \frac{c_s^2 \mathscr{D}_s}{\epsilon}. \tag{24.14}$$

There is no inverse bound independent of ϵ. For an exact example, the two probabilities $\delta_{1/j}$ and $\delta_{2/j}$ have difference with total variation 2, whereas its image under T has total variation $3/j$. They have no atom at zero at any finite j. This example proves the discontinuity of the inverse near zero; it is not a replacement model for the quantum state measures.

The actual finite-box vacuum is simple and the displayed states are centered, so their measures have no atom at zero. This gives injectivity of (24.11) for each finite-box measure here, but no uniform inverse estimate as the box and coupling vary. Equations (24.10)–(24.14) state exactly which original positive-energy data are controlled without that estimate.

24.4. Original depth and the prescribed logarithmic coupling

Retain the full original dyadic geometric tuple

$$L_j = j^2, \quad a_j = \frac{1}{100j}, \quad D_j = L_{j^2} q_{j^2} + 6m_{j^2}^2, \quad \theta_j = \frac{2\pi}{10^4 j^2 D_j}, \quad s_{(n,1),j} = \theta_j n_2. \tag{24.15}$$

Other direction angles are zero. In particular the cusp depth remains j^2, not a coupling-dependent larger depth. The already prescribed logarithmic trajectory is

$$g_j^2 = \frac{1}{\log j}, \qquad \xi_j = \frac{(\log j)^2}{4}, \qquad \kappa_j = \frac{200j}{\log j}, \qquad b_j = 50j \log j, \qquad 2b_j M_j = 100 j M_j \log j. \tag{24.16}$$

All quantities are positive for dyadic $j \geq 2$. Each ψ_j is the actual vacuum at these parameters. In particular $\xi_j \to \infty$; the small-ξ physical-gap theorem does not apply on this trajectory.

The geometric finite sums remain unchanged. Introduce the explicit coefficient constants

$$T_* = \frac{16\pi^2}{3 \cdot 10^8}, \qquad A_* = \frac{16384\pi^2}{3 \cdot 10^8}.$$

They are not new physical parameters. From (22.15) and the all-angle Taylor bounds,

$$\begin{aligned} j^2 T_{1,j} &\longrightarrow T_*, \qquad z_{\max,j} = O(j^{-8}), \\ R_{2,j} &\longrightarrow 2, \qquad \frac{\|\mathsf{I}(s_j^4)\|}{\|\mathsf{I} t_j\|} \leq 3\theta_j^2 L_j^2 = O(j^{-8}). \end{aligned} \tag{24.17}$$

For the ratio limit, set $x = \theta_j n_2$. The inequality $\left|1 - \cos x - x^2/2\right| \leq x^4/24$ gives an entrywise relative error at most $(\theta_j L_j)^2/12$ between t_j and $s_j^2/2$, including equality at zero entries. Since incidence entries are nonnegative, the same multiplicative bounds hold for each face coefficient and its Euclidean norm. This proves $R_{2,j} \to 2$ with no change of weights.

Write $\ell_j = \log j$ only in the following calculation; it is not the physical box length used elsewhere. Substitution of $\xi_j = \ell_j^2/4$ into every term of (24.5)–(24.6) gives

$$\begin{aligned} \frac{a_j^{\mathrm{el}}}{\ell_j} &\longrightarrow 64, \\ \frac{j^2 \eta_j}{\ell_j^3} &\longrightarrow A_*, \\ \frac{j\, \mathscr{D}_j}{\ell_j^3} &\longrightarrow 25600 A_*. \end{aligned} \tag{24.18}$$

Here the second limit comes from

$$64\sqrt{\xi_j}(14 + 128\xi_j) = 448\ell_j + 1024\ell_j^3.$$

Multiplication by $T_{1,j}$ gives the nonzero leading term $1024 T_* = A_*$. Both remaining terms of η_j are $O(\ell_j^3 j^{-8})$. For the first limit,

$$\frac{8(1 + 8\xi_j)}{\sqrt{\xi_j}} = \frac{16}{\ell_j} + 32\ell_j,$$

and $R_{2,j} \to 2$. Finally $\eta_j / a_j^{\mathrm{el}} = O(\ell_j^2 j^{-2}) \to 0$. Keeping $\kappa_j = 200j/\ell_j$ in $\mathscr{D}_j = \kappa_j \eta_j (2a_j^{\mathrm{el}} + \eta_j)$ proves the third limit, including its coefficient 25600. Polynomial powers of $\log j$ divided by positive powers of j tend to zero: put $x = \log j$ and bound e^{rx} below by any higher integer term of its positive power series.

Consequently the following is a proved weak-coupling result on the original, unmodified cusp:

$$\boxed{\left\| \lambda \left(\eta_{\chi_j} - \frac{4}{9}\theta_j^4 \eta_{v_j} \right) \right\|_{\mathrm{TV}} \leq c_j^2 \mathscr{D}_j, \qquad \mathscr{D}_j = O\left(\frac{(\log j)^3}{j} \right) \longrightarrow 0.} \tag{24.19}$$

The coefficient is still $c_j = \xi_j \|\mathsf{I} t_j\|/3$. Its exact leading size is

$$\frac{j^5 c_j}{(\log j)^2} \longrightarrow \frac{4\pi^2}{3\sqrt{5} \cdot 10^8}. \tag{24.20}$$

This follows from the geometric $j^5\|\mathsf{I}t_j\| \to 16\pi^2/(\sqrt{5}\cdot 10^8)$, not from treating ξ as fixed in (22.15). In particular the raw bound in (24.19) is $O((\log j)^7 j^{-11})$, and its error relative to the retained coefficient c_j^2 tends to zero. No unknown total spectral mass has been used as a denominator.

For any fixed $\epsilon > 0$, (24.10) also yields the actual positive-energy state comparison with explicit error coefficient

$$\frac{\|P_{A_j}([\epsilon,\infty))(\chi_j + (2/3)\theta_j^2 v_j)\|}{c_j} \le \sqrt{\kappa_j/\epsilon}\,\eta_j = O_\epsilon\!\left(\frac{(\log j)^{5/2}}{j^{3/2}}\right) \longrightarrow 0. \tag{24.21}$$

This division reports an error relative to an explicit input coefficient; it does not alter either raw state. The bound (24.14) similarly controls every Borel spectral window bounded away from zero, with error $c_j^2\mathscr{D}_j/\epsilon$.

24.5. Physical-time differences with the constant term retained

Let $C_{X,j} = \eta_{X,j}([0,\infty))$, and retain the raw time functions $\Phi_{X,j}(u) = \int e^{-iu\lambda} d\eta_{X,j}$ for real u, and $L_{X,j}(u) = \int e^{-u\lambda} d\eta_{X,j}$ for $u \ge 0$. The first moments are finite at each j, so dominated differentiation and (24.8) give

$$\begin{aligned} &\sup_{u\in\mathbb{R}} |\Phi'_{\chi,j}(u) - \Phi'_{\mathcal{Z},j}(u)| \le c_j^2\mathscr{D}_j, \\ &\sup_{u\ge 0} |L'_{\chi,j}(u) - L'_{\mathcal{Z},j}(u)| \le c_j^2\mathscr{D}_j. \end{aligned} \tag{24.22}$$

At $u = 0$ the derivative in the second line is a right first derivative. Integrating the first derivatives from zero retains the unknown constant term exactly:

$$\begin{aligned} &|(\Phi_{\chi,j}(u) - C_{\chi,j}) - (\Phi_{\mathcal{Z},j}(u) - C_{\mathcal{Z},j})| \le |u| c_j^2\mathscr{D}_j, \\ &|(L_{\chi,j}(u) - C_{\chi,j}) - (L_{\mathcal{Z},j}(u) - C_{\mathcal{Z},j})| \le u c_j^2\mathscr{D}_j. \end{aligned} \tag{24.23}$$

These inequalities can equivalently be proved using $|e^{-iu\lambda} - 1| \le |u|\lambda$ and $0 \le 1 - e^{-u\lambda} \le u\lambda$. On (24.16), every bound divided by c_j^2 vanishes, uniformly on bounded physical-time intervals, and the derivative bounds are uniform on their whole time domains. The measures and constants for $\mathcal{Z}$ are still the exact multiples $(4/9)\theta_j^4$ of those for v_j. No constant term is silently set to zero.

24.6. The fixed-electric-coefficient path tested by the same full bound

For the second prescribed trajectory retain (24.15) and

$$g_j^2 = \frac{\kappa_*}{200j}, \quad \xi_j = \frac{10000 j^2}{\kappa_*^2}, \quad \kappa_j = \kappa_*, \quad b_j = \frac{10000 j^2}{\kappa_*}, \quad 2b_j M_j = \frac{20000 j^2 M_j}{\kappa_*}, \qquad \kappa_* > 0. \tag{24.24}$$

Define $\alpha_* = 10000/\kappa_*^2$ and $F_* = 8192\alpha_*^{3/2} T_*$. Direct substitution into the same bounds, without suppressing the growing magnetic coefficient, gives

$$\begin{aligned} &\frac{a_j^{\mathrm{el}}}{j} \longrightarrow 128\sqrt{\alpha_*}, \\ &\frac{\eta_j}{j} \longrightarrow F_*, \\ &\frac{\mathscr{D}_j}{j^2} \longrightarrow \kappa_* F_* (256\sqrt{\alpha_*} + F_*) > 0. \end{aligned} \tag{24.25}$$

To check the second limit, $64\sqrt{\xi_j}(14 + 128\xi_j)T_{1,j}/j \to 8192\alpha_*^{3/2} T_*$. The $z_{\max}$ and s^4 terms are $O(j^{-5})$ before division by j. The first limit follows from $R_{2,j} \to 2$, and the third retains both terms of $\eta_j(2a_j^{\mathrm{el}} + \eta_j)$.

Thus (24.8) is still an exact valid bound on this path, but its coefficient-relative energy error upper bound grows. This proves failure of this particular estimate to establish vanishing transport there. It is not a lower bound on the actual difference and not a proof that the two state families fail to be related or to converge. No replacement trajectory is introduced to change that outcome.

24.7. Source identities and exact continuation scope

The two parameter tuples are the companion *Original cusp to spatial quantum continuum*, Section11, equations(88) and(91), read in their complete containing section. Only those explicitly displayed parameter choices are used here; no later selected diagonal, free limit, or claimed continuum result from that manuscript is imported. The geometric tuple is its original Section2/2.1, already retained and pinned in Sections20–23. The human Hilbert-space source remains Baez, *Spin Network States in Gauge Theory*, Section2, with the exact inverse-link convention map in Section15. Its full TeX174–321 was reread. The canonical index was queried for spectral moments in lattice gauge theory; the returned routing hits were not used as content evidence. The Casimir differences, bounded spectral-cap approximation, band estimates, measure kernel, inverse and all trajectory coefficients have full proofs above.

Section24 proves transport of the original positive-energy raw state and measure data on the prescribed logarithmic path, including its physical-time derivatives. It does not derive a covariance denominator, a limiting probability, a nonzero limiting energy-weighted measure, or a surviving nonabelian continuum interaction. The finite-box measure map (24.11)–(24.14) identifies precisely the unresolved near-zero inverse control. The fixed-electric-coefficient path remains a distinct unresolved estimate, with the actual limitation of the present bound proved in (24.25).

25. Polynomial conditional anti-concentration and the original weighted covariance

This section replaces the exponential conditional-density estimate used in Section14 by an exact integration-by-parts estimate on the original radius-two (SU(2)) link sphere. The Wilson density is not replaced by a Haar density, a one-plaquette model, or a truncated state. The conditional density below is the conditional density of the actual positive finite-box ground state already fixed in Section1. All constants, link coordinates and the original weight (n_2^2) are retained.

25.1. A rank-three sphere estimate

Write a link as

$$U = x_0 I + i \sum_{\alpha=1}^{3} x_\alpha \sigma_\alpha, \qquad x_0^2 + x_1^2 + x_2^2 + x_3^2 = 1.$$

In the metric and differential-operator convention of (1.2), the coordinate identities are

$$\Delta x_\alpha = -\frac{3}{4} x_\alpha, \qquad \langle \nabla x_\alpha, \nabla x_\beta \rangle = \frac{1}{4}(\delta_{\alpha\beta} - x_\alpha x_\beta). \tag{25.1}$$

Let $v = (v_1, v_2, v_3)$ be any three orthogonal linear coordinates obtained from (x_0, x_1, x_2, x_3) by an $O(4)$ change of coordinates. Let $p > 0$ be a smooth probability density on this sphere and suppose

$$\|\nabla \log p\|_\infty \le K, \qquad K \ge 0. \tag{25.2}$$

Put $m = \int vp$, $V = \int |v - m|^2 p$, and $J = \int \sum_i |\nabla v_i|^2 p$. Since the three rows of the differential of v are orthogonal projections of the four coordinate differentials,

$$\sum_i |\nabla v_i|^2 = \frac{3 - |v|^2}{4} \ge \frac{1}{2}, \qquad \|Dv\|_{\mathrm{op}} \le \frac{1}{2}. \tag{25.3}$$

Integrating the divergence of $p(v_i - m_i)\nabla v_i$, summing in i, and using (25.1), gives the identity

$$J = \frac{3}{4}V - \int \sum_i (v_i - m_i)\langle \nabla v_i, \nabla \log p \rangle p. \tag{25.4}$$

The last term has absolute value at most $K\sqrt{V}/2$, by (25.3) and Cauchy–Schwarz. Hence

$$\frac{1}{2} \le \frac{3}{4}V + \frac{K}{2}\sqrt{V}. \tag{25.5}$$

Solving this displayed quadratic inequality without changing the sphere or the density yields

$$V \ge \sigma(K), \qquad \sigma(K) = \left(\frac{\sqrt{K^2+6} - K}{3} \right)^2 = \frac{4}{(\sqrt{K^2+6}+K)^2}. \tag{25.6}$$

The equality of the two expressions follows from $(\sqrt{K^2+6}-K)(\sqrt{K^2+6}+K) = 6$. In particular, for every fixed vector $B \in \mathbb{R}^3$,

$$\int |B+v|^2 p = V + |B+m|^2 \ge \sigma(K). \tag{25.7}$$

Thus an exterior-dependent translation cannot cancel the conditional fluctuation. The explicit elementary lower estimate

$$\sigma(K) \ge \frac{2}{3(1+K)^2} \tag{25.8}$$

follows from $\sqrt{K^2+6} + K \le \sqrt{6}(1+K)$. The exact root (25.6), rather than (25.8), is used below whenever the sharper coefficient matters.

25.2. Scalar trace variance on the same sphere

Let $f = 2x_0$ after any $O(4)$ change of coordinates. Then

$$\Delta f = -\frac{3}{4} f, \qquad |\nabla f|^2 = 1 - \frac{f^2}{4} = |v|^2. \tag{25.9}$$

Writing $\bar{f} = \int f p$, $J_f = \int |\nabla f|^2 p$, and $V_f = \int (f - \bar{f})^2 p$, one has $J_f \ge V$. A second integration by parts gives

$$J_f = \frac{3}{4} V_f - \int (f - \bar{f}) \langle \nabla f, \nabla \log p \rangle p, \qquad 1 \le \frac{3}{4} y^2 + Ky, \quad y = \sqrt{V_f / J_f}. \tag{25.10}$$

Consequently

$$V_f \ge r(K) J_f \ge r(K)\sigma(K), \qquad r(K) = \left(\frac{2(\sqrt{K^2+3} - K)}{3} \right)^2 = \frac{4}{(\sqrt{K^2+3}+K)^2}. \tag{25.11}$$

The elementary form $r(K) \ge (1+K)^{-2}$ also gives the fully explicit polynomial estimate

$$V_f \ge \frac{2}{3(1+K)^4}. \tag{25.12}$$

No independence of coordinates has been used in (25.10)–(25.12).

25.3. Conditional Wilson gradients in the actual vacuum

Fix an edge e and one incident plaquette p. Let $f(e,p)$ be the unique edge of p parallel to e and opposite to it. Condition on all link variables except $U_{f(e,p)}$, including U_e. The conditional density is

$$p_f(U_f) = \frac{\rho(U_f, U_{\neq f})}{\int_{SU(2)} \rho(V, U_{\neq f})\, dV}, \qquad \rho = \psi^2. \tag{25.13}$$

The denominator is positive because $\psi > 0$. The score estimate of Section10 gives, for every exterior configuration,

$$\|\nabla_f \log p_f\|_\infty = \|\nabla_f \log \rho\|_\infty \le 4r_f\xi \le 16\xi. \tag{25.14}$$

As a function of U_f, the plaquette trace is a trace of a fixed product with U_f or U_f^{-1}. Left or right multiplication and inversion are Haar-preserving orthogonal transformations of the quaternion coordinates. Differentiating in the fixed $T_a = -i\sigma_a/2$ frame therefore gives, up to a fixed orthogonal transformation and signs, the three coordinates v_1, v_2, v_3 of (25.3). In particular the complete derivative is the rank-three coordinate vector in (25.7); no coefficient is discarded.

All other plaquettes containing e are independent of U_f. Thus, with B their conditional contribution,

$$\nabla_e \mathcal{W} = B + \nabla_e W_p, \qquad \mathbb{E}\left[|\nabla_e \mathcal{W}|^2 \mid U_{\neq f}\right] \ge \sigma(4r_f\xi) \ge \sigma(16\xi). \tag{25.15}$$

The first inequality is (25.7); the second uses that (25.6) is decreasing in K. Integrating the conditional inequality gives the actual-vacuum bound

$$\mathbb{E}_\rho |\nabla_e \mathcal{W}|^2 \ge s(\xi), \qquad s(\xi) = \sigma(16\xi) > 0. \tag{25.16}$$

For the same conditional fibre, W_p is a trace of the form $2x_0$, so (25.11) and the law of total variance give

$$\mathrm{Var}_\rho(W_p) \ge r(4r_f\xi)\sigma(4r_f\xi) \ge r(16\xi)\sigma(16\xi). \tag{25.17}$$

The form identity (1.5) and the linkwise derivative identity $\sum_a |X_{e,a} W_p|^2 = 1 - W_p^2/4$ also yield directly for the three-link contribution to the full form

$$q^{(e)}_{H-\mathcal{E}}[(W_p - \mathbb{E}_\rho W_p)\psi] = \kappa\, \mathbb{E}_\rho \sum_{a=1}^{3} |X_{e,a} W_p|^2 \ge \kappa\, \sigma(16\xi). \tag{25.18}$$

Here (q^{(e)}) denotes the summand of the full form containing only the three (T_a)-derivatives on the chosen edge, so that the equality is exact; the full form is at least this nonnegative summand. This is a positive local form estimate, not an assertion about a lowest nonzero eigenvalue of the full Hamiltonian.

25.4. Propagation to the original weighted electric covariance

Retain the original operator $\Gamma = \sum_{e=(n,1)} n_2^2 E_e$, and put $v_\Gamma = Q\Gamma\psi$. The exact commutator calculation in the companion source, with $B_{\mathrm{sk}} = [H_0, \mathcal{W}]$, is

$$[\Gamma, H_0] = 0, \qquad \langle [\Gamma, B_{\mathrm{sk}}] \rangle_\rho = \xi \langle [\mathcal{W}, [\Gamma, \mathcal{W}]] \rangle_\rho = 2\xi S_\Gamma, \tag{25.19}$$

where S_Γ is the sum of the squared Wilson derivatives with the original n_2^2 edge weights, with the incident-plaquette contribution retained. Since B_{sk} is skew-adjoint in the real form used here,

$$2\xi S_\Gamma = 2\,\mathrm{Re}\langle v_\Gamma, B_{\mathrm{sk}}\psi\rangle, \qquad \|v_\Gamma\| \geq \frac{\xi S_\Gamma}{\|B_{\mathrm{sk}}\psi\|}. \tag{25.20}$$

The complete incidence count already proved in the companion calculation gives

$$S_\Gamma \geq \sigma(16\xi) S_{1,L}, \qquad S_{1,L} = \frac{2}{3}L^2(L+1)(2L+1)^2. \tag{25.21}$$

The electric moment bounds of Sections10 and14 give

$$\|B_{\mathrm{sk}}\psi\| \leq M(6+64\xi), \qquad M = 12L^2(2L+1). \tag{25.22}$$

Combining (25.20)–(25.22), while retaining the exact displayed incidence ratio, proves for every $L \geq 2$ and every $\xi > 0$

$$\boxed{\|v_\Gamma\| \geq \frac{\xi\,\sigma(16\xi)\,(L+1)(2L+1)}{18(6+64\xi)} > 0.} \tag{25.23}$$

This is an actual lower bound for the original weighted electric covariance vector. It does not divide by, or replace, the reference coefficient $c = \xi\|It\|/3$ used in Sections23–24.

25.5. The unchanged logarithmic path

On the original path $L_j = j^2$, $\xi_j = (\log j)^2/4$, the argument of σ in (25.23) is $16\xi_j = 4(\log j)^2$. From the exact second form in (25.6),

$$(\log j)^4\sigma(4(\log j)^2) \longrightarrow \frac{1}{16}, \qquad \frac{\xi_j}{6+64\xi_j} \longrightarrow \frac{1}{64}, \qquad \frac{(L_j+1)(2L_j+1)}{j^4} = 2 + \frac{3}{j^2} + \frac{1}{j^4} \longrightarrow 2. \tag{25.24}$$

The factor $1/18$ remains in (25.23), so the product of the three limits and that factor is exactly

$$\frac{1}{16}\cdot\frac{1}{64}\cdot\frac{2}{18} = \frac{1}{9216}.$$

Substitution into the unaltered lower bound (25.23) gives the actual divergence statement

$$\boxed{\liminf_{j\to\infty} (\log j)^4 j^{-4}\,\|v_{\Gamma,j}\| \geq \frac{1}{9216}.} \tag{25.25}$$

and hence $\|v_{\Gamma,j}\| \to \infty$. The phrase "bounded below by" is intentional: (25.23) is a lower estimate, not an asymptotic equality for the covariance.

For the native projected vector $Z_j = (2/3)\theta_j^2 v_{\Gamma,j}$, the same exact original angle gives

$$\|Z_j\| \geq \frac{2}{3}\theta_j^2 \frac{\xi_j\sigma(16\xi_j)(L_j+1)(2L_j+1)}{18(6+64\xi_j)}. \tag{25.26}$$

Using only the already fixed cusp asymptotic $j^6\theta_j \to 2\pi/10^4$, (25.26) implies the retained lower-rate statement

$$\liminf_{j\to\infty} j^8 (\log j)^4 \|Z_j\| \geq \frac{\pi^2}{3456\,10^8}. \tag{25.27}$$

Because Γ is self-adjoint and $\langle\psi, Q\Gamma\psi\rangle = 0$, the vector norm in (25.23) is exactly the standard vacuum variance:

$$\mathrm{Var}_\rho(\Gamma) := \langle\psi,\Gamma^2\psi\rangle - \langle\psi,\Gamma\psi\rangle^2 = \|Q\Gamma\psi\|^2 \geq \left[\frac{\xi\,\sigma(16\xi)(L+1)(2L+1)}{18(6+64\xi)}\right]^2. \tag{25.28}$$

For the scalar native projection $Z = (2/3)\theta^2 Q\Gamma\psi$, its corresponding squared norm is therefore

$$\|Z\|^2 \geq \frac{4}{9}\theta^4\left[\frac{\xi\,\sigma(16\xi)(L+1)(2L+1)}{18(6+64\xi)}\right]^2. \tag{25.29}$$

On the unchanged logarithmic path, (25.27) equivalently gives the exact variance-rate statement

$$\liminf_{j\to\infty} j^{16}(\log j)^8\|Z_j\|^2 \geq \frac{\pi^4}{3456^2\,10^{16}}. \tag{25.30}$$

These are lower bounds for the explicitly defined vacuum covariances. They do not identify either covariance with a spectral gap, and they do not provide the missing inverse estimate for spectral mass accumulating at zero.

Equations (25.16)–(25.30) are local finite-volume statements evaluated on the actual vacuum, followed by an explicit calculation on the prescribed path. They do not establish a nonzero limiting mass, a probability transport through zero energy, a continuum interacting theory, or the Yang–Mills Millennium conclusion. They do establish that the original weighted covariance cannot be discarded as a vanishing denominator on this path, and they replace the earlier exponential lower bound by a polynomial one. The near-zero spectral inverse and the continuum construction remain separate questions addressed only by further proved calculations.

26. An explicit finite-volume spectral window for the weighted covariance

Section25 gives a lower bound for the total spectral weight of the actual vector $Q\Gamma\psi$, but a covariance lower bound alone does not select an energy interval. The next calculation supplies such an interval at each finite L, using the actual Hamiltonian commutator. It is an unconditional finite-volume statement. Its displayed upper endpoint is not asserted to be uniform in L or in the prescribed cusp.

26.1. Spectral measure and an exact first-moment identity

Let

$$A = H - \mathcal{E}, \qquad v_\Gamma = Q\Gamma\psi, \qquad \nu_\Gamma(B) = \langle v_\Gamma, P_A(B)v_\Gamma\rangle \tag{26.1}$$

for Borel sets $B \subset [0,\infty)$, where P_A is the spectral projection of the finite-box operator A. Since $A\psi = 0$ and $Q\Gamma\psi = \Gamma\psi - \langle\psi,\Gamma\psi\rangle\psi$,

$$[H_0,\Gamma] = 0, \qquad [2bMI,\Gamma] = 0, \qquad [H,\Gamma] = -b[\mathcal{W},\Gamma].$$

The first equality holds because H_0 and Γ are real linear combinations of the mutually commuting link Casimirs; the second holds because M is the scalar plaquette count, not an additional operator. Thus, on the smooth finite-box ground state,

$$Av_\Gamma = A\Gamma\psi = [H,\Gamma]\psi = -b[\mathcal{W},\Gamma]\psi. \tag{26.2}$$

The finite product of compact groups is smooth, the potential is smooth, and elliptic regularity gives $\psi \in C^\infty$. Hence $\Gamma\psi$, v_Γ, and the right side of (26.2) are smooth; in particular $v_\Gamma \in \mathrm{Dom}(A)$, and the first spectral moment below is finite. Since $A \geq 0$, the spectral theorem gives

$$\nu_\Gamma([0,\infty)) = \|v_\Gamma\|^2, \qquad \int_{[0,\infty)} \lambda\, d\nu_\Gamma(\lambda) = \langle v_\Gamma, Av_\Gamma\rangle \leq |\langle v_\Gamma, Av_\Gamma\rangle| \leq b\|v_\Gamma\|\,\|[\mathcal{W},\Gamma]\psi\|. \tag{26.3}$$

26.2. Direct commutator estimate with the original weights

For a multiplication function F, the fixed T_a convention gives

$$\begin{aligned}[F, E_e]\psi &= F\left(-\sum_a X_{e,a}^2\psi\right) + \sum_a X_{e,a}^2(F\psi) \\ &= \left(\sum_{a=1}^{3} X_{e,a}^2 F\right)\psi + 2\sum_{a=1}^{3}(X_{e,a}F)X_{e,a}\psi.\end{aligned} \tag{26.4}$$

This is the product rule with the signs from $E_e = -\sum_a X_{e,a}^2$ left unchanged. Let r_e be the number of plaquettes incident to e. Every incident plaquette trace has

$$\left|\sum_a X_{e,a}^2 W_p\right| = \frac{3}{4}|W_p| \le \frac{3}{2}, \qquad \left(\sum_a |X_{e,a}W_p|^2\right)^{1/2} \le 1. \tag{26.5}$$

The triangle inequality over the r_e incident plaquettes and $\langle\psi, E_e\psi\rangle \le 8\xi$ consequently gives

$$[\mathcal{W}, E_e]\psi = \sum_{p\ni e}[W_p, E_e]\psi, \qquad \|[\mathcal{W}, E_e]\psi\| \le r_e\left(\frac{3}{2} + 2\sqrt{8\xi}\right). \tag{26.6}$$

Indeed, (26.4)–(26.5), $\|\psi\| = 1$, and pointwise Cauchy–Schwarz give for each incident p

$$\|[W_p, E_e]\psi\| \le \frac{3}{2}\|\psi\| + 2\left\|\left(\sum_a |X_{e,a}W_p|^2\right)^{1/2}\right\|_\infty \left(\sum_a \|X_{e,a}\psi\|^2\right)^{1/2} \le \frac{3}{2} + 2\sqrt{8\xi}.$$

Only direction-one edges occur in Γ. Define the exact weighted incidence sum

$$R_{1,L} := \sum_{e=(n,1)} n_2^2 r_e. \tag{26.7}$$

Since $2 \le r_e \le 4$,

$$R_{1,L} \le 4S_{1,L}, \qquad S_{1,L} = \sum_{e=(n,1)} n_2^2 = \frac{2}{3}L^2(L+1)(2L+1)^2. \tag{26.8}$$

Using (26.4)–(26.8) without changing any weight yields

$$\|[\mathcal{W}, \Gamma]\psi\| \le R_{1,L}\left(\frac{3}{2} + 2\sqrt{8\xi}\right) \le S_{1,L}\left(6 + 8\sqrt{8\xi}\right). \tag{26.9}$$

Also, put $m_2(\xi) = 6\xi + 128\xi^2$. The part $\|E_e^2\psi\| \le 6\xi + 128\xi^2$ of (13.2) holds for every $\xi > 0$; its subsequent small-coupling estimate by 7ξ is not used here. Self-adjointness on the smooth vector and Cauchy–Schwarz give

$$\|E_e\psi\|^2 = \langle\psi, E_e^2\psi\rangle \le \|\psi\|\,\|E_e^2\psi\| \le m_2(\xi).$$

Consequently

$$\|\Gamma\psi\| \le \sum_{e=(n,1)} n_2^2\|E_e\psi\| \le S_{1,L}\sqrt{m_2(\xi)}. \tag{26.10}$$

26.3. A proved low-energy window

Put

$$b_\Gamma(L,\xi) := \frac{\xi\,\sigma(16\xi)(L+1)(2L+1)}{18(6+64\xi)}, \tag{26.11}$$

so that (25.23) says $\|v_\Gamma\| \geq b_\Gamma(L,\xi) > 0$. Equations (26.3), (26.9), (26.10), and $\|v_\Gamma\| \leq \|\Gamma\psi\|$ give the first-moment upper bound

$$\int \lambda \, d\nu_\Gamma(\lambda) \leq Q_\Gamma(L,\xi) := b \, S_{1,L}^2 \sqrt{m_2(\xi)} \cdot \left(6 + 8\sqrt{8\xi}\right). \tag{26.12}$$

For $R > 0$, positivity of ν_Γ and $\lambda \geq R$ on (R,∞) imply the exact Markov estimate

$$\nu_\Gamma((R,\infty)) \leq \frac{1}{R} \int \lambda \, d\nu_\Gamma(\lambda). \tag{26.13}$$

Choose

$$R_\Gamma(L,\xi) := \frac{2Q_\Gamma(L,\xi)}{b_\Gamma(L,\xi)^2}. \tag{26.14}$$

Then (26.12)–(26.14) prove

$$\boxed{\nu_\Gamma([0, R_\Gamma(L,\xi)]) \geq \|v_\Gamma\|^2 - \frac{1}{2} b_\Gamma(L,\xi)^2 \geq \frac{1}{2} b_\Gamma(L,\xi)^2 > 0.} \tag{26.15}$$

The interval in (26.15) is an actual spectral window for the weighted covariance vector of the full finite-box Hamiltonian. It includes the endpoint zero only as a Borel endpoint; the ground-state simplicity already implies $\nu_\Gamma(\{0\}) = 0$, because $v_\Gamma \perp \psi$.

On the unchanged logarithmic path, (26.11) gives the explicit liminf lower rate

$$\liminf_{j\to\infty} \frac{(\log j)^8}{j^8} \, b_\Gamma(j^2, \xi_j)^2 \geq \frac{1}{9216^2}. \tag{26.16}$$

while (26.12)–(26.14) retain the original b_j, $S_{1,j}$, and ξ_j coefficients. No uniform positive upper endpoint, no uniform spectral gap, and no continuum mass statement follows from (26.15). What is proved is the finite-volume existence of nonzero actual covariance spectral weight in an explicitly computed interval, with its amount and endpoint both retained rather than replaced by a qualitative assertion.